

SCFT & LST bases

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1 Description

LST with P-type endpoint bases with $T \leq 193$

2 List of P-type LSTs

2.1 0 node(827) **there may exist bases violate the tensor decoupling rule**

2.1.1 Side Links with an extra curve(300)

¹ 1,	¹ 21,	² 21
¹ 221,	² 221,	³ 221
¹ 2221,	² 2221,	³ 2221
¹ 22221,	² 22221,	³ 22221
⁵ 22221,	¹ 222221,	² 222221
³ 222221,	⁵ 222221,	¹ 2222221
² 2222221,	³ 2222221,	⁵ 2222221
¹ 22222221,	² 22222221,	³ 22222221
⁵ 22222221,	¹ 222222221,	² 222222221
³ 222222221,	⁵ 222222221,	¹ 2222222221
² 2222222221,	³ 2222222221,	⁵ 2222222221
¹ 22222222221,	² 22222222221,	³ 22222222221
⁵ 22222222221,	² 131,	¹ 131
² 131,	³ 12321,	² 12321
¹ 12321,	² 12321,	³ 12321
³ 123151321,	² 123151321,	¹ 123151321
² 123151321,	³ 123151321,	⁵ 12231513221
³ 12231513221,	² 12231513221,	¹ 12231513221
² 12231513221,	³ 12231513221,	⁵ 12231513221
³ 1315131,	² 1315131,	¹ 1315131
² 1315131,	³ 1315131,	² 1321
¹ 1321,	² 1321,	³ 1321
² 13221,	¹ 13221,	² 13221
³ 13221,	³ 13151321,	² 13151321
¹ 13151321,	² 13151321,	³ 13151321
³ 131513221,	² 131513221,	¹ 131513221

131513221, ²	131513221, ³	131513221 ⁵
1231513221, ³	1231513221, ²	1231513221 ¹
1231513221, ²	1231513221, ³	1231513221 ⁵
^{1 1} 231,	^{3 1} 231,	^{1 2} 231
^{1 3} 231,	^{3 2} 1231,	^{1 2} 1231
^{2 2} 1231,	^{2 3} 1231,	^{1 1} 2321
^{3 1} 2321,	^{1 2} 2321,	^{1 3} 2321
^{1 2} 321,	^{2 2} 321,	^{2 1} 231
^{2 2} 231,	^{1 1} 2313221,	^{2 2} 2313221
2313221, ³	2313221, ⁵	^{1 1} 22313221
22313221, ²	22313221, ³	22313221 ⁵
^{1 1} 313221,	^{2 2} 313221,	^{3 3} 313221
313221, ⁵	^{1 1} 231321,	^{2 2} 231321
231321, ³	231321, ⁵	^{1 1} 2231321
2231321, ²	2231321, ³	2231321 ⁵
^{1 1} 31321,	^{2 2} 31321,	^{3 3} 31321
^{1 1} 23131,	^{2 2} 23131,	^{3 3} 23131
^{1 1} 223131,	^{2 2} 223131,	^{3 3} 223131
223131, ⁵	^{1 1} 3131,	^{2 2} 3131
3131, ³	^{1 1} 2231,	^{2 1} 15131
^{2 1} 15131,	^{1 1} 15131,	^{1 2} 15131
^{1 3} 15131,	^{2 1} 151321,	^{2 1} 151321
^{1 1} 151321,	^{1 2} 151321,	^{1 3} 151321
^{2 1} 1513221,	^{2 1} 1513221,	^{1 1} 1513221
^{1 2} 1513221,	^{1 3} 1513221,	^{1 5} 1513221
^{1 1} 315131,	^{3 1} 315131,	^{1 2} 315131

$315131,$	$3151321,$	3151321
$3151321,$	$3151321,$	3151321
$31513221,$	$31513221,$	31513221
$31513221,$	$31513221,$	23151321
$23151321,$	$23151321,$	23151321
$231513221,$	$231513221,$	231513221
$231513221,$	$2231513221,$	2231513221
$2231513221,$	$2231513221,$	2231513221
$2315131,$	$2315131,$	2315131
$2315131,$	$223151321,$	223151321
$223151321,$	$223151321,$	223151321
$1512321,$	$1512321,$	1512321
$1512321,$	$321513221,$	321513221
$321513221,$	$321513221,$	231513221
$2321513221,$	$2321513221,$	2321513221
$2321513221,$	$32151321,$	32151321
$32151321,$	$32151321,$	23151321
$232151321,$	$232151321,$	232151321
$232151321,$	$231512321,$	231512321
$231512321,$	$231512321,$	2231512321
$2231512321,$	$2231512321,$	2231512321
$31512321,$	$31512321,$	31512321
$31512321,$	$3215131,$	3215131
$3215131,$	$2315131,$	23215131
$23215131,$	$23215131,$	23215131
$151231,$	$151231,$	151231
$151231,$	$512231,$	512231
$512231,$	$215131,$	215131

$215131,$	$215131,$	2151321
$2151321,$	$2151321,$	2151321
$21513221,$	$21513221,$	21513221
$21513221,$	$21513221,$	15131513221
$15131513221,$	$15131513221,$	15131513221
$15131513221,$	$1513151321,$	1513151321
$1513151321,$	$1513151321,$	1513151321
$151315131,$	$151315131,$	151315131
$151315131,$	$51231513221,$	51231513221
$51231513221,$	$51231513221,$	2315131513221
$2315131513221,$	$2315131513221,$	2315131513221
$22315131513221,$	$22315131513221,$	22315131513221
$22315131513221,$	$315131513221,$	315131513221
$315131513221,$	$315131513221,$	5123151321
$5123151321,$	$5123151321,$	5123151321
$231513151321,$	$231513151321,$	231513151321
$231513151321,$	$31513151321,$	31513151321
$31513151321,$	$31513151321,$	512315131
$512315131,$	$512315131,$	3151315131
$3151315131,$	$3151315131,$	3151315131

2.1.2 Noble Molecules with an extra curve(165)

$1322,$	$1322,$	1322
$2313,$	$2313,$	2313
$23132,$	$231322,$	3132
$2132,$	$2132,$	2132
$3123,$	$3123,$	21322
$21322,$	$21322,$	23213
$23213,$	$23213,$	213

$213,$	$211,$	1511
$1 \overset{1,1}{5} 1,$	$1511,$	1511
$15132,$	$15132,$	15132
$1512,$	$1512,$	1512
$1512,$	$23151,$	22315131
$22315131,$	$22315131,$	22315131
$22315131,$	$315132,$	315132
$315132,$	$2315132,$	2315132
$2315132,$	$22315132,$	22315132
$22315132,$	$22315132,$	51322
$51322,$	$51232,$	51232
$31512,$	$31512,$	31512
$31512,$	$231512,$	231512
$231512,$	$2231512,$	2231512
$2231512,$	$2231512,$	2231512
$31513,$	$31513,$	13215132
$13215132,$	$13215132,$	13215132
$223151231,$	$223151231,$	223151231
$223151231,$	$3151231,$	3151231
$3151231,$	$31513,$	31513
$31513,$	$231513,$	2231513
$22215,$	$22215,$	313215
$313215,$	$231315,$	231315
$215132,$	$2215132,$	2215132
$2215132,$	$315123,$	215123
$215123,$	$215123,$	22151322
$22151322,$	$22151322,$	22151322

$\overset{3}{2}21513,$	$\overset{2}{2}21513,$	$\overset{1}{2}21513$
$\overset{3}{2}151232,$	$\overset{2}{2}151232,$	$\overset{1}{2}151232$
$\overset{2}{2}1512,$	$\overset{1}{2}1512,$	$\overset{2}{2}1512$
$\overset{3}{3}1315132,$	$\overset{2}{3}1315132,$	$\overset{1}{3}1315132$
$\overset{5}{3}13151322,$	$\overset{3}{3}13151322,$	$\overset{2}{3}13151322$
$\overset{1}{3}13151322,$	$\overset{3}{3}131513,$	$\overset{2}{3}131513$
$\overset{1}{3}131513,$	$\overset{1}{2}231513151321,$	$\overset{2}{2}231513151321$
$\overset{3}{2}231513151321,$	$\overset{5}{2}231513151321,$	$\overset{5}{1}3151315132$
$\overset{3}{1}3151315132,$	$\overset{2}{1}3151315132,$	$\overset{1}{1}3151315132$
$\overset{5}{1}31513151322,$	$\overset{3}{1}31513151322,$	$\overset{2}{1}31513151322$
$\overset{1}{1}31513151322,$	$\overset{1}{3}1513215,$	$\overset{3}{2}1513215$
$\overset{2}{2}1513215,$	$\overset{1}{2}1513215,$	$\overset{2}{3}1512315$
$\overset{1}{3}1512315,$	$\overset{1}{2}315132215,$	$\overset{2}{2}315132215$
$\overset{3}{2}315132215,$	$\overset{1}{2}2315132215,$	$\overset{2}{2}2315132215$
$\overset{3}{2}2315132215,$	$\overset{5}{2}2315132215,$	$\overset{1}{3}15132215$
$\overset{2}{3}15132215,$	$\overset{3}{3}15132215,$	$\overset{2}{3}32151315$
$\overset{1}{3}32151315,$	$\overset{1}{2}3151315,$	$\overset{5}{2}151315132$
$\overset{3}{2}151315132,$	$\overset{2}{2}151315132,$	$\overset{1}{2}151315132$
$\overset{5}{2}1513151322,$	$\overset{3}{2}1513151322,$	$\overset{2}{2}1513151322$
$\overset{1}{2}1513151322,$	$\overset{3}{2}15131513,$	$\overset{2}{2}15131513$
$\overset{1}{2}15131513,$	$\overset{1}{5}131513215,$	$\overset{2}{5}131513215$

2.1.3 A,D series like LSTs(362)

$$\begin{array}{l} \overbrace{1222 \cdots 2}^n \overset{1}{1} \quad 12 \leq n \leq 192 \\ \overbrace{222 \cdots 2}^n \overset{1}{1} \quad 12 \leq n \leq 192 \end{array}$$

2.2 1 node(969)

2.2.1 Sg(8)

$22313221\epsilon'_7,$	$2321513221\epsilon'_7,$	$2231512321\epsilon'_7$
$315131513221\epsilon'_7,$	$231513151321\epsilon'_7,$	$\overset{1}{2}31513221\epsilon'_7$

$$223151321\epsilon'_7$$

$$1321\mathfrak{so}$$

2.2.2 $^x Sg(57)$

$$\begin{array}{lll} \overset{3}{1321\mathfrak{so}}, & \overset{2}{1321\mathfrak{so}}, & \overset{3}{12321\mathfrak{so}} \\ \overset{2}{12321\mathfrak{so}}, & \overset{5}{12315131\mathfrak{so}}, & \overset{1}{3221\mathfrak{so}} \\ \overset{2}{221\mathfrak{so}}, & \overset{2}{3131\mathfrak{so}} & \\ \overset{1}{2231\mathfrak{so}}, & \overset{1}{512321\mathfrak{so}}, & \overset{1}{23151321\mathfrak{so}} \\ \overset{2}{23151321\mathfrak{so}}, & \overset{1}{223151321\mathfrak{so}}, & \overset{2}{223151321\mathfrak{so}} \\ \overset{1}{3151321\mathfrak{so}}, & \overset{1}{315131\mathfrak{so}}, & \overset{3}{151231\mathfrak{so}} \\ \overset{3}{215131\mathfrak{so}}, & \overset{1}{51315131\mathfrak{so}}, & \overset{2}{231\mathfrak{so}} \\ \overset{2}{151321\mathfrak{so}}, & \overset{1}{151321\mathfrak{so}}, & \overset{2}{1231\mathfrak{so}} \\ \overset{3}{15131\mathfrak{so}} & & \end{array}$$

$$\begin{array}{lll} \overset{5}{1231513221\epsilon_6}, & \overset{2}{313221\epsilon_6}, & \overset{2}{231321\epsilon_6} \\ \overset{2}{22221\epsilon_6}, & \overset{2}{223131\epsilon_6}, & \overset{1}{31513221\epsilon_6} \\ \overset{2}{32151321\epsilon_6}, & \overset{1}{23151321\epsilon_6}, & \overset{2}{31512321\epsilon_6} \\ \overset{1}{22315131\epsilon_6}, & \overset{2}{23215131\epsilon_6}, & \overset{3}{21513221\epsilon_6} \\ \overset{1}{5131513221\epsilon_6}, & \overset{2}{3151315131\epsilon_6}, & \overset{2}{3151321\epsilon_6} \\ \overset{2}{2315131\epsilon_6}, & \overset{3}{1513151321\epsilon_6}, & \overset{3}{1513221\epsilon_6} \end{array}$$

$$\begin{array}{ll} \overset{2}{222221\epsilon'_7}, & \overset{2}{2231321\epsilon'_7} \\ \overset{2}{232151321\epsilon'_7} & \\ \overset{2}{31513221\epsilon'_7}, & \overset{2}{23151321\epsilon'_7} \end{array}$$

$$\begin{array}{lll} \overset{2}{22313221\epsilon_7}, & \overset{2}{2222221\epsilon_7}, & \overset{1}{2231513221\epsilon_7} \\ \overset{2}{2321513221\epsilon_7}, & \overset{2}{2231512321\epsilon_7}, & \overset{2}{315131513221\epsilon_7} \\ \overset{2}{231513151321\epsilon_7}, & \overset{2}{231513221\epsilon_7}, & \overset{2}{223151321\epsilon_7} \end{array}$$

$$\overset{2}{2222222221\epsilon_8}$$

2.2.3 SgS(388)

$22313221\mathfrak{e}_7,$	$I^{\oplus 7}\mathfrak{e}_7,$	$2321513221\mathfrak{e}_7$
$2231512321\mathfrak{e}_7,$	$315131513221\mathfrak{e}_7,$	$231513151321\mathfrak{e}_7$
$231513221\mathfrak{e}_7,$	$223151321\mathfrak{e}_7,$	$3,5\begin{smallmatrix}2,3\\\otimes\end{smallmatrix}\mathfrak{e}_7$
$4,5\begin{smallmatrix}2,3\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$5,5\begin{smallmatrix}2,3\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$313221\mathfrak{e}_7$
$231321\mathfrak{e}_7,$	$I^{\oplus 5}\mathfrak{e}_7,$	$32151321\mathfrak{e}_7$
$31512321\mathfrak{e}_7,$	$21513221\mathfrak{e}_7,$	$5123151321\mathfrak{e}_7$
$3151321\mathfrak{e}_7,$	$1513151321\mathfrak{e}_7,$	$1513221\mathfrak{e}_7$
$2,4\begin{smallmatrix}2,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$3,4\begin{smallmatrix}2,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$4,4\begin{smallmatrix}2,4\\\otimes\end{smallmatrix}\mathfrak{e}_7$
$5,4\begin{smallmatrix}2,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$I^{\oplus 4}\mathfrak{e}_7,$	$31321\mathfrak{e}_7$
$2151321\mathfrak{e}_7,$	$2321\mathfrak{e}_7,$	$1512321\mathfrak{e}_7$
$151321\mathfrak{e}_7,$	$3,5\begin{smallmatrix}3,3\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$4,5\begin{smallmatrix}3,3\\\otimes\end{smallmatrix}\mathfrak{e}_7$
$5,5\begin{smallmatrix}3,3\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$313221\mathfrak{e}_7,$	$231321\mathfrak{e}_7$
$I^{\oplus 5}\mathfrak{e}_7,$	$32151321\mathfrak{e}_7,$	$31512321\mathfrak{e}_7$
$21513221\mathfrak{e}_7,$	$5123151321\mathfrak{e}_7,$	$3151321\mathfrak{e}_7$
$1513151321\mathfrak{e}_7,$	$1513221\mathfrak{e}_7,$	$3,4\begin{smallmatrix}3,4\\\otimes\end{smallmatrix}\mathfrak{e}_7$
$4,4\begin{smallmatrix}3,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$5,4\begin{smallmatrix}3,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$I^{\oplus 4}\mathfrak{e}_7$
$31321\mathfrak{e}_7,$	$2151321\mathfrak{e}_7,$	$2321\mathfrak{e}_7$
$1512321\mathfrak{e}_7,$	$151321\mathfrak{e}_7,$	$I^{\oplus 3}\mathfrak{e}_7$
$4,4\begin{smallmatrix}4,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$5,4\begin{smallmatrix}4,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$I^{\oplus 4}\mathfrak{e}_7$
$31321\mathfrak{e}_7,$	$2151321\mathfrak{e}_7,$	$2321\mathfrak{e}_7$
$1512321\mathfrak{e}_7,$	$151321\mathfrak{e}_7,$	$I^{\oplus 3}\mathfrak{e}_7$
$5,4\begin{smallmatrix}5,4\\\otimes\end{smallmatrix}\mathfrak{e}_7,$	$I^{\oplus 4}\mathfrak{e}_7,$	$31321\mathfrak{e}_7$
$2151321\mathfrak{e}_7,$	$2321\mathfrak{e}_7,$	$1512321\mathfrak{e}_7$
$151321\mathfrak{e}_7,$	$I^{\oplus 3}\mathfrak{e}_7,$	$513151321\mathfrak{e}_7$

$$\begin{array}{lll}
\begin{array}{l}
3,3 \overset{2,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
3,3 \overset{2,3}{\otimes} \\
\bigcirc \mathfrak{e}_6, \\
215131 \overset{2,3}{\otimes} \mathfrak{e}_6, \\
3,3 \overset{3,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
3,3 \overset{3,3}{\otimes} \\
\bigcirc \mathfrak{e}_6, \\
215131 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
4,3 \overset{4,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
I^{\oplus 3} \overset{4,3}{\otimes} \mathfrak{e}_6, \\
1 \overset{4,3}{\otimes} \\
15131 \mathfrak{e}_6, \\
3,3 \overset{5,3}{\otimes} \\
\bigcirc \mathfrak{e}_6, \\
215131 \overset{5,3}{\otimes} \mathfrak{e}_6, \\
321 \overset{3221}{\mathfrak{e}_6}, \\
51315131 \overset{321}{\mathfrak{e}_6}, \\
3,3 \overset{3,3}{\otimes} \\
\bigcirc \mathfrak{e}_6, \\
215131 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
31321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
2151321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
1 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
1 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
151321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
215131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
1513221 \overset{31}{\mathfrak{e}_6}, \\
1 \overset{3131}{\mathfrak{e}_6}, \\
15131 \overset{3131}{\mathfrak{e}_6}, \\
5131 \overset{3151321}{\mathfrak{e}_6}, \\
15131 \overset{51315131}{\mathfrak{e}_6},
\end{array}
&
\begin{array}{l}
4,3 \overset{2,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
I^{\oplus 3} \overset{2,3}{\otimes} \mathfrak{e}_6, \\
1 \overset{2,3}{\otimes} \\
15131 \mathfrak{e}_6, \\
4,3 \overset{3,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
I^{\oplus 3} \overset{3,3}{\otimes} \mathfrak{e}_6, \\
1 \overset{3,3}{\otimes} \\
15131 \mathfrak{e}_6, \\
5,3 \overset{4,3}{\otimes} \\
\otimes \mathfrak{e}_6, \\
3131 \overset{4,3}{\otimes} \mathfrak{e}_6, \\
I^{\oplus 2} \overset{4,4}{\otimes} \mathfrak{e}_6, \\
I^{\oplus 3} \overset{5,3}{\otimes} \mathfrak{e}_6, \\
1 \overset{5,3}{\otimes} \\
15131 \mathfrak{e}_6, \\
15131 \overset{3221}{\mathfrak{e}_6}, \\
151321 \overset{321}{\mathfrak{e}_6}, \\
I^{\oplus 3} \overset{3,3}{\otimes} \mathfrak{e}_6, \\
1 \overset{3,3}{\otimes} \\
15131 \mathfrak{e}_6, \\
23131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
512315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
1512321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
I^{\oplus 3} \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
1 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
15131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
3131 \overset{3131}{\mathfrak{e}_6}, \\
315131 \overset{51321}{\mathfrak{e}_6}, \\
215131 \overset{215131}{\mathfrak{e}_6}, \\
151321 \overset{15131}{\mathfrak{e}_6},
\end{array}
&
\begin{array}{l}
5,3 \overset{2,3}{\otimes} \\
\otimes \mathfrak{e}_6 \\
3131 \overset{2,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 2} \overset{2,4}{\otimes} \mathfrak{e}_6 \\
5,3 \overset{3,3}{\otimes} \\
\otimes \mathfrak{e}_6 \\
3131 \overset{3,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 2} \overset{3,4}{\otimes} \mathfrak{e}_6 \\
3,3 \overset{4,3}{\otimes} \\
\bigcirc \mathfrak{e}_6 \\
215131 \overset{4,3}{\otimes} \mathfrak{e}_6 \\
5,3 \overset{5,3}{\otimes} \\
\otimes \mathfrak{e}_6 \\
3131 \overset{5,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 2} \overset{5,4}{\otimes} \mathfrak{e}_6 \\
512321 \overset{321}{\mathfrak{e}_6} \\
51321 \overset{2321}{\mathfrak{e}_6} \\
3131 \overset{3,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 4} \overset{I^{\oplus 2}}{\mathfrak{e}_6} \\
3215131 \overset{I^{\oplus 2}}{\mathfrak{e}_6} \\
1 \overset{I^{\oplus 2}}{\mathfrak{e}_6} \\
2321 \overset{I^{\oplus 2}}{\mathfrak{e}_6} \\
151315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6} \\
3131 \overset{I^{\oplus 3}}{\mathfrak{e}_6} \\
513151321 \overset{31}{\mathfrak{e}_6} \\
215131 \overset{3131}{\mathfrak{e}_6} \\
15131 \overset{512321}{\mathfrak{e}_6} \\
1 \overset{215131}{\mathfrak{e}_6} \\
15131 \overset{15131}{\mathfrak{e}_6}
\end{array}
\end{array}$$

$$I^{\oplus 11} \overset{1}{\mathfrak{e}_8},$$

$$I^{\oplus 8} \overset{2,4}{\otimes} \mathfrak{e}_8,$$

$$2315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8,$$

$$22313221 \overset{3,5}{\otimes} \mathfrak{e}_8,$$

$$2231513221 \overset{2,4}{\otimes} \mathfrak{e}_8$$

$$I^{\oplus 7} \overset{3,5}{\otimes} \mathfrak{e}_8$$

$$\begin{array}{lll}
2321513221 \overset{3,5}{\otimes} \mathfrak{e}_8, & 315131513221 \overset{3,5}{\otimes} \mathfrak{e}_8, & 231513221 \overset{3,5}{\otimes} \mathfrak{e}_8 \\
22313221 \overset{4,5}{\otimes} \mathfrak{e}_8, & I^{\oplus 7} \overset{4,5}{\otimes} \mathfrak{e}_8, & 2321513221 \overset{4,5}{\otimes} \mathfrak{e}_8 \\
315131513221 \overset{4,5}{\otimes} \mathfrak{e}_8, & 231513221 \overset{4,5}{\otimes} \mathfrak{e}_8, & 22313221 \overset{5,5}{\otimes} \mathfrak{e}_8 \\
I^{\oplus 7} \overset{5,5}{\otimes} \mathfrak{e}_8, & 2321513221 \overset{5,5}{\otimes} \mathfrak{e}_8, & 315131513221 \overset{5,5}{\otimes} \mathfrak{e}_8 \\
231513221 \overset{5,5}{\otimes} \mathfrak{e}_8, & 2313221 \overset{2313221}{\mathfrak{e}_8}, & I^{\oplus 6} \overset{2313221}{\mathfrak{e}_8} \\
321513221 \overset{2313221}{\mathfrak{e}_8}, & 51231513221 \overset{2313221}{\mathfrak{e}_8}, & 31513221 \overset{2313221}{\mathfrak{e}_8} \\
15131513221 \overset{2313221}{\mathfrak{e}_8}, & 313221 \overset{22313221}{\mathfrak{e}_8}, & I^{\oplus 5} \overset{22313221}{\mathfrak{e}_8} \\
21513221 \overset{22313221}{\mathfrak{e}_8}, & 1513221 \overset{22313221}{\mathfrak{e}_8}, & I^{\oplus 7} \overset{313221}{\mathfrak{e}_8} \\
2321513221 \overset{313221}{\mathfrak{e}_8}, & 315131513221 \overset{313221}{\mathfrak{e}_8}, & 231513221 \overset{313221}{\mathfrak{e}_8} \\
I^{\oplus 10} \overset{I^{\oplus 2}}{\mathfrak{e}_8}, & I^{\oplus 9} \overset{I^{\oplus 3}}{\mathfrak{e}_8}, & 22315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} \\
2315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8}, & 2231513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8}, & I^{\oplus 8} \overset{I^{\oplus 4}}{\mathfrak{e}_8} \\
I^{\oplus 7} \overset{I^{\oplus 5}}{\mathfrak{e}_8}, & 2321513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8}, & 315131513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} \\
231513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8}, & I^{\oplus 6} \overset{I^{\oplus 6}}{\mathfrak{e}_8}, & 321513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} \\
51231513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8}, & 31513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8}, & 15131513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} \\
21513221 \overset{I^{\oplus 7}}{\mathfrak{e}_8}, & 1513221 \overset{I^{\oplus 7}}{\mathfrak{e}_8}, & 321513221 \overset{321513221}{\mathfrak{e}_8} \\
51231513221 \overset{321513221}{\mathfrak{e}_8}, & 31513221 \overset{321513221}{\mathfrak{e}_8}, & 15131513221 \overset{321513221}{\mathfrak{e}_8} \\
21513221 \overset{2321513221}{\mathfrak{e}_8}, & 1513221 \overset{2321513221}{\mathfrak{e}_8}, & 315131513221 \overset{21513221}{\mathfrak{e}_8} \\
231513221 \overset{1}{\mathfrak{e}_8} \overset{21513221}{\mathfrak{e}_8}, & 51231513221 \overset{51231513221}{\mathfrak{e}_8}, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{51231513221}{\mathfrak{e}_8} \\
15131513221 \overset{51231513221}{\mathfrak{e}_8}, & 1513221 \overset{315131513221}{\mathfrak{e}_8}, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{31513221}{\mathfrak{e}_8} \\
15131513221 \overset{31513221}{\mathfrak{e}_8}, & 1513221 \overset{231513221}{\mathfrak{e}_8}, & 15131513221 \overset{15131513221}{\mathfrak{e}_8}
\end{array}$$

$$\begin{array}{lll}
2313221 \overset{1}{\mathfrak{e}'_7}, & I^{\oplus 6} \overset{1}{\mathfrak{e}'_7}, & 2231321 \overset{1}{\mathfrak{e}'_7} \\
321513221 \overset{1}{\mathfrak{e}'_7}, & 232151321 \overset{1}{\mathfrak{e}'_7}, & 231512321 \overset{1}{\mathfrak{e}'_7} \\
51231513221 \overset{1}{\mathfrak{e}'_7}, & 31513151321 \overset{1}{\mathfrak{e}'_7}, & 31513221 \overset{1}{\mathfrak{e}'_7} \\
23151321 \overset{1}{\mathfrak{e}'_7}, & 15131513221 \overset{1}{\mathfrak{e}'_7}, & \overset{2,3}{2,4} \otimes \overset{2,3}{\mathfrak{e}'_7} \\
\overset{2,3}{3,4} \otimes \overset{2,3}{\mathfrak{e}'_7}, & \overset{2,3}{4,4} \otimes \overset{2,3}{\mathfrak{e}'_7}, & \overset{2,3}{5,4} \otimes \overset{2,3}{\mathfrak{e}'_7} \\
I^{\oplus 4} \overset{2,3}{\mathfrak{e}'_7}, & 31321 \overset{2,3}{\mathfrak{e}'_7}, & 2151321 \overset{2,3}{\mathfrak{e}'_7}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
\overset{1}{2321}\overset{2,3}{\otimes}\mathfrak{e}'_7, \\
\overset{3,3}{\otimes}\overset{2,4}{\otimes}\mathfrak{e}'_7, \\
\overset{4,4}{\otimes}\overset{3,3}{\otimes}\mathfrak{e}'_7, \\
31321\overset{3,3}{\otimes}\mathfrak{e}'_7, \\
1512321\overset{3,3}{\otimes}\mathfrak{e}'_7, \\
I^{\oplus 2}\overset{3,5}{\otimes}\mathfrak{e}'_7, \\
I^{\oplus 3}\overset{5,4}{\otimes}\mathfrak{e}'_7, \\
512321\overset{3221}{\mathfrak{e}'_7}, \\
513151321\overset{321}{\mathfrak{e}'_7}, \\
I^{\oplus 2}\overset{231321}{\mathfrak{e}'_7}, \\
31512321\overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
3151321\overset{1}{\mathfrak{e}'_7}, \\
I^{\oplus 4}\overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
2321\overset{1}{\mathfrak{e}'_7}, \\
3151321\overset{51321}{\mathfrak{e}'_7}, \\
151321\overset{151321}{\mathfrak{e}'_7}
\end{array}
&
\begin{array}{l}
\overset{1512321}{\overset{2,3}{\otimes}}\mathfrak{e}'_7, \\
I^{\oplus 3}\overset{2,4}{\otimes}\mathfrak{e}'_7, \\
\overset{5,4}{\otimes}\overset{3,3}{\otimes}\mathfrak{e}'_7, \\
2151321\overset{3,3}{\otimes}\mathfrak{e}'_7, \\
151321\overset{1}{\overset{3,3}{\otimes}}\mathfrak{e}'_7, \\
I^{\oplus 3}\overset{4,4}{\otimes}\mathfrak{e}'_7, \\
I^{\oplus 2}\overset{5,5}{\otimes}\mathfrak{e}'_7, \\
151321\overset{3221}{\mathfrak{e}'_7}, \\
1513221\overset{321}{\mathfrak{e}'_7}, \\
I^{\oplus 5}\overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
21513221\overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
1513151321\overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
31321\overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
1512321\overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
512321\overset{512321}{\mathfrak{e}'_7},
\end{array}
&
\begin{array}{l}
\overset{151321}{\overset{2,3}{\otimes}}\mathfrak{e}'_7 \\
\overset{3,3}{\otimes}\overset{3,4}{\otimes}\mathfrak{e}'_7 \\
I^{\oplus 4}\overset{3,3}{\otimes}\mathfrak{e}'_7 \\
2321\overset{1}{\overset{3,3}{\otimes}}\mathfrak{e}'_7 \\
I^{\oplus 3}\overset{3,4}{\otimes}\mathfrak{e}'_7 \\
I^{\oplus 2}\overset{4,5}{\otimes}\mathfrak{e}'_7 \\
3221\overset{3221}{\mathfrak{e}'_7} \\
I^{\oplus 2}\overset{313221}{\mathfrak{e}'_7} \\
513221\overset{2321}{\mathfrak{e}'_7} \\
32151321\overset{I^{\oplus 2}}{\mathfrak{e}'_7} \\
5123151321\overset{I^{\oplus 2}}{\mathfrak{e}'_7} \\
1513221\overset{1}{\mathfrak{e}'_7} \\
2151321\overset{I^{\oplus 3}}{\mathfrak{e}'_7} \\
151321\overset{1}{\mathfrak{e}'_7} \\
151321\overset{512321}{\mathfrak{e}'_7}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
\overset{2,3}{\otimes}\overset{1}{\mathfrak{so}}, \\
\overset{5,3}{\otimes}\overset{1}{\mathfrak{so}}, \\
3131\overset{1}{\mathfrak{so}}, \\
215131\overset{1}{\mathfrak{so}}, \\
15131\overset{1}{\mathfrak{so}}, \\
\overset{4,2}{\otimes}\overset{2,2}{\otimes}\mathfrak{so}, \\
\overset{3,2}{\otimes}\overset{3,2}{\otimes}\mathfrak{so}, \\
I^{\oplus 2}\overset{3,2}{\otimes}\mathfrak{so}, \\
I^{\oplus 2}\overset{4,2}{\otimes}\mathfrak{so},
\end{array}
&
\begin{array}{l}
\overset{3,3}{\otimes}\overset{1}{\mathfrak{so}}, \\
\overset{3,3}{\bigcirc}\overset{1}{\mathfrak{so}}, \\
151231\overset{1}{\mathfrak{so}}, \\
231\overset{1}{\mathfrak{so}}, \\
\overset{2,2}{\otimes}\overset{2,2}{\otimes}\mathfrak{so}, \\
231\overset{2}{\overset{2,2}{\otimes}}\mathfrak{so}, \\
\overset{4,2}{\otimes}\overset{3,2}{\otimes}\mathfrak{so}, \\
\overset{4,2}{\otimes}\overset{4,2}{\otimes}\mathfrak{so}, \\
I^{\oplus 2}\overset{2}{\overset{231}{\mathfrak{so}}},
\end{array}
&
\begin{array}{l}
\overset{4,3}{\otimes}\overset{1}{\mathfrak{so}} \\
I^{\oplus 3}\overset{1}{\mathfrak{so}} \\
512231\overset{1}{\mathfrak{so}} \\
1231\overset{2}{\mathfrak{so}} \\
\overset{3,2}{\otimes}\overset{2,2}{\otimes}\mathfrak{so} \\
I^{\oplus 2}\overset{2,2}{\otimes}\mathfrak{so} \\
231\overset{2}{\overset{3,2}{\otimes}}\mathfrak{so} \\
231\overset{2}{\overset{4,2}{\otimes}}\mathfrak{so} \\
31\overset{321}{\mathfrak{so}}
\end{array}
\end{array}$$

$$I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{so}},$$

$$5131 \overset{231}{\mathfrak{so}}$$

$$51231 \overset{31}{\mathfrak{so}},$$

$$15131 \overset{31}{\mathfrak{so}}$$

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$$I^{\oplus 10} \overset{1}{\mathfrak{e}_8} 1,$$

$$2321513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$313221 \overset{2313221}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{2313221}{\mathfrak{e}_8} 1,$$

$$321513221 \overset{313221}{\mathfrak{e}_8} 1,$$

$$15131513221 \overset{313221}{\mathfrak{e}_8} 1,$$

$$2315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 7} \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1,$$

$$231513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1,$$

$$51231513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1,$$

$$21513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{321513221}{\mathfrak{e}_8} 1,$$

$$15131513221 \overset{21513221}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{31513221}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 4} \overset{2,4}{\otimes} \overset{4,2}{\mathfrak{e}_8},$$

$$I^{\oplus 3} \overset{5,5}{\otimes} \overset{4,2}{\mathfrak{e}_8},$$

$$I^{\oplus 3} \overset{313221}{\mathfrak{e}_8} \overset{4,2}{\otimes},$$

$$51231513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes,$$

$$22313221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 5} \overset{2313221}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 4} \overset{22313221}{\mathfrak{e}_8} 1,$$

$$51231513221 \overset{313221}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 9} \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1,$$

$$2231513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1,$$

$$2321513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 6} \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1,$$

$$31513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} 1,$$

$$51231513221 \overset{21513221}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{51231513221}{\mathfrak{e}_8} 1,$$

$$1513221 \overset{15131513221}{\mathfrak{e}_8} 1,$$

$$I^{\oplus 3} \overset{3,5}{\otimes} \overset{4,2}{\mathfrak{e}_8},$$

$$1513221 \overset{3221}{\mathfrak{e}_8} \overset{4,2}{\otimes},$$

$$I^{\oplus 6} \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes,$$

$$31513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes,$$

$$I^{\oplus 7} \overset{2,4}{\otimes} \mathfrak{e}_8 1$$

$$231513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1$$

$$21513221 \overset{2313221}{\mathfrak{e}_8} 1$$

$$I^{\oplus 6} \overset{313221}{\mathfrak{e}_8} 1$$

$$31513221 \overset{313221}{\mathfrak{e}_8} 1$$

$$22315131513221 \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1$$

$$I^{\oplus 8} \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1$$

$$315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1$$

$$321513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1$$

$$15131513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1$$

$$21513221 \overset{321513221}{\mathfrak{e}_8} 1$$

$$31513221 \overset{21513221}{\mathfrak{e}_8} 1$$

$$5131513221 \overset{5131513221}{\mathfrak{e}_8} 1$$

$$\overset{2,4}{\otimes} \overset{2,4}{\mathfrak{e}_8} \overset{4,2}{\otimes}$$

$$I^{\oplus 3} \overset{4,5}{\otimes} \overset{4,2}{\mathfrak{e}_8}$$

$$I^{\oplus 2} \overset{2313221}{\mathfrak{e}_8} \overset{4,2}{\otimes}$$

$$321513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes$$

$$15131513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes$$

$$\begin{array}{lll}
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 34,2}}, & 21513221 \mathfrak{e}_8^{I^{\oplus 34,2}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 34,2}} \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 44,2}}, & I^{\oplus 2} \mathfrak{e}_8^{3,5 \otimes 5,3}, & I^{\oplus 2} \mathfrak{e}_8^{4,5 \otimes 5,3} \\
I^{\oplus 2} \mathfrak{e}_8^{5,5 \otimes 5,3}, & 3221 \mathfrak{e}_8^{3221^{5,3}}, & I^{\oplus 2} \mathfrak{e}_8^{313221^{5,3}} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 25,3}}, & 21513221 \mathfrak{e}_8^{I^{\oplus 25,3}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 25,3}} \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 35,3}}, & I^{\oplus 2} \mathfrak{e}_8^{4,5 \otimes 5,4}, & I^{\oplus 2} \mathfrak{e}_8^{5,5 \otimes 5,4} \\
3221 \mathfrak{e}_8^{3221^{5,4}}, & I^{\oplus 2} \mathfrak{e}_8^{313221^{5,4}}, & I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 25,4}} \\
21513221 \mathfrak{e}_8^{I^{\oplus 25,4}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 25,4}}, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 35,4}} \\
I^{\oplus 2} \mathfrak{e}_8^{5,5 \otimes 5,5}, & 3221 \mathfrak{e}_8^{3221^{5,5}}, & I^{\oplus 2} \mathfrak{e}_8^{313221^{5,5}} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 25,5}}, & 21513221 \mathfrak{e}_8^{I^{\oplus 25,5}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 25,5}} \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 35,5}}, & 313221 \mathfrak{e}_8^{3221} 1223, & I^{\oplus 5} \mathfrak{e}_8^{3221} 1223 \\
21513221 \mathfrak{e}_8^{3221} 1223, & 1513221 \mathfrak{e}_8^{3221} 1223, & 5131513221 \mathfrak{e}_8^{I^{\oplus 3}} 1223 \\
1513221 \mathfrak{e}_8^{I^{\oplus 4}} 1223, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} 1223132, & I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} 1223132 \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} 12231322, & I^{\oplus 2} \mathfrak{e}_8^{313221} 122313, & I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}} 122313 \\
21513221 \mathfrak{e}_8^{I^{\oplus 2}} 122313, & 1513221 \mathfrak{e}_8^{I^{\oplus 2}} 122313, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} 122313 \\
2315131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & 2231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & I^{\oplus 8} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2} \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 2321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 315131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2} \\
231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & 321513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & 31513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & 21513221 \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & 1513221 \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2} \\
21513221 \mathfrak{e}_8^{21513221} I^{\oplus 2}, & 1513221 \mathfrak{e}_8^{21513221} I^{\oplus 2}, & 1513221 \mathfrak{e}_8^{5131513221} I^{\oplus 2} \\
1513221 \mathfrak{e}_8^{1513221} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & 321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & 31513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & 21513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & 1513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3} \\
31513221 \mathfrak{e}_8^{513221} I^{\oplus 3}, & 1513221 \mathfrak{e}_8^{1513221} I^{\oplus 3}, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 4} \\
\\
2313221 \mathfrak{e}_7^1, & I^{\oplus 6} \mathfrak{e}_7^1, & 2231321 \mathfrak{e}_7^1 \\
321513221 \mathfrak{e}_7^1, & 232151321 \mathfrak{e}_7^1, & 231512321 \mathfrak{e}_7^1 \\
51231513221 \mathfrak{e}_7^1, & 31513151321 \mathfrak{e}_7^1, & 31513221 \mathfrak{e}_7^1 \\
23151321 \mathfrak{e}_7^1, & 15131513221 \mathfrak{e}_7^1, & 2,4 \otimes^2,3 \\
& & \otimes \mathfrak{e}_7^1
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^{3,4} \otimes {}^{2,3} \\
\otimes \mathfrak{e}_7 1, \\
I^{\oplus 4} \otimes {}^{2,3} \\
\mathfrak{e}_7 1, \\
{}^1 \otimes {}^{2,3} \\
2321 \mathfrak{e}_7 1, \\
{}^{3,3} \otimes {}^{2,4} \\
\otimes \mathfrak{e}_7 1, \\
{}^{4,4} \otimes {}^{3,3} \\
\otimes \mathfrak{e}_7 1, \\
{}^{3,3} \otimes {}^{3,3} \\
31321 \mathfrak{e}_7 1, \\
{}^{3,3} \otimes {}^{3,3} \\
1512321 \mathfrak{e}_7 1, \\
{}^{3,5} \otimes {}^{3,5} \\
I^{\oplus 2} \mathfrak{e}_7 1, \\
{}^{5,4} \otimes {}^{5,4} \\
I^{\oplus 3} \mathfrak{e}_7 1, \\
512321 \mathfrak{e}_7^{3221} 1, \\
513151321 \mathfrak{e}_7^{321} 1, \\
I^{\oplus 2} \mathfrak{e}_7^{231321} 1, \\
31512321 \mathfrak{e}_7^{I^{\oplus 2}} 1, \\
{}^1 \mathfrak{e}_7^{I^{\oplus 2}} 1, \\
I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 3}} 1, \\
{}^1 \mathfrak{e}_7^{I^{\oplus 3}} 1, \\
3151321 \mathfrak{e}_7^{51321} 1, \\
151321 \mathfrak{e}_7^{151321} 1, \\
321 \mathfrak{e}_7^{3213,2} \otimes, \\
I^{\oplus 2} \mathfrak{e}_7^{3,3} \otimes {}^{3,3}, \\
I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 24,3}} \otimes, \\
I^{\oplus 2} \mathfrak{e}_7^{321} 1223, \\
151321 \mathfrak{e}_7^{I^{\oplus 2}} 123, \\
31321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}, \\
1512321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}, \\
{}^{2,4} \mathfrak{e}_6^1, \\
{}^{5,4} \mathfrak{e}_6^1,
\end{array}
&
\begin{array}{l}
{}^{4,4} \otimes {}^{2,3} \\
\otimes \mathfrak{e}_7 1, \\
{}^{2,3} \otimes {}^{2,3} \\
31321 \mathfrak{e}_7 1, \\
{}^{2,3} \otimes {}^{2,3} \\
1512321 \mathfrak{e}_7 1, \\
{}^{2,4} \otimes {}^{2,4} \\
I^{\oplus 3} \mathfrak{e}_7 1, \\
{}^{5,4} \otimes {}^{3,3} \\
\otimes \mathfrak{e}_7 1, \\
{}^{3,3} \otimes {}^{3,3} \\
2151321 \mathfrak{e}_7 1, \\
{}^{3,3} \otimes {}^{3,3} \\
{}^1 \mathfrak{e}_7^{151321} 1, \\
{}^{4,4} \otimes {}^{4,4} \\
I^{\oplus 3} \mathfrak{e}_7 1, \\
{}^{5,5} \otimes {}^{5,5} \\
I^{\oplus 2} \mathfrak{e}_7 1, \\
151321 \mathfrak{e}_7^{3221} 1, \\
1513221 \mathfrak{e}_7^{321} 1, \\
I^{\oplus 5} \mathfrak{e}_7^{I^{\oplus 2}} 1, \\
21513221 \mathfrak{e}_7^{I^{\oplus 2}} 1, \\
1513151321 \mathfrak{e}_7^{I^{\oplus 2}} 1, \\
31321 \mathfrak{e}_7^{I^{\oplus 3}} 1, \\
1512321 \mathfrak{e}_7^{I^{\oplus 3}} 1, \\
512321 \mathfrak{e}_7^{512321} 1, \\
I^{\oplus 2} \mathfrak{e}_7^{2,3} \otimes {}^{3,2}, \\
I^{\oplus 3} \mathfrak{e}_7^{I^{\oplus 23,2}} \otimes, \\
321 \mathfrak{e}_7^{3213,3} \otimes, \\
I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 24,4}} \otimes, \\
I^{\oplus 3} \mathfrak{e}_7^{321} 123, \\
51321 \mathfrak{e}_7^{I^{\oplus 2}} 1232, \\
2151321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}, \\
{}^1 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}, \\
{}^{3,4} \mathfrak{e}_6^1, \\
I^{\oplus 4} \mathfrak{e}_6^1,
\end{array}
&
\begin{array}{l}
{}^{5,4} \otimes {}^{2,3} \\
\otimes \mathfrak{e}_7 1 \\
{}^{2,3} \otimes {}^{2,3} \\
2151321 \mathfrak{e}_7 1 \\
{}^{2,3} \otimes {}^{2,3} \\
{}^1 \mathfrak{e}_7^{151321} 1 \\
{}^{3,4} \otimes {}^{3,3} \\
\otimes \mathfrak{e}_7 1 \\
{}^{3,3} \otimes {}^{3,3} \\
I^{\oplus 4} \mathfrak{e}_7 1 \\
{}^{3,3} \otimes {}^{3,3} \\
{}^1 \mathfrak{e}_7^{2321} 1 \\
{}^{3,4} \otimes {}^{3,4} \\
I^{\oplus 3} \mathfrak{e}_7 1 \\
{}^{4,5} \otimes {}^{4,5} \\
I^{\oplus 2} \mathfrak{e}_7 1 \\
3221 \mathfrak{e}_7^{3221} 1 \\
I^{\oplus 2} \mathfrak{e}_7^{313221} 1 \\
513221 \mathfrak{e}_7^{2321} 1 \\
32151321 \mathfrak{e}_7^{I^{\oplus 2}} 1 \\
5123151321 \mathfrak{e}_7^{I^{\oplus 2}} 1 \\
{}^1 \mathfrak{e}_7^{1513221} I^{\oplus 2} \\
2151321 \mathfrak{e}_7^{I^{\oplus 3}} 1 \\
{}^1 \mathfrak{e}_7^{151321} I^{\oplus 3} \\
151321 \mathfrak{e}_7^{512321} 1 \\
I^{\oplus 2} \mathfrak{e}_7^{3,3} \otimes {}^{3,2} \\
I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 24,2}} \otimes \\
I^{\oplus 3} \mathfrak{e}_7^{I^{\oplus 23,3}} \otimes \\
I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 24,5}} \otimes \\
512321 \mathfrak{e}_7^{I^{\oplus 2}} 123 \\
I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2} \\
{}^1 \mathfrak{e}_7^{2321} I^{\oplus 2} \\
I^{\oplus 3} \mathfrak{e}_7^{I^{\oplus 3}} I^{\oplus 2} \\
{}^{4,4} \mathfrak{e}_6^1 \\
31321 \mathfrak{e}_6^1
\end{array}
\end{array}$$

$23131\epsilon_6^1 1,$	$3215131\epsilon_6^1 1,$	$2151321\epsilon_6^1 1$
$512315131\epsilon_6^1 1,$	$2321\epsilon_6^1 1,$	$315131\epsilon_6^1 1$
$1512321\epsilon_6^1 1,$	$151315131\epsilon_6^1 1,$	$151321\epsilon_6^1 1$
$2,3 \otimes \epsilon_6^1,$	$3,3 \otimes \epsilon_6^1,$	$4,3 \otimes \epsilon_6^1$
$5,3 \otimes \epsilon_6^1,$	$3,3 \otimes \epsilon_6^1,$	$I^{\oplus 3} \otimes \epsilon_6^1$
$3131\epsilon_6^1,$	$215131\epsilon_6^1,$	$15131\epsilon_6^1$
$I^{\oplus 2} \otimes \epsilon_6^1,$	$I^{\oplus 2} \otimes \epsilon_6^1,$	$I^{\oplus 2} \otimes \epsilon_6^1$
$I^{\oplus 2} \otimes \epsilon_6^1,$	$31^{3221} \epsilon_6 1,$	$321^{321} \epsilon_6 1$
$15131^{321} \epsilon_6 1,$	$5131^{2321} \epsilon_6 1,$	$I^{\oplus 2} \otimes \epsilon_6^1$
$I^{\oplus 3} \epsilon_6^1,$	$3131 \epsilon_6^1,$	$215131 \epsilon_6^1$
$15131 \epsilon_6^1,$	$512321\epsilon_6^1,$	$51315131\epsilon_6^1$
$151321\epsilon_6^1,$	$315131 \epsilon_6^1,$	$15131^{15131} \epsilon_6 1$
$2,2 \otimes 2,2 \otimes \epsilon_6 \otimes,$	$I^{\oplus 2} \otimes 2,2 \otimes \epsilon_6 \otimes,$	$31^{321,2,2} \epsilon_6 \otimes$
$I^{\oplus 2} \epsilon_6^1 \otimes,$	$15131\epsilon_6^1 \otimes,$	$31^{313,2} \epsilon_6 \otimes$
$31^{313,3} \epsilon_6 \otimes,$	$31^{313,4} \epsilon_6 \otimes,$	$31^{313,5} \epsilon_6 \otimes$
$31 \epsilon_6^1 123,$	$31 \epsilon_6^1 \otimes,$	$I^{\oplus 2} \epsilon_6^1 I^{\oplus 2}$
$15131\epsilon_6^1 I^{\oplus 2},$	$31 \epsilon_6^1 I^{\oplus 3},$	$3131\epsilon_6^1 13$
$215131\epsilon_6^1 13,$	$15131\epsilon_6^1 13$	
$3,5 \otimes \epsilon_7^1,$	$4,5 \otimes \epsilon_7^1,$	$5,5 \otimes \epsilon_7^1$
$313221\epsilon_7^1 1,$	$231321\epsilon_7^1 1,$	$I^{\oplus 5} \epsilon_7^1 1$
$32151321\epsilon_7^1 1,$	$31512321\epsilon_7^1 1,$	$21513221\epsilon_7^1 1$
$5123151321\epsilon_7^1 1,$	$3151321\epsilon_7^1 1,$	$1513151321\epsilon_7^1 1$
$1513221\epsilon_7^1 1,$	$2,3 \otimes \epsilon_7^1 1,$	$3,3 \otimes \epsilon_7^1 1$
$I^{\oplus 3} \epsilon_7^1 1,$	$I^{\oplus 2} \epsilon_7^1 1,$	$3,3 \otimes \epsilon_7^1 1$
$I^{\oplus 3} \epsilon_7^1 1,$	$I^{\oplus 2} \epsilon_7^1 1,$	$I^{\oplus 2} \epsilon_7^1 1$
$I^{\oplus 2} \epsilon_7^1 1,$	$321 \epsilon_7^1 1,$	$512321 \epsilon_7^1 1$

$$\begin{aligned}
& 151321 \overset{321}{\mathfrak{c}'_7} 1, \\
& 31321 \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1, \\
& 1512321 \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1, \\
& I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{c}'_7} \otimes, \\
& I^{\oplus 3} \overset{I^{\oplus 2}}{\mathfrak{c}'_7} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 51321 \overset{2321}{\mathfrak{c}'_7} 1, \\
& 2151321 \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1, \\
& 151321 \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1, \\
& I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{c}'_7} \otimes,
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 4} \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1 \\
& 2321 \overset{I^{\oplus 2}}{\mathfrak{c}'_7} 1 \\
& I^{\oplus 3} \overset{I^{\oplus 3}}{\mathfrak{c}'_7} 1 \\
& I^{\oplus 2} \overset{321}{\mathfrak{c}'_7} 123
\end{aligned}$$

$$\begin{aligned}
& \overset{2,2}{\otimes} \overset{1}{\mathfrak{so}} 1, \\
& \overset{2}{231} \overset{1}{\mathfrak{so}} 1,
\end{aligned}$$

$$\begin{aligned}
& \overset{3,2}{\otimes} \overset{1}{\mathfrak{so}} 1, \\
& I^{\oplus 2} \overset{1}{\mathfrak{so}} 1,
\end{aligned}$$

$$\begin{aligned}
& \overset{4,2}{\otimes} \overset{1}{\mathfrak{so}} 1 \\
& 31 \overset{31}{\mathfrak{so}} 1
\end{aligned}$$

2.2.5 SSgIS(183)

$$\begin{aligned}
& I^{\oplus 9} \overset{1}{\mathfrak{c}_8} 1, \\
& I^{\oplus 6} \overset{1}{\mathfrak{c}_8} \overset{4,2}{\otimes}, \\
& 31513221 \overset{1}{\mathfrak{c}_8} \overset{1}{\otimes} \overset{4,2}{\otimes}, \\
& \overset{4,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes}, \\
& I^{\oplus 5} \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes}, \\
& \overset{4,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes}, \\
& I^{\oplus 5} \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes}, \\
& \overset{5,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,5}{\otimes}, \\
& 21513221 \overset{1}{\mathfrak{c}_8} \overset{5,5}{\otimes}, \\
& 22313221 \overset{I^{\oplus 3}}{\mathfrak{c}_8} 1, \\
& 21513221 \overset{1}{\mathfrak{c}_8} 122313, \\
& 2231513221 \overset{I^{\oplus 2}}{\mathfrak{c}_8} 1, \\
& 2321513221 \overset{I^{\oplus 3}}{\mathfrak{c}_8} 1, \\
& I^{\oplus 6} \overset{1}{\mathfrak{c}_8} I^{\oplus 4}, \\
& 31513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 4}, \\
& 21513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 5},
\end{aligned}$$

$$\begin{aligned}
& 22315131513221 \overset{1}{\mathfrak{c}_8} 1, \\
& 321513221 \overset{1}{\mathfrak{c}_8} \overset{4,2}{\otimes}, \\
& 15131513221 \overset{1}{\mathfrak{c}_8} \overset{4,2}{\otimes}, \\
& \overset{5,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes}, \\
& 21513221 \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes}, \\
& \overset{5,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes}, \\
& 21513221 \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes}, \\
& 313221 \overset{1}{\mathfrak{c}_8} \overset{5,5}{\otimes}, \\
& 1513221 \overset{1}{\mathfrak{c}_8} \overset{5,5}{\otimes}, \\
& 313221 \overset{1}{\mathfrak{c}_8} 122313, \\
& 1513221 \overset{1}{\mathfrak{c}_8} 122313, \\
& I^{\oplus 8} \overset{I^{\oplus 2}}{\mathfrak{c}_8} 1, \\
& 315131513221 \overset{I^{\oplus 3}}{\mathfrak{c}_8} 1, \\
& 321513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 4}, \\
& 15131513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 4}, \\
& 1513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 5},
\end{aligned}$$

$$\begin{aligned}
& 2313221 \overset{1}{\mathfrak{c}_8} \overset{4,2}{\otimes} \\
& 51231513221 \overset{1}{\mathfrak{c}_8} \overset{4,2}{\otimes} \\
& \overset{3,5}{\otimes} \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes} \\
& 313221 \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes} \\
& 1513221 \overset{1}{\mathfrak{c}_8} \overset{5,3}{\otimes} \\
& 313221 \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes} \\
& 1513221 \overset{1}{\mathfrak{c}_8} \overset{5,4}{\otimes} \\
& I^{\oplus 5} \overset{1}{\mathfrak{c}_8} \overset{5,5}{\otimes} \\
& I^{\oplus 4} \overset{1}{\mathfrak{c}_8} 1223132 \\
& I^{\oplus 5} \overset{1}{\mathfrak{c}_8} 122313 \\
& 2315131513221 \overset{I^{\oplus 2}}{\mathfrak{c}_8} 1 \\
& I^{\oplus 7} \overset{I^{\oplus 3}}{\mathfrak{c}_8} 1 \\
& 231513221 \overset{I^{\oplus 3}}{\mathfrak{c}_8} 1 \\
& 51231513221 \overset{1}{\mathfrak{c}_8} I^{\oplus 4} \\
& I^{\oplus 5} \overset{1}{\mathfrak{c}_8} I^{\oplus 5} \\
& 21513221 \overset{1}{\mathfrak{c}_8} 12231512
\end{aligned}$$

$$1513221 \underset{1}{\mathfrak{e}_8} 12231512,$$

$$2,4 I^{\oplus 3,4,2} \underset{1}{\otimes} \mathfrak{e}_8 \otimes,$$

$$5,5 I^{\oplus 2,4,2} \underset{1}{\otimes} \mathfrak{e}_8 \otimes,$$

$$I^{\oplus 5} \underset{1}{\mathfrak{e}_8} I^{\oplus 2,4,2} \otimes,$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 3,4,2} \otimes,$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 2,5,4} \otimes,$$

$$I^{\oplus 3} \underset{1}{\mathfrak{e}_8} I^{\oplus 3,5,5} \otimes,$$

$$1513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} 1223,$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 2} 122313,$$

$$2321513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$I^{\oplus 6} \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$31513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$21513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 4},$$

$$1513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} 1223151,$$

$$1513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 3},$$

$$I^{\oplus 4} \underset{2,4}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$I^{\oplus 2} \underset{4,5}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$1513221 \underset{3221}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$I^{\oplus 6} \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$31513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$21513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$I^{\oplus 4} \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 3},$$

$$1513221 \underset{1}{\mathfrak{e}_8} 1223151315,$$

$$3,5 I^{\oplus 2,4,2} \underset{1}{\otimes} \mathfrak{e}_8 \otimes,$$

$$3221 \underset{2,4}{\mathfrak{e}_8} 1223,$$

$$21513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2,4,2} \otimes,$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 2,5,3} \otimes,$$

$$I^{\oplus 3} \underset{1}{\mathfrak{e}_8} I^{\oplus 3,5,4} \otimes,$$

$$I^{\oplus 4} \underset{3221}{\mathfrak{e}_8} 1223,$$

$$I^{\oplus 2} \underset{1}{\mathfrak{e}_8} I^{\oplus 3} 1223132,$$

$$I^{\oplus 3} \underset{1}{\mathfrak{e}_8} I^{\oplus 3} 122313,$$

$$315131513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$321513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$15131513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$1513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 4},$$

$$I^{\oplus 5} \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 3},$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 4},$$

$$I^{\oplus 3} \underset{2,4}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$I^{\oplus 2} \underset{5,5}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$I^{\oplus 2} \underset{2313221}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$321513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$15131513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2},$$

$$1513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2},$$

$$I^{\oplus 3} \underset{I^{\oplus 3}}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 3},$$

$$1513221 \underset{1}{\mathfrak{e}_8} 1223151 \underset{1}{1}$$

$$4,5 I^{\oplus 2,4,2} \underset{1}{\otimes} \mathfrak{e}_8 \otimes$$

$$313221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2,4,2} \otimes$$

$$1513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2,4,2} \otimes$$

$$I^{\oplus 3} \underset{1}{\mathfrak{e}_8} I^{\oplus 3,5,3} \otimes$$

$$I^{\oplus 4} \underset{1}{\mathfrak{e}_8} I^{\oplus 2,5,5} \otimes$$

$$5131513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} 1223$$

$$I^{\oplus 2} \underset{1}{\mathfrak{e}_8} I^{\oplus 2} 12231322$$

$$I^{\oplus 7} \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2}$$

$$231513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2}$$

$$51231513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2}$$

$$I^{\oplus 5} \underset{1}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 4}$$

$$31513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 2} 122315$$

$$21513221 \underset{1}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 3}$$

$$I^{\oplus 2} \underset{2,4}{\mathfrak{e}_8} I^{\oplus 2,4,2} \otimes$$

$$I^{\oplus 2} \underset{3,5}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2}$$

$$I^{\oplus 2} \underset{3221}{\mathfrak{e}_8} I^{\oplus 3} 1223$$

$$I^{\oplus 2} \underset{313221}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2}$$

$$51231513221 \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 2}$$

$$I^{\oplus 5} \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 3} I^{\oplus 2}$$

$$I^{\oplus 4} \underset{I^{\oplus 2}}{\mathfrak{e}_8} I^{\oplus 2} I^{\oplus 4}$$

$$\begin{array}{c} 1 \\ 151321\epsilon'_7 1, \\ 1 \end{array}$$

$$\begin{array}{c} 1 \\ 321\epsilon'_7 123, \\ 1 \end{array}$$

$$\begin{array}{c} 2,3I^{\oplus 2} \\ \otimes \epsilon'_7 1, \\ 1 \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \\ I^{\oplus 2} \epsilon'_7 1, \\ 1 \end{array}$$

$$\begin{array}{c} 3,3I^{\oplus 2} \\ \otimes \epsilon'_7 1 \\ 1 \end{array}$$

$$\begin{array}{c} I^{\oplus 2} \\ I^{\oplus 2} \epsilon'_7 I^{\oplus 2} \\ 1 \end{array}$$

$$\begin{array}{c} 1 \\ 1\mathfrak{so}1 \\ 1 \end{array}$$

2.3 2 node(1504)

2.3.1 gLg(4)

$$\begin{array}{c} 3,3 \\ \mathfrak{so} \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{c} 3,3 \\ \mathfrak{so} \bigcirc \mathfrak{so} \end{array}$$

$$\begin{array}{c} 3,5 \\ \mathfrak{so} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{c} 5,5 \\ \mathfrak{e}_6 \otimes \mathfrak{e}_6 \end{array}$$

2.3.2 111gLg(3)

$$\begin{array}{c} 1 \\ 1\epsilon'_7 \otimes \mathfrak{so} \\ 1 \end{array} \begin{array}{c} 4,3 \end{array}$$

$$\begin{array}{c} 1 \\ 1\epsilon'_7 \otimes \mathfrak{e}_6 \\ 1 \end{array} \begin{array}{c} 4,5 \end{array}$$

$$\begin{array}{c} 1 \\ 1\epsilon'_7 \otimes \mathfrak{so} \\ 1 \end{array} \begin{array}{c} 3,3 \end{array}$$

2.3.3 SgLg(115)

$$513151321\epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 3,2 \end{array}$$

$$2231\mathfrak{so} \otimes \epsilon_7, \begin{array}{c} 2,4 \end{array}$$

$$\begin{array}{c} 2,4 \\ \otimes \epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 3,3 \end{array}$$

$$\begin{array}{c} 5,4 \\ \otimes \epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 3,3 \end{array}$$

$$2151321\epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 3,3 \end{array}$$

$$\begin{array}{c} 1 \\ 151321\epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 3,3 \end{array}$$

$$I^{\oplus 3} \epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 4,3 \end{array}$$

$$1513221\epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 3,2 \end{array}$$

$$512321\epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 4,2 \end{array}$$

$$\begin{array}{c} 3,4 \\ \otimes \epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 3,3 \end{array}$$

$$I^{\oplus 4} \epsilon_7 \otimes \mathfrak{so}, \begin{array}{c} 3,3 \end{array}$$

$$\begin{array}{c} 1 \\ 2321\epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 3,3 \end{array}$$

$$\begin{array}{c} 2,3 \\ \otimes \epsilon_7 \otimes \mathfrak{so}, \end{array} \begin{array}{c} 4,3 \end{array}$$

$$I^{\oplus 2} \epsilon_7 \otimes \mathfrak{so} \begin{array}{c} 5,3 \end{array}$$

$$3221\epsilon_7 \otimes \mathfrak{so} \begin{array}{c} 4,2 \end{array}$$

$$151321\epsilon_7 \otimes \mathfrak{so} \begin{array}{c} 4,2 \end{array}$$

$$\begin{array}{c} 4,4 \\ \otimes \epsilon_7 \otimes \mathfrak{so} \end{array} \begin{array}{c} 3,3 \end{array}$$

$$31321\epsilon_7 \otimes \mathfrak{so} \begin{array}{c} 3,3 \end{array}$$

$$1512321\epsilon_7 \otimes \mathfrak{so} \begin{array}{c} 3,3 \end{array}$$

$$\begin{array}{c} 3,3 \\ \otimes \epsilon_7 \otimes \mathfrak{so} \end{array} \begin{array}{c} 4,3 \end{array}$$

$$\begin{aligned}
& 3221\mathfrak{e}_6 \otimes \mathfrak{so}, \\
& 51315131\mathfrak{e}_6 \otimes \mathfrak{so}, \\
& 231\mathfrak{so} \otimes \mathfrak{e}_6, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{so}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
& 2231\mathfrak{so} \otimes \mathfrak{e}_6, \\
& 151321\mathfrak{e}_6 \otimes \mathfrak{so}, \\
& 15131\mathfrak{e}_6 \otimes \mathfrak{so}, \\
& \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\
& \otimes \mathfrak{e}_6 \bigcirc \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
& 512321\mathfrak{e}_6 \otimes \mathfrak{so} \\
& 321\mathfrak{e}_6 \otimes \mathfrak{so} \\
& 31\mathfrak{so} \otimes \mathfrak{e}_6 \\
& I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{so} \\
& I^{\oplus 2}\mathfrak{e}_6 \bigcirc \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
& 31513221\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\
& 2315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\
& 315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\
& 151321\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\
& 321\mathfrak{e}_7 \otimes \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
& 22315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\
& 2321\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\
& 3221\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\
& \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6, \\
& I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
& 3151321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\
& 2321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\
& 512321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\
& \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \\
& I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
& 3221\mathfrak{e}'_7 \otimes \mathfrak{so}, \\
& 151321\mathfrak{e}'_7 \otimes \mathfrak{so}, \\
& \otimes \mathfrak{e}'_7 \otimes \mathfrak{so}, \\
& I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
& 2231\mathfrak{so} \otimes \mathfrak{e}'_7, \\
& 321\mathfrak{e}'_7 \otimes \mathfrak{so}, \\
& \otimes \mathfrak{e}'_7 \otimes \mathfrak{so}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
& 512321\mathfrak{e}'_7 \otimes \mathfrak{so} \\
& 231\mathfrak{so} \otimes \mathfrak{e}'_7 \\
& I^{\oplus 3}\mathfrak{e}'_7 \otimes \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
& 3151321\mathfrak{e}'_7 \otimes \mathfrak{e}_6, \\
& 2321\mathfrak{e}'_7 \otimes \mathfrak{e}_6, \\
& 15131\mathfrak{e}_6 \otimes \mathfrak{e}'_7, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
& 2315131\mathfrak{e}_6 \otimes \mathfrak{e}'_7, \\
& 315131\mathfrak{e}_6 \otimes \mathfrak{e}'_7, \\
& 321\mathfrak{e}'_7 \otimes \mathfrak{e}_6, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
& 2321\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \\
& 321\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \\
& I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
& 23151321\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& 321\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& 2321\mathfrak{e}_6 \bigcirc \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
& 2321\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& 15131\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}_6, \\
& 315131\mathfrak{e}_6 \bigcirc \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
& 315131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \\
& \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}_6
\end{aligned}$$

$$231513221\mathfrak{e}_7 \otimes \mathfrak{e}'_7,$$

$$223151321\mathfrak{e}'_7 \otimes \mathfrak{e}_7,$$

$$3151321\mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$\begin{array}{l} 2321e_7 \otimes^{5,4} e'_7 \\ 321e_7 \otimes^{5,5} e'_7 \end{array}$$

$$\begin{array}{l} 321513221\mathbf{e}_8 \otimes^{5,5} \mathbf{e}_6 \\ 15131513221\mathbf{e}_8 \otimes^{5,5} \mathbf{e}_6 \end{array}$$

$$\begin{array}{l} 321513221e_8 \otimes^{5,3} \mathfrak{so} \\ 15131513221e_8 \otimes^{5,3} \mathfrak{so} \end{array}$$

$$3150 \overset{2,2}{\otimes} 50$$

$$3151321\mathbf{e}_7^{4,5} \otimes \mathbf{e}_7$$

$$321e'_7 \overset{4,5}{\otimes} e'_7$$

$$\begin{array}{l}
\begin{array}{ccc} 2,3 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 1223151 \end{array} \\
\begin{array}{ccc} 2,4 & 3,3 & 4,4 \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & \otimes \end{array} \\
\begin{array}{ccc} 2,4 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 12313 \end{array} \\
\begin{array}{ccc} 2,4 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 1232151 \end{array} \\
\begin{array}{ccc} 3,3 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 1223151 \end{array} \\
\begin{array}{ccc} 3,4 & 3,3 & 4,5 \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & \otimes \end{array} \\
\begin{array}{ccc} 3,4 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 1231512 \end{array} \\
\begin{array}{ccc} 3,4 & 3,3 & 1 \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 123151 \end{array} \\
\begin{array}{ccc} I^{\oplus 4} & 3,3 & 4,4 \\ \mathbf{e}_7 & \otimes \mathbf{e}_7 & \otimes \end{array} \\
\begin{array}{ccc} 1 & 3,3 & 4,4 \\ 2321 & \mathbf{e}_7 & \otimes \mathbf{e}_7 \otimes \end{array} \\
\begin{array}{ccc} 5,4 & 3,3 & 4,5 \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & \otimes \end{array} \\
\begin{array}{ccc} 5,4 & 3,3 & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & 1231512 \end{array}
\end{array}$$

$$\begin{aligned}
& 321\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 12315, \\
& 3131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \bigcirc, \\
& I^{\oplus 2}\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 123215, \\
& I^{\oplus 3}\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 I^{\oplus 3}, \\
& I^{\oplus 3}\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 13151, \\
& 3131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 13151, \\
& 15131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 131512, \\
& 1\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_6 13151, \\
& I^{\oplus 2}\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}, \\
& 1\mathbf{e}_6 \overset{4,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}, \\
& 1\mathbf{e}_6 \overset{3,3}{\bigcirc} \mathbf{e}_6 123, \\
& I^{\oplus 2}\mathbf{e}_6 \overset{3,3}{\bigcirc} \mathbf{e}_6 \overset{2,2}{\otimes},
\end{aligned}$$

$$\begin{aligned}
& \overset{3,3}{\bigcirc} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 \overset{3,3}{\bigcirc}, \\
& \overset{3,3}{\bigcirc} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 131512, \\
& I^{\oplus 2} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 13151315, \\
& I^{\oplus 3} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 1313, \\
& 3131 \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 1313, \\
& 51321 \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 13151, \\
& \overset{1}{15131} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc} \mathfrak{e}_6 \overset{1}{13151}, \\
& \overset{2,2}{\bigcirc} \mathfrak{e}_6 \otimes \overset{3,3}{\bigcirc} \mathfrak{e}_6 \otimes \overset{2,2}{\bigcirc}, \\
& 31 \mathfrak{e}_6 \otimes \overset{3,3}{\bigcirc} \mathfrak{e}_6 1315, \\
& 31 \mathfrak{e}_6 \otimes \overset{4,3}{\bigcirc} \mathfrak{e}_6 1, \\
& 1 \mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6 13151, \\
& I^{\oplus 2} \mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6 I^{\oplus 2},
\end{aligned}$$

$$\begin{array}{l}
I^{\oplus 3} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,3} \bigcirc \\
\bigcirc \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2} 13151^1 \\
I^{\oplus 2} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 123151 \\
I^{\oplus 3} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 131512 \\
3131 \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 131512 \\
215131 \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 131512 \\
1 \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{3,3} 123 \\
I^{\oplus 2} \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \\
1 \mathfrak{e}_6^{4,3} \otimes \mathfrak{e}_6^{2,2} \otimes \\
1 \mathfrak{e}_6^{4,4} \otimes \mathfrak{e}_6^1 \\
\otimes \mathfrak{e}_6^{2,2} \bigcirc \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \\
31 \mathfrak{e}_6^{3,3} \bigcirc \mathfrak{e}_6^{3,3} 1315
\end{array}$$

$$\begin{aligned}
& \overset{3,3}{\otimes} \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 122315123, \\
& \overset{3,3}{\otimes} \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 12231513151, \\
& 151321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \overset{5,3}{\otimes} \mathbf{e}_8, \\
& 151321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \overset{5,4}{\otimes} \mathbf{e}_8, \\
& 151321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \overset{5,5}{\otimes} \mathbf{e}_8, \\
& 3221 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 12231512, \\
& 512321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 122313, \\
& I^{\oplus 3} \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 I^{\oplus 6}, \\
& I^{\oplus 3} \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 12231513215, \\
& 3151321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 I^{\oplus 4}, \\
& 51321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 1223151315, \\
& 151321 \mathbf{e}_7 \otimes \overset{4,5}{\otimes} \mathbf{e}_8 12231512, \\
& 321 \mathbf{e}_7 \otimes \overset{5,5}{\otimes} \overset{5,3}{\otimes} \mathbf{e}_8, \\
& I^{\oplus 2} \mathbf{e}_7 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 1223132, \\
& 321 \mathbf{e}_7 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 12231512, \\
& I^{\oplus 2} \mathbf{e}_7 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 I^{\oplus 6}, \\
& I^{\oplus 2} \mathbf{e}_7 \otimes \overset{5,5}{\otimes} \overset{1}{\otimes} \mathbf{e}_8 12231513,
\end{aligned}$$

$$\begin{aligned}
& \otimes_{\mathfrak{e}_6}^{2,2} \otimes_{\mathfrak{e}_8}^{3,5} 1223132, \\
& \otimes_{\mathfrak{e}_6}^{2,2} \otimes_{\mathfrak{e}_8}^{3,5} 12231513215, \\
& 2321\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} \mathfrak{e}_8 \otimes^{4,2}, \\
& 15131\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} \mathfrak{e}_8 \otimes^{5,3}, \\
& 321\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} \mathfrak{e}_8 \otimes^{5,5}, \\
& 321\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} 122313, \\
& 321\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} 12231512, \\
& I^{\oplus 2}\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} I^{\oplus 6}, \\
& I^{\oplus 2}\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} 12231513215, \\
& 2315131\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} \mathfrak{e}_8 I^{\oplus 3}, \\
& 5131\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} \mathfrak{e}_8 1223151315, \\
& 1\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{4,5} \mathfrak{e}_8 1223132.
\end{aligned}$$

$$\begin{aligned}
& 3,3 \quad 4,5 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 12231513215, \\
& 3221\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes, \\
& 3221\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes, \\
& 3221\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes, \\
& 3221\mathbf{e}_7 \otimes \mathbf{e}_8 122313, \\
& 3221\mathbf{e}_7 \otimes \mathbf{e}_8 1223151, \\
& 151321\mathbf{e}_7 \otimes \mathbf{e}_8 122313, \\
& I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 122315123, \\
& I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 12231513, \\
& 512321\mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 5}, \\
& 512321\mathbf{e}_7 \otimes \mathbf{e}_8 12231512, \\
& 151321\mathbf{e}_7 \otimes \mathbf{e}_8 1223151, \\
& 321\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes, \\
& 321\mathbf{e}_7 \otimes \mathbf{e}_8 122313, \\
& 321\mathbf{e}_7 \otimes \mathbf{e}_8 1223151, \\
& I^{\oplus 2} \mathbf{e}_7 \otimes \mathbf{e}_8 122315123, \\
& I^{\oplus 2} \mathbf{e}_7 \otimes \mathbf{e}_8 12231513151
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,2 \\ \otimes \mathbf{e}_6 \end{smallmatrix} \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} I^{\oplus 6}, \\
& \begin{smallmatrix} 2,2 \\ \otimes \mathbf{e}_6 \end{smallmatrix} \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 12231513, \\
& 315131\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 4,2 \\ \mathbf{e}_8 \end{smallmatrix}, \\
& 321\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,4 \\ \mathbf{e}_8 \end{smallmatrix}, \\
& 15131\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ \mathbf{e}_8 \end{smallmatrix}, \\
& 15131\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 122313, \\
& 321\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 1223151, \\
& I^{\oplus 2} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 122315123, \\
& I^{\oplus 2} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 12231513, \\
& 315131\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} I^{\oplus 4}, \\
& 15131\mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \mathbf{e}_8 \end{smallmatrix} 12231512, \\
& 1\mathbf{e}_6 \otimes \begin{smallmatrix} 4,5 \\ \mathbf{e}_8 \end{smallmatrix} I^{\oplus 6},
\end{aligned}$$

$$\begin{array}{l}
{}^{3,3} \otimes \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 12231513 \\
512321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 \otimes {}^{5,3} \\
512321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 \otimes {}^{5,4} \\
512321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 \otimes {}^{5,5} \\
3221 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 I^{\oplus 5} \\
I^{\oplus 3} \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 1223132 \\
223151321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 I^{\oplus 2} \\
23151321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 I^{\oplus 3} \\
I^{\oplus 3} \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 12231513151 \\
151321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 I^{\oplus 5} \\
512321 \mathbf{e}_7 \otimes {}^{4,5} \mathbf{e}_8 1223151 \\
2321 \mathbf{e}_7 \otimes {}^{5,5} \mathbf{e}_8 \otimes {}^{4,2} \\
321 \mathbf{e}_7 \otimes {}^{5,5} \mathbf{e}_8 \otimes {}^{5,5} \\
321 \mathbf{e}_7 \otimes {}^{5,5} \mathbf{e}_8 I^{\oplus 5} \\
2321 \mathbf{e}_7 \otimes {}^{5,5} \mathbf{e}_8 I^{\oplus 4} \\
I^{\oplus 2} \mathbf{e}_7 \otimes {}^{5,5} \mathbf{e}_8 12231513215
\end{array}$$

$$\begin{array}{l}
{}^{2,2} \otimes \mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 122315123 \\
{}^{2,2} \otimes \mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 12231513151 \\
321\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 \otimes {}^{5,3} \\
15131\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 \otimes {}^{5,4} \\
I^{\oplus 2} \mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 1223132 \\
321\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 I^{\oplus 5} \\
2321\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 I^{\oplus 4} \\
22315131\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 I^{\oplus 2} \\
I^{\oplus 2} \mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 12231513151 \\
15131\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 I^{\oplus 5} \\
15131\mathfrak{e}_6 \otimes {}^{3,5} \otimes \mathfrak{e}_8 1223151 \\
1\mathfrak{e}_6 \otimes {}^{4,5} \otimes \mathfrak{e}_8 122315123
\end{array}$$

$$\begin{array}{l} 1\epsilon_6^{4,5} \otimes \epsilon_8 12231513151 \\ 31\epsilon_6^{4,5} \otimes \epsilon_8^{5,5} \\ 31\epsilon_6^{4,5} \otimes \epsilon_8 12231512 \end{array}$$

$$\begin{array}{l}
1\epsilon_6^{3,3} \otimes \epsilon_7' 123215 \\
\otimes \epsilon_6^{2,2} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7^{3,2} \\
I^{\oplus 2} \epsilon_6^{3,3} \otimes \epsilon_7'^{3,2} \otimes \epsilon_7^{3,2} \\
5131\epsilon_6^{3,3} \otimes \epsilon_7' 123 \\
31\epsilon_6^{3,3} \otimes \epsilon_7' 12315 \\
15131\epsilon_6^{3,4} \otimes \epsilon_7' 1 \\
\otimes \epsilon_6^{2,2} \otimes \epsilon_7'^{3,5} \otimes \epsilon_7' 1 \\
31\epsilon_6^{4,4} \otimes \epsilon_7' 1
\end{array}$$

$$\begin{array}{l} 321\mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \\ 321\mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 122313 \\ 321\mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 12231\overset{1}{5}1 \\ I^{\oplus 2}\mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 122315123 \\ I^{\oplus 2}\mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 12231513151 \\ 1\mathbf{e}'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 122315123 \\ 1\mathbf{e}'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 12231513151 \end{array}$$

$$\begin{array}{l} 2313221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 122315123 \\ 2313221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513151 \\ I^{\oplus 6} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513215 \\ 321513221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 122315123 \\ 321513221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513151 \\ 15131513221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513215 \\ 15131513221\mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_8^{2,4} 12231322, \\
& \mathbf{1so} \otimes \mathbf{e}_8^{2,4} 122315131513, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{5,3}, \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} 1223132, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} I^{\oplus 5}, \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} 12231513215, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} 12231512,
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_8^{2,4} I^{\oplus 7}, \\
& \mathbf{1so} \otimes \mathbf{e}_8^{2,4} 122315132, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{5,4}, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} 122313, \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} I^{\oplus 6}, \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} 12231513, \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} 1223151
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_8^{2,4} 1223151232 \\
& 2231\mathbf{so} \otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{4,2} \\
& 231\mathbf{so} \otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{5,5} \\
& 2231\mathbf{so} \otimes \mathbf{e}_8^{2,4} I^{\oplus 4} \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} 122315123 \\
& 31\mathbf{so} \otimes \mathbf{e}_8^{2,4} 12231513151
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{4,2}, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{4,5}, \\
& 2231\mathbf{so} \otimes \mathbf{e}_7^{2,3} 1, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} 1232151, \\
& 31\mathbf{so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{3,3}, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,4} \otimes \mathbf{e}_7^{3,2}, \\
& 231\mathbf{so} \otimes \mathbf{e}_7^{2,4} 1,
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{4,3}, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} I^{\oplus 4}, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} 1231512, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} 123151, \\
& 231\mathbf{so} \otimes \mathbf{e}_7^{2,3} I^{\oplus 2}, \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,4} \otimes \mathbf{e}_7^{3,3}, \\
& 31\mathbf{so} \otimes \mathbf{e}_7^{2,4} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{4,4} \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} 12313 \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,3} 1232 \\
& 31\mathbf{so} \otimes \mathbf{e}_7^{2,3} \otimes \mathbf{e}_7^{3,2} \\
& 31\mathbf{so} \otimes \mathbf{e}_7^{2,3} I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}_7^{2,4} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}^{1,1} 123, \\
& \mathbf{so}^{2,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,2}, \\
& \mathbf{so}^{2,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^2 132, \\
& \mathbf{so}^{3,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,4}, \\
& \mathbf{so}^{4,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,4}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^2 132, \\
& \mathbf{1so} \otimes \mathbf{so}^{2,2} 1
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}^{1,1} 13215, \\
& \mathbf{so}^{2,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,3}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,2}, \\
& \mathbf{so}^{3,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^2 132, \\
& \mathbf{so}^{4,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^2 132, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}^{1,1} 13151 \\
& \mathbf{so}^{2,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,4} \\
& \mathbf{so}^{3,2} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,3} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,3} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}^{1,1} \otimes \mathbf{so}^{2,4} \\
& 31\mathbf{so} \otimes \mathbf{so}^{1,1} 1315
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,2}, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,5}, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} 1313, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} 13151, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,3} \otimes \mathbf{e}_6^{2,2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,3}, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} \bigcirc, \\
& 231\mathbf{so} \otimes \mathbf{e}_6^{2,2} 1, \\
& 31\mathbf{so} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2}, \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,3} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,4} \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}_6^{2,2} 131512 \\
& 31\mathbf{so} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} \\
& 31\mathbf{so} \otimes \mathbf{e}_6^{2,3} 1
\end{aligned}$$

$$1\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_6 1$$

$$1\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 1231513,$$

$$I^{\oplus 3} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,2}{\otimes},$$

$$I^{\oplus 2} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 1223,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 123151,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}'_7 I^{\oplus 2},$$

$$\overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,2}{\otimes},$$

$$\overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes},$$

$$321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 12315,$$

$$I^{\oplus 3} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 I^{\oplus 3},$$

$$1\mathfrak{e}'_7 \overset{5,4}{\otimes} \mathfrak{e}'_7 I^{\oplus 2},$$

$$\overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes}$$

$$I^{\oplus 3} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7 123215$$

$$1\mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}'_7 123$$

$$1\mathfrak{e}'_7 \overset{5,5}{\otimes} \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,2}{\otimes},$$

$$231\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}'_7 1,$$

$$31\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes},$$

$$31\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}'_7 I^{\oplus 3}$$

$$1\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}'_7 I^{\oplus 2}$$

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$$3221\mathfrak{e}_8 \overset{2,4}{\otimes} \overset{4,2}{\otimes} \mathfrak{so},$$

$$1513221 \overset{I^{\oplus 3}}{\otimes} \overset{4,2}{\otimes} \mathfrak{so},$$

$$\overset{5,5}{\otimes} \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$21513221 \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_8 \overset{I^{\oplus 2}}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_8 \overset{3221}{\otimes} \overset{4,2}{\otimes} \mathfrak{so},$$

$$\overset{3,5}{\otimes} \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$313221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{\otimes} 1513221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{I^{\oplus 3}}{\otimes} \overset{5,3}{\otimes} \mathfrak{so}$$

$$5131513221 \overset{I^{\oplus 2}}{\otimes} \overset{4,2}{\otimes} \mathfrak{so}$$

$$\overset{4,5}{\otimes} \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 5} \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_8 \overset{2,4}{\otimes} \overset{5,3}{\otimes} \mathfrak{so}$$

$$31513221 \mathfrak{e}_8 \overset{I^{\oplus 2}}{\otimes} \overset{5,3}{\otimes} \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8 \overset{I^{\oplus 2}}{\otimes} \overset{5,4}{\otimes} \mathfrak{e}_6,$$

$$\overset{5,5}{\otimes} \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$21513221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 4} \mathfrak{e}_8 \overset{I^{\oplus 2}}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$5131513221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,4}{\otimes} \mathfrak{e}_6,$$

$$\overset{3,5}{\otimes} \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$313221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\otimes} 1513221 \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{I^{\oplus 3}}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{3221}{\otimes} \overset{5,4}{\otimes} \mathfrak{e}_6$$

$$\overset{4,5}{\otimes} \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 5} \mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_8 \overset{2,4}{\otimes} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$3221 \mathfrak{e}_7 \overset{1}{\otimes} \overset{3,2}{\otimes} \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \overset{321}{\otimes} \overset{3,2}{\otimes} \mathfrak{so},$$

$$\overset{3,3}{\otimes} \overset{1}{\otimes} \overset{3,3}{\otimes} \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \overset{1}{\otimes} \overset{4,3}{\otimes} \mathfrak{so},$$

$$512321 \mathfrak{e}_7 \overset{1}{\otimes} \overset{3,2}{\otimes} \mathfrak{so},$$

$$321 \mathfrak{e}_7 \overset{1}{\otimes} \overset{4,2}{\otimes} \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7 \overset{1}{\otimes} \overset{3,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{\otimes} 1\mathfrak{e}_7 \overset{5,3}{\otimes} \mathfrak{so}$$

$$151321 \mathfrak{e}_7 \overset{1}{\otimes} \overset{3,2}{\otimes} \mathfrak{so}$$

$$\overset{2,3}{\otimes} \overset{1}{\otimes} \overset{3,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \overset{I^{\oplus 2}}{\otimes} \overset{3,3}{\otimes} \mathfrak{so}$$

$$321\mathfrak{e}_6^1 \otimes \mathfrak{so}^{2,2},$$

$$31\mathfrak{e}_6^{I^{\oplus 2} 2,2} \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6^1 \bigcirc \mathfrak{so}^{3,3}$$

$$15131\mathfrak{e}_6^1 \otimes \mathfrak{so}^{2,2},$$

$$31\mathfrak{e}_6^1 \otimes \mathfrak{so}^{3,2},$$

$$31\mathfrak{e}_6^{\otimes 2,2} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{so}^{3,3}$$

$$3151321\mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{3,3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{4,4},$$

$$I^{\oplus 2} \mathfrak{e}_7^{2321} \otimes \mathfrak{e}_6^{3,3},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{4,5},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{4,3}$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{5,5}$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7',$$

$$1513221\mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes \mathfrak{e}_7'$$

$$5131513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7'^{5,5},$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221} \otimes \mathfrak{e}_7'^{5,5}$$

$$321\mathfrak{e}_7'^1 \otimes \mathfrak{so}^{3,2},$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 \otimes \mathfrak{so}^{3,3},$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{so}^{4,3}$$

$$2321\mathfrak{e}_7'^1 \otimes \mathfrak{e}_6^{3,3},$$

$$31\mathfrak{e}_6^1 \otimes \mathfrak{e}_7'^{3,5},$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_6^{4,5}$$

$$23151321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{3,3},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{4,4},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{4,5}$$

$$2315131\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2},$$

$$31\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,4},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,5}$$

$$231513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7,$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes \mathfrak{e}_7$$

$$223151321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{3,3},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{4,5}$$

$$231\mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

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$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 123151322,$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$I^{\oplus 5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} 1223$$

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$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6 I^{\oplus 3}, \\
& 15131 \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 5131 \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6 13151, \\
& 31 \mathbf{e}_6^{2,2 \otimes 2,2} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 31 \mathbf{e}_6^{31 \ 2,2 \ 3,4} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{I^{\oplus 2} \ 2,2} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 31 \mathbf{e}_6^{31 \ 2,2} \otimes \mathbf{e}_6 131512, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \bigcirc \mathbf{e}_6^{2,2} \otimes, \\
& 1 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1231513, \\
& 1 \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 1223151, \\
& 3,3 \ 1 \ 3,3 \ 3,2 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' \otimes, \\
& 2,3 \ 1 \ 3,3 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' I^{\oplus 3}, \\
& 321 \mathbf{e}_7'^{1 \ 3,3 \ 3,3} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3 \ 4,3} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1223, \\
& 321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 12315, \\
& I^{\oplus 2} \mathbf{e}_7^{2321 \ 3,3} \otimes \mathbf{e}_7' 1, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 123215, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 1232, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 1232151, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3 \ 3,2} \otimes \mathbf{e}_7' \otimes, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 3}, \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' 123, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_7' 1, \\
& 1 \mathbf{e}_6^{1 \ 3,3 \ 4,2} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6 1313, \\
& I^{\oplus 2} \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6^{1 \ 2,2} 13151, \\
& 31 \mathbf{e}_6^{2,2 \otimes 2,2 \ 2,2} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{31 \ 2,2 \ 3,2} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{31 \ 2,2 \ 3,5} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{31 \ 2,2} \otimes \mathbf{e}_6 I^{\oplus 3}, \\
& 31 \mathbf{e}_6^{31 \ 2,2} \otimes \mathbf{e}_6^{1 \ 2,2} 13151, \\
& 31 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_6 1, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \bigcirc \mathbf{e}_6 I^{\oplus 2}, \\
& 3151321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1, \\
& 2,3 \ 1 \ 3,3 \ 3,2 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' \otimes, \\
& 321 \mathbf{e}_7'^{1 \ 3,3 \ 3,2} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3 \ 4,2} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' \otimes, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3 \ 4,4} \otimes \mathbf{e}_7 \otimes, \\
& 3221 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& 51321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 123, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 I^{\oplus 4}, \\
& 512321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 123151, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 123151, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3 \ 3,3} \otimes \mathbf{e}_7' \otimes, \\
& 1 \mathbf{e}_7'^{1 \ 4,4 \ 3,2} \otimes \mathbf{e}_7 \otimes, \\
& 321 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' 1, \\
& 1 \mathbf{e}_7'^{1 \ 4,5} \otimes \mathbf{e}_7 I^{\oplus 2}, \\
& 1 \mathbf{e}_7^{1 \ 5,5} \otimes \mathbf{e}_7' 1, \\
& 1 \mathbf{e}_6^{1 \ 3,3 \ 4,3} \otimes \mathbf{e}_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}_6^{1 \ 2,2} \otimes \mathbf{e}_6 131512, \\
& 31 \mathbf{e}_6^{I^{\oplus 2} \ 2,2 \ 2,2} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{31 \ 2,2 \ 3,3} \otimes \mathbf{e}_6 \otimes, \\
& 31 \mathbf{e}_6^{31 \ 2,2 \ 3,3} \bigcirc, \\
& 31 \mathbf{e}_6^{31 \ 2,2} \otimes \mathbf{e}_6 1313, \\
& 1 \mathbf{e}_6^{1 \ 3,3 \ 2,2} \otimes \mathbf{e}_6 \otimes, \\
& 1 \mathbf{e}_6^{1 \ 3,4} \otimes \mathbf{e}_6 1, \\
& 31 \mathbf{e}_6^{1 \ 3,3} \bigcirc \mathbf{e}_6 1, \\
& 1 \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 123151315, \\
& 2,3 \ 1 \ 3,3 \ 3,3 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' \otimes, \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3 \ 3,2} \otimes \mathbf{e}_7' \otimes, \\
& 3,3 \ 1 \ 3,3 \ 3,3 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' \otimes, \\
& 3,3 \ 1 \ 3,3 \\
& \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3 \ 4,5} \otimes \mathbf{e}_7 \otimes, \\
& 321 \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 2321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 12313, \\
& I^{\oplus 2} \mathbf{e}_7'^{1 \ 3,3} \otimes \mathbf{e}_7 1231512, \\
& 151321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_7^{321 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& 1 \mathbf{e}_7'^{1 \ 4,4 \ 3,3} \otimes \mathbf{e}_7 \otimes, \\
& 1 \mathbf{e}_7'^{1 \ 4,4} \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 1 \mathbf{e}_7^{1 \ 5,4} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& 1 \mathbf{e}_6^{1 \ 3,3 \ 4,4} \otimes \mathbf{e}_7 \otimes
\end{aligned}$$

$$51231513221\mathfrak{e}_8 \otimes \mathfrak{so}1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{\otimes 2,4} \otimes \mathfrak{so}1,$$

$$\mathfrak{e}_8^{3,5} \otimes \mathfrak{so}13,$$

$$I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_8^{313221} \otimes \mathfrak{so}1,$$

$$21513221 I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{so}1,$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{so}1322,$$

$$21513221 \mathfrak{e}_8 \otimes \mathfrak{so}13,$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1322,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes \mathfrak{so}13$$

$$31513221 \mathfrak{e}_8 \otimes \mathfrak{so}1,$$

$$\mathfrak{e}_8^{2,4} \otimes \mathfrak{so}132,$$

$$I^{\oplus 2} \mathfrak{e}_8^{4,5} \otimes \mathfrak{so}1,$$

$$\mathfrak{e}_8^{5,5} \otimes \mathfrak{so}13,$$

$$313221 \mathfrak{e}_8 \otimes \mathfrak{so}13,$$

$$1513221 I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{so}1,$$

$$I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{so}132,$$

$$1513221 \mathfrak{e}_8 \otimes \mathfrak{so}13,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}132,$$

$$15131513221 \mathfrak{e}_8 \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_8^{3,5} \otimes \mathfrak{so}1$$

$$\mathfrak{e}_8^{4,5} \otimes \mathfrak{so}13$$

$$3221 \mathfrak{e}_8^{3221} \otimes \mathfrak{so}1$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{so}13$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{so}13$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}13$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' \otimes,$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_61,$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes,$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7'1,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' \otimes,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7'1,$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_6123$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_613151$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_61$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7'1223,$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7'123151,$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7'1$$

$$2321\mathfrak{e}_7' \otimes \mathfrak{e}_7'1,$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7' I^{\oplus 3},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7'123215$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$\mathfrak{e}_7^{2,3} \otimes \mathfrak{so}1,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{so}132,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{so}1,$$

$$\mathfrak{e}_7^{3,3} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 2} 3,2} \otimes \mathfrak{so}1,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{so}13$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{so}13$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{so}132,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{so} I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes$$

$$31\mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,2,2} \otimes \mathfrak{so}13$$

$$1e_7' \otimes_{\mathbb{Z}} \mathfrak{sl}$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} I^{\oplus 3} \mathfrak{e}_8 I^{\oplus 3}$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151315, \\
& I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_8^{4,5} 1^5, 4, \\
& I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_8^{4,5} 122313, \\
& I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_8^{4,5} I^{\oplus 5}, \\
& I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151, \\
& 2321 \epsilon_7^1 \otimes \epsilon_8^{4,5} I^{\oplus 2}, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1^5, 3, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 122313, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1223151, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& 321\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{1,4,2}, \\
& I^{\oplus 2}\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{1,5,5}, \\
& 321\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 I^{\oplus 4}, \\
& I^{\oplus 2}\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 12231512, \\
& I^{\oplus 2}\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 \otimes, \\
& I^{\oplus 2}\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 I^{\oplus 2}, \\
& 1\mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{1,5,4}, \\
& 1\mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8 I^{\oplus 5}, \\
& 1\mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8 \otimes,
\end{aligned}$$

$$\begin{array}{l}
I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{1, 5, 3} \\
1 \epsilon_7^{1, 4, 5} \otimes I^{\oplus 3} \epsilon_8 \text{ 1223} \\
2321 \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{1, 4, 5} I^{\oplus 3} \\
1 \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{1, 4, 5} I^{\oplus 2} \text{ 1223151} \\
321 \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{1, 4, 5} I^{\oplus 3} I^{\oplus 2} \\
I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{1, 4, 5} I^{\oplus 3} I^{\oplus 3} \\
1 \epsilon_7^{1, 5, 5} \otimes \epsilon_8^{1, 5, 5} \\
1 \epsilon_7^{1, 5, 5} \otimes \epsilon_8^{1, 5, 5} \text{ 12231512} \\
1 \epsilon_7^{1, 5, 5} \otimes \epsilon_8^{1, 5, 5} I^{\oplus 2} I^{\oplus 4}
\end{array}$$

$$\begin{aligned}
& 1\mathbf{e}_7 \otimes \mathbf{e}_7^{1,3,3} 1231513, \\
& I^{\oplus 3} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{1,3,2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{1,2} 1223, \\
& I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{1,2} 123215, \\
& \otimes \mathbf{e}_7^{2,3,1} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} I^{\oplus 2}, \\
& I^{\oplus 3} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{1,2} I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1,4,4} \otimes \mathbf{e}_7^{1,2} 1,
\end{aligned}$$

$$\begin{aligned} & \otimes_{\mathbf{e}_7}^{2,3,1} \otimes_{\mathbf{e}_7}^{3,3,1} \otimes_{\mathbf{e}_7}^{3,2}, \\ & \otimes_{\mathbf{e}_7}^{3,3,1} \otimes_{\mathbf{e}_7}^{3,3,1} \otimes_{\mathbf{e}_7}^{1,3,3}, \\ & 321\mathbf{e}_7^1 \otimes_{\mathbf{e}_7}^{3,3,1} \mathbf{e}_7 12315, \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes_{\mathbf{e}_7}^{3,3,1} \mathbf{e}_7 123151, \\ & \otimes_{\mathbf{e}_7}^{3,3,1} \otimes_{\mathbf{e}_7}^{3,3,1} I^{\oplus 2} \mathbf{e}_7 I^{\oplus 2}, \\ & I^{\oplus 2} \mathbf{e}_7^2 \otimes_{\mathbf{e}_7}^{3,3,1} \mathbf{e}_7 I^{\oplus 2}, \\ & 1\mathbf{e}_7^1 \otimes_{\mathbf{e}_7}^{5,4,1} \mathbf{e}_7 1, \end{aligned}$$

$$\begin{array}{c}
\begin{array}{ccccc}
2,3 & 1 & 3,3 & 1 & 3,3 \\
\otimes \epsilon_7 & \otimes & \epsilon_7 & \otimes & \epsilon_7
\end{array} \\
I^{\oplus 3} \epsilon_7^1 \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} \\
1 \epsilon_7^1 \otimes \epsilon_7^{3,3} I^{\oplus 2} 1232 \\
I^{\oplus 3} \epsilon_7^1 \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} I^{\oplus 3} 1 \\
I^{\oplus 2} \epsilon_7^1 \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} I^{\oplus 2} 123 \\
1 \epsilon_7^1 \epsilon_7^{4,4} \otimes \epsilon_7^1 123 \\
1 \epsilon_7^1 \epsilon_7^{5,5} \otimes \epsilon_7^1 1
\end{array}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{1,5,3} \otimes, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{1,2,2,3,1,3}, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{1,2,2,3,1,5,1}, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{I^{\oplus 2}}, \\ & 31_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{1,5,4}, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8 I^{\oplus 5}, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{I^{\oplus 24,2}}, \\ & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} I^{\oplus 3}, \end{aligned}$$

$$\begin{array}{l} 1^{1,3,5} \mathbf{1e}_6 \otimes \mathbf{e}_8 \otimes 1^{5,5} \\ 1^{1,3,5} \mathbf{1e}_6 \otimes \mathbf{e}_8 12231512 \\ 31^{1,3,5} \mathbf{e}_6 \otimes \mathbf{e}_8 \otimes 1^{4,2} \\ 31^{1,3,5} \mathbf{e}_6 \otimes \mathbf{e}_8 I^{\oplus 4} \end{array}$$

$$\begin{array}{l} 1\mathfrak{e}'_7 \otimes^{3,3} 1\mathfrak{e}_7 1223, \\ 1\mathfrak{e}'_7 \otimes^{3,3} 1\mathfrak{e}_7 123151, \end{array}$$

$$2321 \mathbf{e}'_7 \otimes \mathbf{e}_7 1, \\ I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}_7 \otimes,$$

$$1\epsilon_7' \otimes^{3,3} 1\epsilon_7 123215$$

$$I^{\oplus 2} \epsilon_7' \otimes^{3,3} \epsilon_7 \otimes^{3,3}$$

$$1\epsilon'_7 \otimes \epsilon_7^{3,3} I^{\oplus 2} 123,$$

$$I^{\oplus 2} \epsilon'_7 \otimes \epsilon_7^{3,3} I^{\oplus 2},$$

$$321\epsilon'_7 \otimes \epsilon_7^{3,3} I^{\oplus 2} 1,$$

$$1\epsilon'_7 \otimes \epsilon_7^{4,4} I^{\oplus 2} 1,$$

$$I^{\oplus 2} \epsilon'_7 \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$1\epsilon'_7 \otimes \epsilon_7^{4,5} 1$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} 123,$$

$$I^{\oplus 2} \epsilon_6 \otimes \epsilon_6^{2,2} 123,$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} I^{\oplus 2} 13,$$

$$31\epsilon_6 \otimes \epsilon_6^{2,2} I^{\oplus 2} 1,$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} 13151,$$

$$31\epsilon_6 \otimes \epsilon_6^{2,2} 1,$$

$$31\epsilon_6 \otimes \epsilon_6^{2,2} 1315,$$

$$1\epsilon_6 \otimes \epsilon_6^{3,3} 1,$$

$$\epsilon_6^{2,2} \otimes \epsilon_6^{2,2} 123,$$

$$I^{\oplus 2} \epsilon_6 \otimes \epsilon_6^{2,2} I^{\oplus 2} 1$$

$$31\epsilon_6 \otimes \epsilon_6^{2,2} 123,$$

$$1\epsilon_6 \bigcirc \epsilon_6^{3,3} 1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 15,3,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 122313,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 1223151,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} I^{\oplus 3}$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 15,4,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} I^{\oplus 5},$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} I^{\oplus 24,2},$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 15,5$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} 12231512$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} I^{\oplus 2}$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} 13,2,$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} I^{\oplus 2},$$

$$31\epsilon_6 \otimes \epsilon_7^{3,4} 1,$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} 13,3,$$

$$31\epsilon_6 \otimes \epsilon_7^{3,3} I^{\oplus 2},$$

$$1\epsilon_6 \otimes \epsilon_7^{3,5} 1$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} I^{\oplus 3}$$

$$1\epsilon_6 \otimes \epsilon_7^{3,4} I^{\oplus 2}$$

$$1\epsilon'_7 \otimes \epsilon'_7^{3,3} 123,$$

$$I^{\oplus 2} \epsilon'_7 \otimes \epsilon'_7^{3,3} I^{\oplus 2} 1,$$

$$1\epsilon'_7 \otimes \epsilon'_7^{4,4} 1$$

$$1\epsilon_6 \otimes \epsilon'_7^{3,3} I^{\oplus 2} 1,$$

$$31\epsilon_6 \otimes \epsilon'_7^{3,3} 1,$$

$$1\epsilon_6 \otimes \epsilon'_7^{3,4} 1$$

$$150 \otimes 501$$

2.4 3 node(1907)

2.4.1 gLg(11)

$$50 \otimes \epsilon_7^{2,3} \otimes 50^{4,2},$$

$$50 \otimes \epsilon_7^{3,3} \otimes 50^{3,3}$$

$$50 \otimes \epsilon_8^{3,5} \otimes \epsilon_6^{5,5}$$

$$\mathfrak{so} \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7'$$

$$\mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$\mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7' \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{so} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7'$$

$$\mathfrak{so} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$\mathfrak{so} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{e}_6 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

2.4.2 $\mathfrak{gLg}(8)$

$$\mathfrak{so} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} I^{\oplus 3} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{so} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} I^{\oplus 2} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7'$$

$$\mathfrak{so} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} I^{\oplus 2} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$\mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 1 \ 3,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{so} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 1 \ 2,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{e}_7' \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \ 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \ 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \ 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

2.4.3 SgLgLg(170)

$$\begin{array}{l}
513221e_7^{3,3} \otimes e_7^{3,2} \otimes 50, \\
2,3 \quad 3,3 \quad 4,2 \\
\otimes e_7 \otimes e_7 \otimes 50, \\
2,4 \quad 3,3 \quad 3,3 \\
\otimes e_7 \otimes e_7 \otimes 50, \\
5,4 \quad 3,3 \quad 3,3 \\
\otimes e_7 \otimes e_7 \otimes 50, \\
2151321e_7^{3,3} \otimes e_7^{3,3} \otimes 50, \\
1 \quad 3,3 \quad 3,3 \\
151321e_7 \otimes e_7 \otimes 50
\end{array}$$

$$\begin{aligned}
& I^{\oplus 2} \epsilon_7^{4,4} \otimes \epsilon_7^{3,2} \otimes \mathfrak{so}, \\
& \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \mathfrak{so}, \\
& \otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \otimes \mathfrak{so}, \\
& I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \mathfrak{so}, \\
& 2321 \epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \mathfrak{so},
\end{aligned}$$

$$\begin{array}{l}
1\epsilon_7^{\otimes 5,4} \otimes \epsilon_7^{\otimes 3,2} \otimes \mathfrak{so} \\
I^{\oplus 3} \epsilon_7^{\otimes 3,3} \otimes \epsilon_7^{\otimes 4,2} \otimes \mathfrak{so} \\
\otimes \epsilon_7^{\otimes 4,4} \otimes \epsilon_7^{\otimes 3,3} \otimes \mathfrak{so} \\
31321\epsilon_7^{\otimes 3,3} \otimes \epsilon_7^{\otimes 3,3} \otimes \mathfrak{so} \\
1512321\epsilon_7^{\otimes 3,3} \otimes \epsilon_7^{\otimes 3,3} \otimes \mathfrak{so}
\end{array}$$

$$\begin{array}{l} 3,5 \quad 5,4 \quad 3,2 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ 313221\mathfrak{e}_8 \quad 5,4 \quad 3,2 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ 1513221\mathfrak{e}_8 \quad 5,4 \quad 3,2 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ I^{\oplus 3}\mathfrak{e}_7 \quad 4,5 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{ccc} 4,5 & 5,4 & 3,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \\ I^{\oplus 5} \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \\ 2,3 & 4,5 & 5,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so}, \\ I^{\oplus 2} \mathfrak{e}_7 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{ccccc} 5,5 & 5,4 & 3,2 & & \\ \otimes e_8 & \otimes e_7 & \otimes s_0 & & \\ 21513221e_8 & \otimes e_7 & \otimes s_0 & & \\ 3,3 & 4,5 & 5,3 & & \\ \otimes e_7 & \otimes e_8 & \otimes s_0 & & \end{array}$$

$$\begin{array}{l} 3,5 \quad 5,3 \quad 2,2 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\ 313221 \mathfrak{e}_8 \quad 5,3 \quad 2,2 \\ 1513221 \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\ 1 \\ 1513221 \mathfrak{e}_8 \quad 5,3 \quad 2,2 \\ I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{ccc} 4,5 & 5,3 & 2,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \\ I^{\oplus 5} \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \\ 31\mathfrak{so} & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6, \\ 1\mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{ccccc} 5,5 & 5,3 & 2,2 & & \\ \otimes e_8 & \otimes e_6 & \otimes s_0 & & \\ 21513221e_8 & \otimes e_6 & \otimes s_0 & & \\ 2,2 & 3,5 & 5,3 & & \\ \otimes e_6 & \otimes e_8 & \otimes s_0 & & \end{array}$$

$$\begin{array}{l} \begin{array}{ccc} 2,3 & 3,3 & 2,2 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \end{array} \\ I^{\oplus 2} \mathfrak{e}_7 \otimes \begin{array}{cc} 4,3 & 2,2 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \end{array} \\ 1 \mathfrak{e}_6 \otimes \begin{array}{cc} 3,4 & 3,2 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{array} \\ I^{\oplus 2} \mathfrak{e}_6 \otimes \begin{array}{cc} 3,3 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} \end{array} \end{array}$$

$$\begin{array}{l} 3,3 \quad 3,3 \quad 2,2 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\ \\ 5,3 \quad 2,2 \\ 1\mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\ \\ 3,3 \quad 4,2 \\ 1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \\ 5131 \mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array}$$

$$\begin{aligned} & 3151321\mathbf{e}_7 \otimes^{3,3} \mathbf{e}_6 \otimes^{2,2} \mathbf{e}_6, \\ & 2321\mathbf{e}_7 \otimes^{4,3} \mathbf{e}_6 \otimes^{2,2} \mathbf{e}_6, \\ & 15131\mathbf{e}_6 \otimes^{2,2} \mathbf{e}_6 \otimes^{3,5} \mathbf{e}_7, \\ & 1\mathbf{e}_6 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,4} \mathbf{e}_6 \end{aligned}$$

$$\begin{array}{l} 2315131\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7, \\ 315131\epsilon_6^{2,2} \otimes \epsilon_6^{3,4} \otimes \epsilon_7, \\ 31\epsilon_6^{3,3} \otimes \epsilon_7^{4,3} \otimes \epsilon_6, \end{array}$$

$$\begin{array}{l} 2321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7 \\ 321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_7 \\ 31\mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6 \end{array}$$

$$23151321e_7 \overset{3,3}{\otimes} e_7' \overset{3,3}{\otimes} e_7,$$

$$3221e_7 \overset{3,3}{\otimes} e_7 \overset{4,4}{\otimes} e'_7,$$

$$2321e'_7 \otimes^{3,3} e_7 \otimes^{4,4} e_7$$

$$\begin{array}{c} 2321e_7^{4,4} \otimes e_7^{3,3} \otimes e_7', \\ \otimes e_7^{2,3} \otimes e_7^{3,3} \otimes e_7^{4,5} \otimes e_7', \\ I^{\oplus 3} e_7^{3,3} \otimes e_7^{4,5} \otimes e_7' \end{array}$$

$$\begin{array}{c} 512321e_7 \otimes^{3,3} e_7 \otimes^{4,4} e'_7, \\ \otimes^{3,3} e_7 \otimes^{3,3} e_7 \otimes^{4,5} e'_7, \end{array}$$

$$\begin{array}{c} 151321e_7 \otimes^{3,3} e_7 \otimes^{4,4} e'_7 \\ 321e'_7 \otimes^{3,3} e_7 \otimes^{4,5} e_7 \end{array}$$

$$\begin{array}{l} \begin{array}{ccc} 2,3 & 3,3 & 3,2 \\ \otimes e_7 & \otimes e_7' & \otimes s_0, \end{array} \\ 51321e_7' \otimes \begin{array}{cc} 3,3 & 3,2 \\ \otimes e_7 & \otimes s_0, \end{array} \\ \begin{array}{ccc} 2,3 & 3,3 & 3,3 \\ \otimes e_7' & \otimes e_7 & \otimes s_0, \end{array} \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 3,3 & & 3,3 & & 3,2 \\ \otimes & e_7 & \otimes & e'_7 & \otimes \text{ so,} \end{array} \\ 1e'_7 \otimes \begin{array}{cc} 4,4 & 3,2 \\ \otimes & \otimes \end{array} \text{ so,} \\ \begin{array}{ccccc} 3,3 & & 3,3 & & 3,3 \\ \otimes & e'_7 & \otimes & e_7 & \otimes \text{ so,} \end{array} \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so} \\ I^{\oplus 2} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2} \otimes \mathfrak{so} \\ I^{\oplus 3} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{so} \end{array}$$

$$\begin{aligned} & 3221e_7^{3,3} \otimes e_7^{4,3} \otimes e_6, \\ & \otimes e_7^{2,3} \otimes e_7^{3,3} \otimes e_6^{4,4}, \\ & I^{\oplus 3}e_7^{3,3} \otimes e_7^{4,4} \otimes e_6, \end{aligned}$$

$$\begin{array}{c} 512321e_7^{3,3} \otimes e_7^{4,3} \otimes e_6, \\ \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_6^{4,4}, \\ 31e_6^{3,3} \otimes e_7^{4,5} \otimes e_7 \end{array}$$

$$\begin{array}{c} 151321e_7 \otimes^{3,3} e_7 \otimes^{4,3} e_6 \\ 321e_7 \otimes^{4,4} e_7 \otimes^{3,3} e_6 \end{array}$$

$$\begin{aligned} & 2321\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7', \\ & 321\epsilon_6^{2,2} \otimes \epsilon_6^{3,4} \otimes \epsilon_7', \\ & I^{\oplus 2}\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_7', \end{aligned}$$

$$\begin{array}{l} 2321e'_7 \otimes^{3,3} e_6 \otimes^{2,2} e_6, \\ 15131e_6 \otimes^{2,2} e_6 \otimes^{3,4} e'_7, \\ 31e_6 \otimes^{3,3} e'_7 \otimes^{3,3} e_6 \end{array}$$

$$\begin{array}{ccccc} 315131e_6 & \overset{2,2}{\otimes} & e_6 & \overset{3,3}{\otimes} & e'_7 \\ \overset{2,2}{\otimes} & e_6 & \overset{2,2}{\otimes} & e_6 & \overset{3,5}{\otimes} & e'_7 \end{array}$$

$$\begin{array}{l} 3221e_7^{3,3} \otimes e_7'^{3,3} \otimes e_6, \\ 321e_7'^{3,3} \otimes e_7^{4,3} \otimes e_6, \\ 1e_6^{3,3} \otimes e_7^{4,5} \otimes e_7' \end{array}$$

$$512321e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_6, \\ I^{\oplus 2} e_7' \otimes^{3,3} e_7 \otimes^{4,4} e_6,$$

$$\begin{array}{c} 151321e_7 \otimes^{3,3} e'_7 \otimes^{3,3} e_6 \\ 31e_6 \otimes^{3,3} e_7 \otimes^{4,4} e'_7 \end{array}$$

$$\begin{array}{l} 3221e_7 \otimes^{3,2} so \otimes^{1,1} so, \\ 151321e_7 \otimes^{3,2} so \otimes^{1,1} so, \\ 1so \otimes^{2,3} e_7 \otimes^{3,3} so \end{array}$$

$$\begin{array}{l} 2231\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,3} \otimes \mathfrak{e}_7, \\ 321\mathfrak{e}_7^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{ccccc} 512321e_7 & \overset{3,2}{\otimes} & 50 & \overset{1,1}{\otimes} & 50 \\ 23150 & \overset{1,1}{\otimes} & 50 & \overset{2,4}{\otimes} & e_7 \end{array}$$

$$5131513221 \mathbf{e}_8 \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}_6, \\ \otimes^{3,3} \mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 \otimes^{5,5} \mathbf{e}_6,$$

$$\begin{aligned} & 51321e_7 \otimes^{4,5} e_8 \otimes^{5,4} e_6, \\ & I^{\oplus 3}e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_6, \end{aligned}$$

$$\begin{array}{ccc} 2,3 & 4,5 & 5,5 \\ \otimes e_7 & \otimes e_8 & \otimes e_6 \\ I^{\oplus 2} e_7 & 5,5 & 5,5 \\ & \otimes e_8 & \otimes e_6 \end{array}$$

$$\begin{array}{l} 321\epsilon_6 \otimes^{2,2} \mathfrak{so} \otimes^{1,1} \mathfrak{so}, \\ 31\mathfrak{so} \otimes^{1,1} \mathfrak{so} \otimes^{2,3} \epsilon_6, \end{array}$$

$$\begin{array}{l} 231s_0 \otimes^{1,1} s_0 \otimes^{2,2} e_6, \\ 31e_6 \otimes^{3,2} s_0 \otimes^{1,1} s_0, \end{array}$$

$$\begin{array}{ccccc} & & 2,2 & & 1,1 \\ 15131e_6 & \otimes & 50 & \otimes & 50 \\ & & 1,1 & & 2,4 \\ 150 & \otimes & 50 & \otimes & e_6 \end{array}$$

$$\begin{array}{lll}
321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6, & 15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6, & \begin{array}{c} 2,2 \quad 2,2 \quad 3,4 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \end{array} \\
I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6, & 321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6, & 15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6 \\
\\
2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & 321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\
51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & \begin{array}{c} 1 \\ 31513221\mathfrak{e}_8 \end{array} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & 15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\
\\
2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, & 321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \\
51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, & \begin{array}{c} 1 \\ 31513221\mathfrak{e}_8 \end{array} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, & 15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \\
\\
51321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, & \begin{array}{c} 2,2 \quad 2,2 \quad 3,2 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \end{array} & I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \\
1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, & 1\mathfrak{e}_6 \bigcirc \mathfrak{e}_6 \otimes \mathfrak{so} & \\
\\
5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & \begin{array}{c} 2,2 \quad 3,5 \quad 5,5 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array} & I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\
1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 & & \\
\\
3151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, & 3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, & 512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \\
151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 & & \\
\\
I^{\oplus 2}\mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, & 1\mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_7' \otimes \mathfrak{so} \\
\\
5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7', & I^{\oplus 2}\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, & 1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\
\\
321\mathfrak{e}_7' \otimes \mathfrak{so} \otimes \mathfrak{so}, & 231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7', & 31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7' \\
\\
3151321\mathfrak{e}_7(\otimes \mathfrak{e}_7')^2, & 321\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7', & I^{\oplus 2}\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \\
\\
I^{\oplus 2}\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, & & 1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so}
\end{array}$$

$$I^{\oplus 2} \mathbf{e}_7 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_7 I^{\oplus 2}$$

$$\begin{array}{l}
\begin{array}{ccccc} 2,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 122315123 & \end{array} \\
\begin{array}{ccccc} 2,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513151 & \end{array} \\
\begin{array}{ccccc} 3,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 122315123 & \end{array} \\
\begin{array}{ccccc} 3,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513151 & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & 5,5 & \\ 51321\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & \otimes & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & \\ I^{\oplus 3}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & I^{\oplus 6} & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & 1 \\ I^{\oplus 3}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513 & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & \\ 51321\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231512 & \end{array} \\
\begin{array}{ccccc} & 5,5 & 5,5 & & \\ I^{\oplus 2}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & I^{\oplus 6} & \end{array} \\
\begin{array}{ccccc} & 5,5 & 5,5 & & 1 \\ I^{\oplus 2}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513 & \end{array}
\end{array}$$

$$\begin{array}{l}
\begin{array}{ccccc}
2,2 & 2,2 & 3,5 & 5,3 & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes & \\
\end{array} \\
\begin{array}{ccccc}
2,2 & 2,2 & 3,5 & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & 122313 & \\
\end{array} \\
\begin{array}{ccccc}
2,2 & 2,2 & 3,5 & & 1 \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & 1223151 & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
2,2 & 3,5 & 5,3 & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
2,2 & 3,5 & & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & 122313 & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
2,2 & 3,5 & & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & I^{\oplus 5} & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
2,2 & 3,5 & & & 1 \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & 1223151 & \\
\end{array} \\
5131 \begin{array}{ccccc}
2,2 & 3,5 & & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & 1223151 & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
3,5 & 5,3 & 2,2 & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & \otimes & \\
\end{array} \\
I^{\oplus 2} \begin{array}{ccccc}
3,5 & 5,4 & & & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & 1 & \\
\end{array}
\end{array}$$

$$\begin{array}{cccc}
2,2 & 3,3 & 3,3 & 4,3 \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \\
2,2 & 3,3 & 3,3 & \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & I^{\oplus 4} \\
2,2 & 3,3 & 3,3 & 1 \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & 1232 \\
5131\mathfrak{e}_6 & 3,3 & 3,3 & 3,2 \\
& \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes
\end{array}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 1,$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 123,$$

$$3 1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$3 1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes,$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes,$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 I^{\oplus 3}$$

$$1 \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$3 1 \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}'_7 1,$$

$$1 \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}'_7 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes,$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 123,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$3 1 \mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 1$$

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$$5 1 3 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$3,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$4,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$3 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$2 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$5 1 3 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$3,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$4,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$3 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$2 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$3,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$4,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$5,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$3 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$2 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1 1 5 1 3 2 2 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$3 2 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$3 1 \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\begin{matrix} 2,3 & 3,3 & 1 & 3,2 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{matrix}$$

$$\begin{matrix} 3,3 & 3,3 & 1 & 3,2 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{matrix}$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7'$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\begin{matrix} 2,2 & 2,2 & 1 & 2,2 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \end{matrix}$$

$$I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$2321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$315131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$3151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$2321\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$23151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7'$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

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$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151315,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 5},$$

$$21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151,$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 122313,$$

$$313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151,$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 12231512$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 122313$$

$$I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151315$$

$$I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151$$

$$1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223151315,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes,$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2}$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 123,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1,$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' 123,$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1$$

$$31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,3},$$

$$1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,4},$$

$$31\epsilon_6^1 \otimes \epsilon_6^{2,2} \bigcirc \epsilon_6^{3,3}$$

$$5131\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{2,2} \otimes \mathfrak{so},$$

$$1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,2} \otimes \mathfrak{so}$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7',$$

$$I^{\oplus 2}\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_7'$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,3} \otimes \epsilon_6,$$

$$I^{\oplus 2}\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_6$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_6,$$

$$1\epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_6$$

$$31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,4} \otimes \epsilon_7',$$

$$1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_7'$$

$$31513221\epsilon_8^1 \otimes \epsilon_6^{5,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6,$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_6$$

$$2321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7,$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_7$$

$$2321\epsilon_7^1 (\otimes \epsilon_7^{3,3})^2,$$

$$1\epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_7'$$

$$2321\epsilon_7^1 \otimes \epsilon_6^{3,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6,$$

$$31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_7$$

$$513221\epsilon_8^{I^{\oplus 2}} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7'$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$31513221\epsilon_8^1 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7'$$

$$\begin{aligned}
& I^{\oplus 4} \mathbf{e}_8^{I^{\oplus 2}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{1223132}, \\
& I^{\oplus 4} \mathbf{e}_8^{I^{\oplus 2}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8 I^{\oplus 6}, \\
& I^{\oplus 4} \mathbf{e}_8^{I^{\oplus 2}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{12231513}, \\
& 513221 \mathbf{e}_8^{I^{\oplus 2}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{12231512}, \\
& I^{\oplus 3} \mathbf{e}_8^{I^{\oplus 3}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{122315123}, \\
& I^{\oplus 3} \mathbf{e}_8^{I^{\oplus 3}, 5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{12231513151}
\end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8 1223132, \\ & I^{\oplus 4} \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8 122315123, \\ & I^{\oplus 4} \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8 12231513151, \\ & 513221 \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8 1223151, \\ & I^{\oplus 3} \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8^{5,5} \otimes \mathfrak{c}_8 12231513215, \end{aligned}$$

$$\begin{aligned}
& 513221 I_8^{\oplus 2, 5, 5} \otimes I_8^{5, 5} \otimes I_8^{5, 5} 122313 \\
& I^{\oplus 4} I_8^{\oplus 2, 5, 5} \otimes I_8^{5, 5} \otimes I_8^{5, 5} 12231513215 \\
& 513221 I_8^{\oplus 2, 5, 5} \otimes I_8^{5, 5} \otimes I_8^{5, 5} I^{\oplus 5} \\
& I^{\oplus 3} I_8^{\oplus 3, 5, 5} \otimes I_8^{5, 5} \otimes I_8^{5, 5} I^{\oplus 6} \\
& I^{\oplus 3} I_8^{\oplus 3, 5, 5} \otimes I_8^{5, 5} \otimes I_8^{5, 5} 12231513^1
\end{aligned}$$

$$\begin{aligned}
& \overset{3,5}{\otimes} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& 313221 \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& \overset{1}{1513221} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& \overset{4,5}{\otimes} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& I^{\oplus 5} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& \overset{3,5}{\otimes} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 I^{\oplus 3}, \\
& I^{\oplus 2} \overset{1}{\mathfrak{e}}_7 \otimes \overset{4,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 1223132, \\
& I^{\oplus 2} \overset{1}{\mathfrak{e}}_7 \otimes \overset{4,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 122315123, \\
& I^{\oplus 2} \overset{1}{\mathfrak{e}}_7 \otimes \overset{4,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 12231513151, \\
& \overset{1}{1513221} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 I^{\oplus 3}, \\
& I^{\oplus 4} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 \otimes, \\
& I^{\oplus 2} \overset{2,4}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 I^{\oplus 3}, \\
& 513221 \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 I^{\oplus 2}, \\
& I^{\oplus 3} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,4}{\mathfrak{e}}_7 I^{\oplus 3}, \\
& \overset{1}{1} \overset{1}{\mathfrak{e}}_7 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 122315123, \\
& \overset{1}{1} \overset{1}{\mathfrak{e}}_7 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 12231513151, \\
& \overset{5,5}{\otimes} \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_7 I^{\oplus 2}, \\
& 513221 \overset{1}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_7 1, \\
& I^{\oplus 2} \overset{2,4}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_8 \otimes \overset{5,5}{\mathfrak{e}}_7 I^{\oplus 2},
\end{aligned}$$

[illegible]

[illegible]

$$\begin{array}{cccc} 2,3 & 1 & 3,3 & 3,3 & 4,2 \\ \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7, \\ 2,3 & 1 & 3,3 & 3,3 & 4,5 \\ \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7, \end{array}$$

$$\begin{array}{l} \begin{array}{cccc} 2,3 & 1 & 3,3 & 3,3 & 4,3 \\ \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}_7 & \otimes \end{array}, \\ \begin{array}{cccc} 2,3 & 1 & 3,3 & 3,3 \\ \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}_7 & I^{\oplus 4} \end{array}, \end{array}$$

$$\begin{array}{cccc} 2,3 & 1 & 3,3 & 3,3 & 4,4 \\ \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7 & \otimes e_7 \end{array}$$

$$\begin{aligned} & 1\overset{1}{\mathbf{c}}_7'\overset{4,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_81223132, \\ & 1\overset{1}{\mathbf{c}}_7'\overset{4,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_812231513215, \\ & \overset{3,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,4}{\otimes}\overset{5,4}{\mathbf{c}}_7'I^{\oplus 2}, \\ & 313221\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,4}{\otimes}\overset{5,4}{\mathbf{c}}_7'I^{\oplus 2}, \\ & 21513221\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,4}{\otimes}\overset{5,4}{\mathbf{c}}_7'I^{\oplus 2}. \end{aligned}$$

$$\begin{aligned} & 1\overset{1}{e}_7^{\overset{4,5}{\otimes}} \otimes \overset{5,5}{e}_8 \otimes \overset{5,5}{e}_8 I^{\oplus 6}, \\ & 1\overset{1}{e}_7^{\overset{4,5}{\otimes}} \otimes \overset{5,5}{e}_8 \otimes \overset{5,5}{e}_8 12231\overset{1}{5}13, \\ & \overset{4,5}{\otimes} \overset{5,5}{e}_8 \otimes \overset{5,5}{e}_8 \otimes \overset{5,4}{e}_7' I^{\oplus 2}, \\ & I^{\oplus 5} \overset{1}{e}_8 \otimes \overset{5,5}{e}_8 \otimes \overset{5,4}{e}_7' I^{\oplus 2}, \\ & 1\overset{1}{5}13221\overset{1}{e}_8 \otimes \overset{5,5}{e}_8 \otimes \overset{5,4}{e}_7' I^{\oplus 2}, \end{aligned}$$

$$\begin{array}{l}
1_{\mathcal{C}_7}^{1,4,5} \otimes \mathcal{C}_8^{5,5} \otimes \mathcal{C}_8^{122315123} \\
1_{\mathcal{C}_7}^{1,4,5} \otimes \mathcal{C}_8^{5,5} \otimes \mathcal{C}_8^{12231513151} \\
\otimes \mathcal{C}_8^{5,5,1} \otimes \mathcal{C}_8^{5,5} \otimes \mathcal{C}_7^{5,4} I^{\oplus 2} \\
513221 \otimes \mathcal{C}_8^{I^{\oplus 2}} \otimes \mathcal{C}_8^{5,5} \otimes \mathcal{C}_7^{5,4} 1 \\
I^{\oplus 2} \otimes \mathcal{C}_8^{2,4} \otimes \mathcal{C}_8^{5,5} \otimes \mathcal{C}_7^{5,4} I^{\oplus 2}
\end{array}$$

$$\begin{aligned} & I^{\oplus 4} \epsilon_8^{\oplus 2, 5, 5} \otimes \epsilon_8^{\oplus 5, 4} \otimes \epsilon_7^{\oplus 2}, \\ & \otimes \epsilon_8^{\oplus 4, 5, 1} \otimes \epsilon_8^{\oplus 5, 5} \otimes \epsilon_7^{\oplus 5, 5} \otimes \epsilon_7^{\oplus 1}, \\ & I^{\oplus 5} \epsilon_8^{\oplus 1} \otimes \epsilon_8^{\oplus 5, 5} \otimes \epsilon_8^{\oplus 5, 5} \otimes \epsilon_7^{\oplus 1}, \\ & I^{\oplus 2} \epsilon_8^{\oplus 2, 4} \otimes \epsilon_8^{\oplus 5, 5} \otimes \epsilon_8^{\oplus 5, 5} \otimes \epsilon_7^{\oplus 1}, \end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} \epsilon_8^{1 \oplus 3, 5, 5} \otimes \epsilon_8^{5, 4} \otimes \epsilon_7' I^{\oplus 2}, \\
& \otimes \epsilon_8^{5, 5} \otimes \epsilon_8^{5, 5} \otimes \epsilon_7' 1, \\
& 21513221 \epsilon_8^1 \otimes \epsilon_8^{5, 5} \otimes \epsilon_7' 1, \\
& I^{\oplus 4} \epsilon_8^{1 \oplus 2, 5, 5} \otimes \epsilon_8^{5, 5} \otimes \epsilon_7' 1,
\end{aligned}$$

$$\begin{array}{l} 3,5 \quad 1 \quad 5,5 \quad 5,5 \\ \otimes \mathbf{e}_8 \otimes \mathbf{e}_8 \otimes \mathbf{e}'_7 1 \\ 313221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}'_7^{5,5} 1 \\ 1^1 1513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}'_7^{5,5} 1 \\ I^{\oplus 3} \mathbf{e}_8^{1 \oplus 3} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}'_7^{5,5} 1 \end{array}$$

$$\begin{aligned}
& 1\overset{1}{\mathbf{e}}_7^{\prime 3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,3}, \\
& 1\overset{1}{\mathbf{e}}_7^{\prime 3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{122313}, \\
& 1513221\overset{1}{\mathbf{e}}_8^{5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{\prime 1}, \\
& I^{\oplus 2}\overset{1}{\mathbf{e}}_8^{3221\ 5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{\prime 1}, \\
& I^{\oplus 4}\overset{1}{\mathbf{e}}_8^{5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{\prime 1} I^{\oplus 2}, \\
& 1\overset{1}{\mathbf{e}}_7^{\prime 4,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{3,2}, \\
& I^{\oplus 2}\overset{1}{\mathbf{e}}_7^{4,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{\prime 1} I^{\oplus 2}, \\
& 1\overset{1}{\mathbf{e}}_7 \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{\prime 1} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& 1\overset{1}{c}'_7 \otimes \overset{3,3}{c}_7 \otimes \overset{4,5}{c}_8 \otimes \overset{5,4}{c}_8, \\
& 1\overset{1}{c}'_7 \otimes \overset{3,3}{c}_7 \otimes \overset{4,5}{c}_8 I^{\oplus 5}, \\
& 1\overset{1}{c}'_7 \otimes \overset{3,3}{c}_7 \otimes \overset{4,5}{c}_8 1223151, \\
& I^{\oplus 3} \overset{1}{c}_8 \otimes \overset{5,4}{c}_7 \otimes \overset{3,3}{c}'_7 123, \\
& I^{\oplus 2} \overset{1}{c}_8 \otimes \overset{5,4}{c}_7 \otimes \overset{3,3}{c}'_7 123, \\
& 1\overset{1}{c}'_7 \otimes \overset{4,5}{c}_8 \otimes \overset{5,4}{c}_7 \otimes \overset{3,3}{c}_7, \\
& 1\overset{1}{c}'_7 \otimes \overset{4,5}{c}_8 \otimes \overset{5,5}{c}_7 I^{\oplus 2}, \\
& 1\overset{1}{c}_7 \otimes \overset{5,5}{c}_8 \otimes \overset{5,5}{c}'_7 1
\end{aligned}$$

$$\begin{array}{l}
1^{1,3,3} \quad 4,5 \quad 5,5 \\
1\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon_8 \otimes \\
1\epsilon'_7 \quad 1^{1,3,3} \quad 4,5 \\
1\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon_8 12231512 \\
2,4 \quad 1^{5,4} \quad 3,3 \\
\otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon'_7 I^{\oplus 2} \\
I^{\oplus 2} \quad 1^{5,4} \quad 3,3 \\
\epsilon_8 \otimes \epsilon_7 \otimes \epsilon'_7 1232 \\
I^{\oplus 3} \quad I^{\oplus 2} \quad 5,4 \quad 3,3 \\
\epsilon_8 \otimes \epsilon_7 \otimes \epsilon'_7 I^{\oplus 2} \\
1^{4,5} \quad 5,4 \\
1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 3} \\
I^{\oplus 2} \quad 1^{4,5} \quad 5,5 \\
\epsilon_7 \otimes \epsilon_8 \otimes \epsilon'_7 1
\end{array}$$

$$\begin{aligned}
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2}, \\
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5}, \\
& 51321_{\mathfrak{e}_7}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_{61}, \\
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1232151}, \\
& \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2}, \\
& \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{I^{\oplus 2}}, \\
& I^{\oplus 2} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2}, \\
& 1_{\mathfrak{e}_7}^{1,3,3} \otimes \mathfrak{e}_7^{4,3} \otimes \mathfrak{e}_{613},
\end{aligned}$$

$$\begin{aligned}
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3} \otimes, \\
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{\oplus 4}, \\
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1231512}, \\
& 1_{\mathfrak{e}_6}^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{123151}, \\
& I^{\oplus 3} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes, \\
& I^{\oplus 3} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{\oplus 2}, \\
& I^{\oplus 2} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{\oplus 2}, \\
& 1_{\mathfrak{e}_7}^{1,4,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^1
\end{aligned}$$

$$\begin{array}{l}
1^{1,3,3} \quad 3,3 \quad 4,4 \\
1\mathbf{e}_6 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \\
1^{1,3,3} \quad 3,3 \quad 3,3 \\
1\mathbf{e}_6 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 12313 \\
1^{1,3,3} \quad 3,3 \quad 3,3 \quad 1 \\
1\mathbf{e}_6 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 1232 \\
2,3 \quad 1^{1,3,3} \quad 3,3 \quad 2,2 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_6 \otimes \\
2,3 \quad 1^{1,3,3} \quad 3,3 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_6 I^{\oplus 2} \\
I^{\oplus 2} \quad 1^{1,3,3} \quad 3,3 \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_6 1315 \\
I^{\oplus 2} \quad 1^{1,3,3} \quad 4,3 \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_6 1
\end{array}$$

$$\begin{aligned} & 3,5 \quad 1 \quad 4,2 \quad 1,1 \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & 313221 \quad 1 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & 1513221 \quad 1 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & I^{\oplus 4} \quad 1 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & I^{\oplus 3} \quad 1 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}132, \\ & I^{\oplus 3} \quad 1 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}13 \end{aligned}$$

$$\begin{aligned} & 4,5 \quad 1,4,2 \quad 1,1 \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & I^{\oplus 5} 1,4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & I^{\oplus 2} 2,4 \quad 4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1, \\ & I^{\oplus 2} 1,4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1322, \\ & I^{\oplus 4} 1,4,2 \quad 1,1 \\ & \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}13. \end{aligned}$$

$$\begin{array}{l} \begin{array}{ccc} 5,5 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 1 \\ 21513221 \mathfrak{e}_8 \begin{array}{ccc} 1 & 4,2 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 1 \\ \begin{array}{ccc} 2,4 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 13 \\ I^{\oplus 3} \mathfrak{e}_8 \begin{array}{ccc} I^{\oplus 3} & 4,2 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 1 \\ I^{\oplus 2} \mathfrak{e}_8 \begin{array}{ccc} I^{\oplus 2} & 4,2 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 132 \end{array}$$

$$\begin{array}{c}
I^{\oplus 3} \textcolor{red}{e}_8^{I^{\oplus 2} \textcolor{red}{5}, 4} \otimes \textcolor{red}{e}_7 \otimes \textcolor{red}{e}_6 \textcolor{red}{1}^{3, 3} \\
\textcolor{red}{1} \textcolor{red}{e}_6 \otimes \textcolor{red}{e}_8^{3, 5} \otimes \textcolor{red}{e}_7^{5, 4} \otimes \textcolor{red}{e}_6^{3, 2} \\
\textcolor{red}{1} \textcolor{red}{e}_7^{1, 4, 5} \otimes \textcolor{red}{e}_8^{5, 3} \otimes \textcolor{red}{e}_6 \textcolor{red}{1315} \\
\textcolor{red}{1} \textcolor{red}{e}_7^{1, 5, 5} \otimes \textcolor{red}{e}_8^{5, 3} \otimes \textcolor{red}{e}_6^{2, 2} \\
I^{\oplus 2} \textcolor{red}{e}_7^{1, 4, 5} \otimes \textcolor{red}{e}_8^{5, 4} \otimes \textcolor{red}{e}_6 \textcolor{red}{1}
\end{array}$$

$$\begin{array}{l} 1^{1,3,3} \otimes 6^{2,2} \otimes 6^{123} \\ 1^{1,3,3} \otimes 6^{2,2} \otimes 6^{13151} \\ 31^{1,2,2} \otimes 6^{3,3} \otimes 7 I^{\oplus 2} \\ 1^{1,4,3} \otimes 6^{2,2} \otimes 6 I^{\oplus 2} \\ 1^{1,3,3} \otimes 7^{3,3} \otimes 6^{2,2} \end{array}$$

$$\begin{array}{l}
4,5 \quad 1 \quad 5,5 \quad 4,2 \\
\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}13 \\
313221 \mathfrak{e}_8^1 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13 \\
21513221 \mathfrak{e}_8^1 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13 \\
I^{\oplus 4} \mathfrak{e}_8^{1 \oplus 2} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13
\end{array}$$

$$\begin{array}{l} 1e_7^{1,3,3} \otimes e_7^{3,3} \otimes e_7^{3,3}123 \\ 1e_7^{1,3,3} \otimes e_7^{3,3} \otimes e_7^{3,3}12315 \\ I^{\oplus 2}e_7^{1,3,3} \otimes e_7^{3,3} \otimes e_7^{3,3}I^{\oplus 2} \end{array}$$

$$\begin{array}{l} I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^1 \\ 1 \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{1315} \\ 1 \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{4,3} \otimes \mathbf{e}_6^1 \end{array}$$

$$150 \overset{1}{\otimes} \overset{1,1}{50} \overset{1,1}{50} \overset{2,4}{\otimes}$$

$$1\epsilon_7' \overset{1}{\otimes} \overset{4,5}{\epsilon_8} \overset{5,3}{\otimes} \epsilon_6 I^{\oplus 2}$$

$$1\epsilon_7' \overset{1}{\otimes} \overset{3,3}{\epsilon_6} \overset{2,2}{\otimes} \epsilon_6 I^{\oplus 2}$$

$$I^{\oplus 3} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$31 \overset{31}{\mathfrak{e}_6} \overset{2,2}{\otimes} \overset{2,2}{\mathfrak{e}_6} \otimes \mathfrak{so}1$$

$$1e_7 \overset{1}{\otimes} \overset{4,2}{so} \overset{1,1}{\otimes} so1$$

$$1e_7^{1,5,5} \otimes e_8^{4,2} \otimes so_{13}$$

$$1 \overset{1}{e}'_7 \otimes_{4,5} e_8 \otimes_{5,5} e'_7 1$$

$$31\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes^{1,3,3} \mathfrak{e}_7 \otimes^{3,2} \mathfrak{so}1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} \overset{4,2}{\otimes} 5013$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7'} \overset{3,3}{\otimes} e_7' I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes I^{\oplus 3} \mathbf{e}_7 \mathbf{1}$$

$$\begin{array}{ccccccc} 1 & 2,2 & 3,5 & I^{\oplus 3} & & & \\ 1\mathbf{e}_6 & \otimes \mathbf{e}_6 & \otimes \mathbf{e}_8 & I^{\oplus 2} & & & \\ & 1 & 3,5 & 5,3 & 1 & & \\ & 1\mathbf{e}_6 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_6 & 1 & & \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6^{1,2,2} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{1,2,2}$$

$$31 \mathbf{e}_6^{31,2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} \mathbf{e}_6^1$$

$$1e_7' \otimes^{3,3} e_7 \otimes^{4,5} e_8 \otimes^{I^{\oplus 3}} I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \overset{1}{\mathbf{e}_6} \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,4}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} I^{\oplus 2} \mathbf{e}'_7 1$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_7' \overset{3,3}{\otimes} \mathbf{e}_7' \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7' 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} \otimes \epsilon_7^{5,4} \epsilon_7^1$$

$$1\epsilon_6^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} I^{\oplus 2} \epsilon_7^1$$

$$1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,3} \epsilon_7^1$$

2.4.10 SSgLIgLg(25)

$$1513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,4} \epsilon_7^1,$$

$$I^{\oplus 2} \epsilon_8^{3221} \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,4} \epsilon_7^1,$$

$$2,4 \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,5} \epsilon_7^1$$

$$I^{\oplus 4} \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,5} \epsilon_7^1,$$

$$I^{\oplus 3} \epsilon_8^{I^{\oplus 2}} \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,5} \epsilon_7^1$$

$$1513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_6^{5,3} \epsilon_6^1,$$

$$I^{\oplus 2} \epsilon_8^{3221} \otimes \epsilon_8^{5,5} \otimes \epsilon_6^{5,3} \epsilon_6^1,$$

$$2,4 \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_6^{5,4} \epsilon_6^1$$

$$I^{\oplus 4} \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_6^{5,4} \epsilon_6^1,$$

$$I^{\oplus 3} \epsilon_8^{I^{\oplus 2}} \otimes \epsilon_8^{5,5} \otimes \epsilon_6^{5,4} \epsilon_6^1$$

$$2,4 \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_0^{I^{\oplus 2} 4,2} \epsilon_0^1,$$

$$I^{\oplus 4} \epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_0^{I^{\oplus 2} 4,2} \epsilon_0^1,$$

$$513221\epsilon_8^{I^{\oplus 2}} \otimes \epsilon_8^{5,5} \otimes \epsilon_0^{4,2} \epsilon_0^1$$

$$I^{\oplus 3} \epsilon_8^{I^{\oplus 2}} \otimes \epsilon_8^{5,5} \otimes \epsilon_0^{I^{\oplus 2} 4,2} \epsilon_0^1$$

$$31513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,4} \epsilon_7^1,$$

$$1513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,5} \epsilon_7^1,$$

$$I^{\oplus 2} \epsilon_8^{3221} \otimes \epsilon_8^{5,5} \otimes \epsilon_7^{5,5} \epsilon_7^1$$

$$I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_0^{3,2} \epsilon_0^1$$

$$321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_6^{3,3} \epsilon_6^1$$

$$2321\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \epsilon_7^1$$

$$1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_0^{I^{\oplus 2} 4,2} \epsilon_0^1$$

$$1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_7^{5,5} \epsilon_7^1$$

$$1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_6^{5,4} \epsilon_6^1$$

$$1\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_0^{3,2} \epsilon_0^1$$

$$1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_0^{2,2} \epsilon_0^1$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'^{1,3,3},$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'^{1,3,3} I^{\oplus 2}$$

$$\begin{matrix} 2,4 & 1 & 5,5 & 1 & 5,3 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & 1, \end{matrix}$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 13,$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 13$$

$$\begin{matrix} 2,4 & 1 & 5,5 & 1 & 5,4 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7' & 1, \end{matrix}$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2},$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1,$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 13$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_7 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

2.4.12 SSgLIgLIgS(22)

$$\begin{array}{lll}
1\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 122315132, & \begin{array}{c} 2,4\ 1\ 5,5\ 1\ 5,5\ 1\ 4,2 \\ \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \end{array}, & I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \\
3221\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 122315, & I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 12231513, & I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 1223151 \\
I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 4}, & \begin{array}{c} 2,4\ 1\ 5,5\ 1\ 5,5\ 1\ 5,5\ I^{\oplus 3} \\ \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \end{array} I^{\oplus 2}, & I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2} 1223 \\
1513221\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2}, & I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2}, & I^{\oplus 2} I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2} 1223 \\
I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2} & &
\end{array}$$

$$\begin{array}{lll}
1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1\ 4,2}, & 1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1} I^{\oplus 4}, & 1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2}
\end{array}$$

$$\begin{array}{ll}
1\epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} 123, & I^{\oplus 2} \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} I^{\oplus 2} 1
\end{array}$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7^{1\ 4,5\ 1\ 5,4\ 1} 1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} I^{\oplus 2} 1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} 1$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6^{1\ 2,2\ 1\ 2,2\ 1} 1$$

2.5 4 node(1730)

2.5.1 gLgLG(16)

$$\mathfrak{so}(\otimes \epsilon_6)^{2\ 2,2\ 2\ 3,3} \otimes \epsilon_6, \quad \mathfrak{so}(\otimes \epsilon_6)^{2\ 2,2\ 2\ 3,3} \bigcirc \epsilon_6$$

$$\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8^{3,5\ 5,5\ 5,5} \otimes \epsilon_6$$

$$\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8^{3,5\ 5,5\ 5,3} \otimes \mathfrak{so}$$

$$\mathfrak{so}(\otimes \epsilon_7)^{2\ 3,3\ 2\ 3,3} \otimes \mathfrak{so}$$

$$50 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_7$$

$$50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e_7'$$

$$50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix} e_6$$

$$50 \otimes^{2,3} e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_6$$

$$\mathfrak{so}\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6\right)^2 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7$$

$$\mathfrak{so}(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{3,4}{\otimes} \mathfrak{e}'_7$$

$$50 \overset{1,1}{\otimes} 50 \overset{2,3}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} \otimes \mathfrak{e}_6 \otimes \mathfrak{so}^{2,2}$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^2 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$e_6 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7'$$

$$\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

2.5.2 gLIgLgLg(11)

$$e_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7'$$

$$e_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,4}{\otimes} e_6$$

$$e'_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{4,2}{so} \overset{1,1}{\otimes} so$$

$$e'_7 \overset{4,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7$$

$$\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_6$$

$$e_6 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

2.5.3 gLIgLIgIg(3)

$$50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_7$$

$$50 \otimes^{2,4} e_8 \otimes^{5,5} e_8 \otimes^{5,4} e_7'$$

$$50 \otimes_{\mathbb{Z}} e_8 \otimes_{\mathbb{Z}} e_8 \otimes_{\mathbb{Z}} e_6$$

2.5.4 SgLgLgLg(175)

$$\begin{matrix} 3,5 & 5,5 & 5,4 & 3,3 \\ \otimes & \otimes & \otimes & \otimes \\ e_8 & e_8 & e_7 & e_6, \end{matrix}$$

$$313221\mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6,$$

$$1513221e_8 \overset{5,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_6,$$

$$I^{\oplus 3} \mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$\begin{array}{cccc} 4,5 & 5,5 & 5,4 & 3,3 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array}$$

$$I^{\oplus 5} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6,$$

$$\begin{array}{cccc} 2,3 & 4,5 & 5,5 & 5,5 \\ \otimes e_7 & \otimes e_8 & \otimes e_8 & \otimes e_6, \end{array}$$

$$I^{\oplus 2} \mathbf{e}_7 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^2 \bigotimes^{5,5} \mathbf{e}_6$$

$$\begin{array}{cccc} 5,5 & 5,5 & 5,4 & 3,3 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 \end{array}$$

$$21513221 \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6$$

$$\begin{array}{cccc} 3,3 & 4,5 & 5,5 & 5,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 \end{array}$$

$$1e_7^{4,4} \otimes e_7^{3,3} \otimes e_7^{3,2} \otimes so,$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$${}^{3,4} \otimes \mathfrak{e}_7 ({}^{3,3} \otimes \mathfrak{e}_7)^2 {}^{3,3} \otimes \mathfrak{so}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$231513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2,$$

$$223151321\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_8,$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$1513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \bigcirc \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \bigcirc \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}$$

$$I^{\oplus 3}\mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7,$$

$$15131\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7,$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$51321\mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{4,2}{\otimes} \mathbf{so}$$

$$5131\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

$$1\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,2}{\otimes} \mathbf{so}$$

$$150 \overset{2,4}{\otimes} e_8 \overset{5,3}{\otimes} e_6 \overset{2,2}{\otimes} 50$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6$$

$$31\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_7$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,3}{\otimes}\mathfrak{e}_6$$

$$1e'_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_6$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \oplus \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1e'_7 \left(\overset{5,5}{\otimes} e_8 \right)^2 \overset{5,5}{\otimes} e_6$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,3}{\otimes} so$$

$$31e_6 \overset{3,3}{\otimes} e_7 (\overset{3,3}{\otimes} e_7')^2$$

$$150 \overset{2,4}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \overset{5,3}{\otimes} 50$$

$$1e_6 \overset{3,3}{\otimes} e_7 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e_6$$

$$1\mathfrak{so}(\overset{1,1}{\otimes} \mathfrak{so})^2 \overset{2,4}{\otimes} \mathfrak{e}'_7$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_7'$$

$$5131513221 \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7'$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} 50$$

$$1e'_7 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e_7 \overset{3,2}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$\begin{array}{l} \overset{5,3}{\otimes} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes}, \\ 3131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes}, \\ \overset{3,3}{\bigcirc} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,3}{\bigcirc}, \\ \overset{3,3}{\bigcirc} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \\ I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 1313, \\ 3131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 1313, \\ 215131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \end{array}$	$\begin{array}{l} \overset{3,3}{\bigcirc} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes}, \\ \overset{5,3}{\otimes} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \\ I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,3}{\bigcirc}, \\ \overset{3,3}{\bigcirc} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 13151, \\ I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \\ 3131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \\ \overset{1}{15131}\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 131512, \end{array}$	$\begin{array}{l} I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \\ \overset{5,3}{\otimes} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \\ 3131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,3}{\bigcirc} \\ I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 I^{\oplus 3} \\ I^{\oplus 3} \mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \\ 3131\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \\ \overset{1}{15131}\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \end{array}$
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$$I^{\oplus 2} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 I^{\oplus 2}$$

$$\begin{array}{ccccccccccc} 2,2 & 3,5 & 5,5 & 5,5 & & & & & & & \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & & & & & & & \\ & & & & 122315123 & & & & & & \end{array}$$

$$\begin{array}{cccccccc} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & 12231513151 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} 12231513151$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$1e_6 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231513151$$

$$1e'_7 \otimes^{3,3} e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5}$$

$$1e_7' \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$I^{\oplus 2} \mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 \otimes^{5,5} \mathbf{e}_8 \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_7 I^{\oplus 2}$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes}$$

$$1e_6 \overset{2,2}{\otimes} e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$31\mathbf{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 I^{\oplus 4}$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{c} 2,2 \\ \otimes \mathfrak{so} \end{array} \left(\begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \right)^2 \begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \begin{array}{c} 2,4 \\ \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{ccccccc} 3,2 & & (1,1) & & 1,1 & & 2,3 \\ \otimes \mathfrak{so} & & (\otimes \mathfrak{so})^2 & & \otimes \mathfrak{so} & & \otimes \end{array}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \otimes \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix}$$

$$\begin{matrix} 2,2 & (2,2) & 3,5 & 4,2 \\ \otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_6)^2 & \otimes \mathfrak{e}_8 & \otimes \end{matrix}$$

$$\begin{aligned} & 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix}, \\ & 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix}, \\ & 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 1313, \end{aligned}$$

$$\begin{aligned} & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 \overset{3,3}{\otimes}, \\ & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 \bigcirc, \\ & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 131512, \end{aligned}$$

$$\begin{array}{l} 150 \binom{2,2}{\otimes \mathfrak{e}_6}^2 \otimes \mathfrak{e}_6 \binom{3,4}{\otimes} \\ 150 \binom{2,2}{\otimes \mathfrak{e}_6}^2 \otimes \mathfrak{e}_6 I^{\oplus 3} \\ 150 \binom{2,2}{\otimes \mathfrak{e}_6}^2 \otimes \mathfrak{e}_6 13151^1 \end{array}$$

$$\begin{aligned} & 1\epsilon_7'(\overset{3,3}{\otimes}\epsilon_7)^2\overset{4,5}{\otimes}\epsilon_81223151, \\ & 2321\epsilon_7'(\overset{3,3}{\otimes}\epsilon_7)^2\overset{4,5}{\otimes}\epsilon_8I^{\oplus 2}, \\ & 1\epsilon_7'\overset{4,5}{\otimes}\epsilon_8\overset{5,4}{\otimes}\epsilon_7\overset{3,3}{\otimes}\epsilon_7I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 \otimes, \\ & I^{\oplus 2} e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 I^{\oplus 4}, \\ & 1 e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 \binom{5,5}{\otimes} e_7 1 \end{aligned}$$

$$321e'_7(\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_8 I^{\oplus 3}$$

$$1e'_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 I^{\oplus 2}$$

$$\begin{aligned} & 1\epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7 I^{\oplus 2}, \\ & \otimes \epsilon_6^{2,2} (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_7'^{3,3} \otimes, \\ & I^{\oplus 2} \epsilon_6^{3,3} (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_7'^{3,3} \otimes, \end{aligned}$$

$$\begin{aligned} & 31e_6^{3,3} \otimes e_7^{3,3} \otimes e_7'^{3,3} \otimes e_7 1, \\ & \otimes e_6^{2,2} (\otimes e_7^{3,3})^2 \otimes e_7' I^{\oplus 3}, \\ & I^{\oplus 2} e_6^{3,3} (\otimes e_7^{3,3})^2 \otimes e_7' I^{\oplus 3} \end{aligned}$$

$$\begin{array}{c} \begin{array}{ccccccc} 2,2 & & 3,3 & & 3,3 & & 3,2 \\ \otimes & \mathbf{e}_6 & (\otimes \mathbf{e}_7)^2 & \otimes & \mathbf{e}'_7 & \otimes & \\ I^{\oplus 2} & \mathbf{e}_6 & (\otimes \mathbf{e}_7)^2 & \otimes & \mathbf{e}'_7 & \otimes & \end{array} \end{array}$$

$$\begin{aligned} & 1\mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 123, \\ & \otimes \mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 I^{\oplus 2}, \\ & I^{\oplus 2} \mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 1, \end{aligned}$$

$$\begin{array}{c} 321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_71, \\ I^{\oplus 2}\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_7I^{\oplus 2}, \\ 1\mathbf{e}_6\overset{2,2}{\otimes}\mathbf{e}_6\overset{3,3}{\otimes}\mathbf{e}_7\overset{3,3}{\otimes}\mathbf{e}_61 \end{array}$$

$$\begin{array}{l} 15131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_71 \\ \overset{2,2}{\otimes}\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,4}{\otimes}\mathbf{e}_71 \end{array}$$

$$\begin{array}{l} 1\overset{1,1}{s}\overset{2,4}{o}\overset{5,5}{\otimes}\overset{2,4}{e}\overset{5,5}{\otimes}\overset{5,5}{e}1223132, \\ 1\overset{1,1}{s}\overset{2,4}{o}\overset{5,5}{\otimes}\overset{2,4}{e}\overset{5,5}{\otimes}\overset{5,5}{e}12231513215, \\ 31\overset{2,4}{s}\overset{5,5}{\otimes}\overset{5,5}{e}\overset{4,2}{\otimes}\overset{5,5}{e}\overset{4,2}{\otimes}\overset{5,5}{s}013 \end{array}$$

$$150^{1,1} \otimes 50^{2,4} \otimes e_8^{5,5} \otimes e_8 I^{\oplus 6},$$

$$150^{1,1} \otimes 50^{2,4} \otimes e_8^{5,5} \otimes e_8 12231^1 513,$$

$$\begin{array}{l} 1\overset{1,1}{50} \otimes 5\overset{2,4}{0} \otimes \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\ 1\overset{1,1}{50} \otimes 5\overset{2,4}{0} \otimes \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{aligned} & 1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2}, \\ & 31\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7 I^{\oplus 2}, \\ & I^{\oplus 2} \epsilon_6^{(3,3)} (\otimes \epsilon_7)^2 \otimes \epsilon_6^{3,3} I^{\oplus 2} \end{aligned}$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3},$$

$$\otimes \epsilon_6^{2,2} (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_6^{3,3} \otimes,$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7 I^{\oplus 3}$$

$$I^{\oplus 2} \epsilon_6^{3,3} (\otimes \epsilon_7)^2 \otimes \epsilon_6^{2,2}$$

$$\begin{array}{cccc} 2,2 & 3,5 & 5,5 & 5,4 \\ \otimes e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 I^{\oplus 2}, \\ I^{\oplus 2} e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 1, \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$\begin{array}{cccc} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 1 \\ 1e_6 & 4,5 & 5,5 & 5,5 \\ & \otimes e_8 & \otimes e_8 & \otimes e'_7 1 \end{array}$$

$$3150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 1223132,$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 6},$$

$$3150 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 122315123$$

$$3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513215, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513151$$

$$\begin{array}{lll} 150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} 12315, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\ 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7} I^{\oplus 2} & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(4,2)}, & 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 4}, & 31\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 3} \\ 1\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 I^{\oplus 2}, & 31\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 1 & \end{array}$$

$$\begin{array}{lll}
150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 1315, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 I^{\oplus 2} \\
3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_6 1 & &
\end{array}$$

$$\begin{array}{lll} 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,2}, & 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3}, & 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7 I^{\oplus 3} \\ 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7 I^{\oplus 2} & & \end{array}$$

$$I^{\oplus 2} e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,4} e_7' I^{\oplus 2}, \quad I^{\oplus 2} e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5} e_7' 1, \quad 1 e_7' (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_7' 1,$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix}, \quad 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}, \quad 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} I^{\oplus 3}$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} e_7' \otimes, \quad 150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' \otimes, \quad 150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,3}{\otimes} 6 \overset{2,2}{\otimes}, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,3}{\otimes} 6 I^{\oplus 2}, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,4}{\otimes} 6 1$$

$$150 \binom{1,1}{\otimes 50}^2 \otimes \epsilon_7 I^{\oplus 2}, \quad 3150 \binom{1,1}{\otimes 50}^2 \otimes \epsilon_7 1, \quad 150 \binom{1,1}{\otimes 50}^2 \otimes \epsilon_7 1$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 I^{\oplus 2}, \quad 31\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 1, \quad 1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7 1$$

$$\begin{array}{cc} \begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array} (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1, & I^{\oplus 2} \mathbf{e}_6 (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1 \end{array}$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,4 \\ e_8 \end{smallmatrix} \otimes e_7' I^{\oplus 2}, \quad 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes e_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{so}^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_7^{2,3} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,4} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}$$

2.5.6 SgLgLIgLg(57)

$$\begin{array}{lll}
5131513221e_8 \otimes e_8 \otimes e_8 \otimes e_7, & \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, & \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7 \\
\begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 5,4 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7 \end{array} \\
\begin{array}{c} 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & 313221e_8 \otimes e_8 \otimes e_8 \otimes e_7, & I^{\oplus 5}e_8 \otimes e_8 \otimes e_8 \otimes e_7 \\
\begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ 21513221e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 1 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ 1513221e_8 \otimes e_8 \otimes e_8 \otimes e_7 \end{array} &
\end{array}$$

$$\begin{array}{lll} \otimes_{\epsilon_8}^{3,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4}, & \otimes_{\epsilon_8}^{4,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4}, & \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4} \\ 313221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4}, & I^{\oplus 5}\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4}, & 21513221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4} \\ {}^11513221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{\epsilon'_7}^{5,4} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 1 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} & \begin{array}{l} 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} & \begin{array}{l} 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\ \\ \\ \\ \end{array} \end{array}$$

$$I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}, \quad 31 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}, \quad 1 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1 \epsilon_7^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$\begin{array}{lll} 3221\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} \otimes \epsilon_7', & 2321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} \otimes \epsilon_7, & 512321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} \otimes \epsilon_7' \\ 151321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} \otimes \epsilon_7' & & \end{array}$$

$$\begin{array}{ccc} \begin{array}{cccc} 2,3 & 3,3 & 3,3 & 1,3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array} & \begin{array}{cccc} 3,3 & 3,3 & 3,3 & 1,3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array} & I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,4}, \quad 1 \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,5}, \quad 1 \mathbf{e}_7^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,4}$$

$$321e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^{1,5,4} \otimes e_7, \quad I^{\oplus 2}e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^{1,5,5} \otimes e_7, \quad 1e_7^{5,5} \otimes e_8^{5,5} \otimes e_8^{1,5,5} \otimes e_7$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^1 \mathfrak{e}_6, \quad 31 \mathfrak{e}_6 \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^1 \mathfrak{e}_7'.$$

$$321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2}, \quad 15131\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4},$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$I^{\oplus 2} \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$321\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$3151321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2}} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,4}$$

$$1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}$$

$$1\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_6^{3,3}$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

2.5.7 SgLIgLgLg(28)

$$\begin{array}{l}
\begin{array}{l}
3,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\
\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \\
313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \\
1 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\
1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so},
\end{array}
\end{array}$$

2.5.8 SgLIgLgLgS(93)

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'1, \quad 1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 3}, \quad I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 3}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2}, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 I^{\oplus 2}, \quad 31\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2}, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1$$

2.5.9 SgLIgLIgIg(3)

$$\begin{array}{cccc} 2,4 & 5,5 & 1 & 5,5 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so}, \end{array}$$

$$I^{\oplus 4} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$1e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{4,2} so$$

2.5.10 SgLIgLIgLgS(1)

$$150 \overset{2,4}{\otimes} \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_8} \overset{4,2}{\otimes} 501$$

2.5.11 SSgLgLgLg(76)

$${}^{3,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^2 {}^{5,5} \otimes \mathfrak{e}_6,$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^2 {}^{5,5} \otimes \mathfrak{e}_6,$$

$$\begin{array}{c} 5,5 \\ \otimes \end{array} \mathbf{e}_8 \begin{array}{c} (5,5) \\ \otimes \end{array} \mathbf{e}_8)^2 \begin{array}{c} 5,5 \\ \otimes \end{array} \mathbf{e}_6$$

$$313221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^2\otimes^{5,5}\mathbf{e}_6,$$

$$I^{\oplus 5} \mathbf{e}_8 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^2 \bigotimes^{5,5} \mathbf{e}_6,$$

$$21513221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^2\otimes^{5,5}\mathbf{e}_6$$

$$1513221\mathbf{e}_8^1(5,5\otimes\mathbf{e}_8)^2\otimes\mathbf{e}_6,$$

$$I^{\oplus 2} \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^2 \otimes^{5,5} \mathfrak{e}_6,$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_6 \end{smallmatrix}$$

$$I^{\oplus 3} I^{\oplus 3} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_6 \end{smallmatrix}$$

$$\begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \right)^2 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{matrix} 4,5 & 1 & 5,5 & 5,3 \\ \otimes & \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^2 & \otimes \mathfrak{so}, \end{matrix}$$

$${}^{5,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^2 {}^{5,3} \otimes \mathfrak{so}$$

$$313221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^2\otimes^{5,3}\mathfrak{so},$$

$$I^{\oplus 5} \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes^{5,3} \mathfrak{so},$$

$$21513221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^2 \otimes^{5,3} \mathbf{so}$$

$${}^1_{15}13221\mathfrak{e}_8({}^{5,5}\otimes\mathfrak{e}_8)^2\otimes{}^{5,3}\mathfrak{so},$$

$$I^{\oplus 2} \otimes_{\mathfrak{e}_8} \left(\otimes_{\mathfrak{e}_8} \right)^2 \otimes_{\mathfrak{so}_8} \mathfrak{so}_8.$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$I^{\oplus 3} I^{\oplus 3} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \oplus 50$$

$$\begin{array}{cccc} 2,4 & 1,5,5 & 5,4 & 3,3 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array}$$

$$I^{\oplus 4} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,4 \\ \mathfrak{e}_7 \end{smallmatrix} \otimes \begin{smallmatrix} 3,3 \\ \mathfrak{e}_6 \end{smallmatrix},$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 2} \begin{smallmatrix} 1 \\ \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_6,$$

$$1 \mathbf{e}_7^{1(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_6^{5,5}$$

$$\begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_7 \end{smallmatrix} \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_7 \end{smallmatrix} \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^2 \otimes^{3,3} \mathfrak{so},$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes_{\mathbf{e}_7}^{3,3} \right)^2 \bigotimes_{\mathbf{e}_7}^{3,3} \mathbf{so}$$

$$I^{\oplus 2} I^{\oplus 2} \begin{smallmatrix} 3,3 \\ \otimes \\ \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \\ \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \\ \mathfrak{so} \end{smallmatrix}$$

$$3221 \overset{1}{e}_8 \otimes \overset{5,4}{e}_7 \otimes \overset{3,3}{e}_7 \otimes \overset{3,3}{e}_6,$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{4,5}{\otimes} \overset{5,4}{e_8} \otimes e_6,$$

$$1e_7 \otimes e_8 \otimes e_7 \otimes e_6$$

$$2321 \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{\prime 3,3} \otimes \mathbf{e}_7,$$

$$I^{\oplus 2} \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7',$$

$$1e_7' \begin{pmatrix} 1 & 3,3 \\ \otimes & e_7 \end{pmatrix}^2 \begin{pmatrix} 4,5 \\ \otimes & e_7 \end{pmatrix}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, \quad I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6, \quad 1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_6^{4,3}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4}$$

$$321 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4}$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,3}$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_6^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \bigcirc \mathfrak{e}_6^{3,3}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'},$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} (\otimes \mathfrak{so})^{1,1},$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{4,3}$$

$$513221 I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6$$

$$31513221 \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_6^{1,3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_8^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

[illegible]

$$31\epsilon_6^{31}(\otimes \epsilon_6)^2 \otimes^{2,2} \epsilon_6 I^{\oplus 3}$$

$$31\epsilon_6^{31}(\otimes \epsilon_6)^2 \otimes^{2,2} \epsilon_6 13151^1$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 122315123 \\
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 12231513151 \\
& \otimes \epsilon_8^{5,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,4} I^{\oplus 2} \\
& 21513221\epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,4} I^{\oplus 2} \\
& I^{\oplus 4} \epsilon_8^{I^{\oplus 2}} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,4} I^{\oplus 2} \\
& \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,5} 1 \\
& I^{\oplus 5} \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,5} 1 \\
& I^{\oplus 2} \epsilon_8^{\otimes 2,4} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\overset{1,3,3}{\mathbf{e}_7} \otimes \overset{4,5}{\mathbf{e}_7} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \\
& 1\overset{1,3,3}{\mathbf{e}_7} \otimes \overset{4,5}{\mathbf{e}_7} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \mathbf{e}_8 12231512 \\
& 2,4\overset{1}{\mathbf{e}_8} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \overset{5,4}{\mathbf{e}_8} \otimes \overset{3,3}{\mathbf{e}_7} \otimes \mathbf{e}_7 I^{\oplus 2} \\
& I^{\oplus 2} \overset{1}{\mathbf{e}_8} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \overset{5,4}{\mathbf{e}_7} \otimes \overset{3,3}{\mathbf{e}_7} 1232 \\
& I^{\oplus 3} \overset{1}{\mathbf{e}_8} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \overset{5,4}{\mathbf{e}_7} \otimes \overset{3,3}{\mathbf{e}_7} I^{\oplus 2} \\
& I^{\oplus 2} \overset{1}{\mathbf{e}_7} \otimes \overset{4,5}{\mathbf{e}_8} \otimes \overset{5,5}{\mathbf{e}_8} \otimes \overset{5,4}{\mathbf{e}_7} I^{\oplus 3} \\
& 1\overset{1}{\mathbf{e}_7} (\otimes \mathbf{e}_8)^2 \overset{5,4}{\mathbf{e}_7} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
I^{\oplus 3} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,2} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1223} \\
2321 & \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_8^{4,5} I^{\oplus 2} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{123151} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \\
1\overline{e}7 & \mathbf{e}_8^{1,4,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,4} \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 12313 \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 1232151 \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6^{2,2} \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2} \\
& I^{\oplus 2} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2}
\end{aligned}$$

$$\begin{array}{l}
1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 1232 \\
I^{\oplus 3} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \\
I^{\oplus 2} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 13151 \\
I^{\oplus 2} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}
\end{array}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 5} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 2} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2, 5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7^{3, 3} \otimes \mathbf{e}'_7 1 \\ 1 \mathbf{e}'_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}'_7 1 \end{array}$$

$$\begin{array}{cccc}
5,5 & 1 & 5,5 & 4,2 & 1,1 \\
\otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1 \\
21513221 & \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1 \\
I^{\oplus 4} & \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1
\end{array}$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} \overset{5,5}{\otimes} \overset{5,4}{\mathbf{e}_8} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 3} \otimes \mathbf{e}_6^{2, 2}, \\ & 1 \mathbf{e}_6^{1, 3, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_7 I^{\oplus 2}, \\ & 1 \mathbf{e}_7^{1, (5, 5)^2} \otimes \mathbf{e}_6 1^{5, 4} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6 I^{\oplus 2},$$

$$1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_6 I^{\oplus 2},$$

$$1^1 e_7 (\otimes e_8)^2 \otimes e_6 \otimes e_8$$

$$\begin{aligned} & 1e_7^1 (\otimes e_7)^{3,3} \otimes e_8^{4,5} \otimes e_8^{4,2}, \\ & 1e_7^1 (\otimes e_7)^{3,3} \otimes e_8^{4,5} I^{\oplus 4}, \\ & I^{\oplus 2} e_8^{5,4} \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_7^1 I^{\oplus 2}, \end{aligned}$$

$$\begin{array}{l} 3221 \overset{1}{e_8} \otimes \overset{5,4}{e_7} \otimes \overset{3,3}{e_7} \otimes \overset{3,3}{e'_7 1}, \\ I^{\oplus 2} \overset{1}{e_8} \otimes \overset{5,4}{e_7} \otimes \overset{3,3}{e_7} \otimes \overset{3,3}{e'_7 123}, \\ 1 \overset{1}{e_7} \otimes \overset{3,3}{e_7} \otimes \overset{4,5}{e_8} \otimes \overset{5,4}{e'_7 1}, \end{array}$$

$$\begin{array}{ccccccc}
1 & 5,4 & 3,3 & 3,3 & & & \\
1e_8 \otimes e_7 \otimes e_7 \otimes e'_7 & 1232 & & & & & \\
I^{\oplus 3}e_8 \otimes e_7 \otimes e_7 \otimes e'_7 & I^{\oplus 2} & & & & & \\
1e_7 \otimes e_8 \otimes e_7 \otimes e'_7 & 1 & & & & &
\end{array}$$

$$\begin{array}{cccc} 2,4 & 1 & 5,5 & 5,3 & 2,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & 1, \\ I^{\oplus 3} \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & 13, \\ 1 & 3,5 & 5,5 & 5,3 & \\ 1 \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 & I^{\oplus 2}, \end{array}$$

$$\begin{array}{l} I^{\oplus 4} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6 1, \\ I^{\oplus 2} I^{\oplus 2} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6 13, \\ 1 \mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_6 1 \end{array}$$

$$\begin{array}{cccc} 2,4 & 1 & 4,2 & 1,1 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so}1, \\ I^{\oplus 3} & \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so}1, \\ I^{\oplus 2} & \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so}13 \end{array}$$

$$\begin{aligned} I^{\oplus 4} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1, \\ I^{\oplus 2} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{132}, \end{aligned}$$

$$1e_8 \otimes^{1,4,2} so \otimes^{1,1} so \otimes^{1,1} so 1322$$

$$I^{\oplus 3} e_8 \otimes^{1,4,2} so \otimes^{1,1} so \otimes^{1,1} so 13$$

$$1e_7' \otimes^{1,3,3} e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7 I^{\oplus 2},$$

$$I^{\oplus 2} e_7' \otimes^{1,3,3} (e_7)^2 \otimes^{3,3} e_7' \otimes^{3,2},$$

$$1e_7^{1,3,3} \otimes e_7'^{3,3} \otimes e_7^{3,3} \otimes e_7'^{\oplus 2},$$

$$I^{\oplus 2} e_7'^1 (\otimes e_7)^2 \otimes e_7^{3,3} \otimes e_7'^{3,3},$$

$$I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7'^{3,3} \otimes \mathbf{e}_7'^1$$

$$I^{\oplus 2} \mathbf{e}_7'^1 (\otimes \mathbf{e}_7^{3,3})^2 \otimes \mathbf{e}_7'^3 I^{\oplus 3}$$

$$1e_7^1 \otimes^{3,3} e_7' \otimes^{3,3} e_7 \otimes^{3,3} e_6 1, \\ 1e_6^1 (\otimes^{3,3} e_7)^2 \otimes^{3,3} e_7' I^{\oplus 3},$$

$$1\epsilon_6^1(\otimes \epsilon_7)^2 \otimes \epsilon_7' \otimes, \\ I^{\oplus 2} \epsilon_7'(\otimes \epsilon_7)^2 \otimes \epsilon_6 \otimes,$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^2{}^{3,3} \otimes \mathbf{e}_7'{}^{3,3} \otimes I^{\oplus 2} \mathbf{e}_7'^1(\otimes \mathbf{e}_7)^2{}^{3,3} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$\begin{aligned} & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^2 \otimes^{1,1} \mathfrak{so} \otimes^{2,2}, \\ & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^2 \otimes^{1,1} \mathfrak{so} 1^2 3^2, \end{aligned}$$

$$\begin{aligned} & 1_{\mathfrak{so}} \binom{1,1}{\otimes \mathfrak{so}}^2 \binom{1,1}{\otimes \mathfrak{so}} \binom{2,3}{\otimes}, \\ & 1_{\mathfrak{so}} \binom{1,1}{\otimes \mathfrak{so}}^2 \binom{1,1}{\otimes \mathfrak{so}} I^{\oplus 2} \end{aligned}$$

$$150 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ 50 \end{smallmatrix}^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix}$$

$$\begin{array}{ccccccc} & 1 & 4,5 & & 5,5 & & 5,3 & & 2,2 \\ 1e_7' & \otimes & e_8 & \otimes & e_8 & \otimes & e_6 & \otimes & , \\ & 1 & 3,5 & & 5,5 & & 5,5 & & \\ 1e_6 & \otimes & e_8 & \otimes & e_8 & \otimes & e_7' & 1, & \end{array}$$

$$\begin{array}{cccc} 1 & 3,5 & 5,5 & 5,4 \\ 1e_6 & \otimes e_8 & \otimes e_8 & \otimes e_7' I^{\oplus 2}, \\ \\ 1 & 4,5 & 5,5 & 5,4 \\ 1e_7' & \otimes e_8 & \otimes e_8 & \otimes e_6 1 \end{array}$$

$$1e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,3} e_6 I^{\oplus 2}$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_6} \overset{2,2}{\otimes} \overset{2,2}{e_6} \overset{2,2}{\otimes} \overset{2,2}{e_6},$$

$$1\mathbf{e}_6 \binom{2,2}{\otimes} \mathbf{e}_6^2 \otimes \mathbf{e}_7 I^{\oplus 2},$$

$$1\mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_6} \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}_6} \overset{2,2}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{3,3}{\otimes}\mathfrak{e}_71,$$

$$1\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{3,4}{\otimes}\mathfrak{e}_71$$

$$\overset{2,3}{\otimes}\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}{}^2\overset{3,2}{\otimes}\mathfrak{so}1,$$

$$\overset{3,3}{\otimes}\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}{}^2\overset{3,2}{\otimes}\mathfrak{so}1,$$

$$I^{\oplus 3}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 2}I^{\oplus 2}\overset{3,3}{\mathfrak{e}_7}{}^2\overset{3,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 3}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}_7}\overset{3,3}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_61,$$

$$I^{\oplus 2}I^{\oplus 2}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_61,$$

$$I^{\oplus 2}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}_7}\overset{3,3}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_613$$

$$1\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_61$$

$$I^{\oplus 2}\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}\overset{3,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_61,$$

$$1\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_613,$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6I^{\oplus 2}$$

$$\overset{2,2}{\otimes}\mathfrak{e}_6\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}_6}{}^2\overset{2,2}{\otimes}\mathfrak{so}1,$$

$$I^{\oplus 2}\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{2,2}{\otimes}\mathfrak{so}1,$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{2,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 2}\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}13,$$

$$1\mathfrak{e}_7(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{4,2}{\otimes}\mathfrak{so}13$$

$$I^{\oplus 2}\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}1,$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}_8}\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7'I^{\oplus 2},$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_7'1$$

$$I^{\oplus 2}\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}\overset{4,5}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 2}\mathfrak{e}_7'\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}{}^2\overset{3,2}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}13$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_61$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7' \otimes \epsilon_7'1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_6 \otimes \epsilon_61$$

$$1\epsilon_7 \otimes \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}13$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_7)^2 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_6)^2 \otimes \epsilon_7'1$$

2.5.13 SSgLgIgS(149)

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 I^{\oplus 5},$$

$$21513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$1513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

[illegible]

$$\begin{array}{lll}
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 122313, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 5}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 12231512 \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 1223151, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 24,2}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 2} \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 3} & & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 4}
\end{array}$$

$$\begin{array}{lll}
1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2} I^{\oplus 4}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2} I^{\oplus 2} \\
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{1,4,2} I^{\oplus 2}, & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7^{1,4,2}, & 1_{\mathbf{e}_7}^{1,5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7^{5,5} \otimes \mathbf{e}_7^{1,4,2}
\end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1223, & 2321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1, & 321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2} \\ I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 3} 1, & I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2}, & 1\epsilon_7^{(3,3)} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,4} 1\epsilon_7 \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} (2,2) \\ (\otimes \mathfrak{e}_6) \end{array} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} 2,2 \\ \otimes \mathfrak{e}_6 \end{array}, & I^{\oplus 2} \begin{array}{l} 1 \\ \mathfrak{e}_6 \end{array} \begin{array}{l} (2,2) \\ (\otimes \mathfrak{e}_6) \end{array} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} 2,2 \\ \otimes \mathfrak{e}_6 \end{array}, & I^{\oplus 2} \begin{array}{l} 1 \\ \mathfrak{e}_6 \end{array} \begin{array}{l} (2,2) \\ (\otimes \mathfrak{e}_6) \end{array} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} 2,2 \\ \otimes \mathfrak{e}_6 \end{array} 1 \\ \begin{array}{l} 31 \\ \mathfrak{e}_6 \end{array} \begin{array}{l} (2,2) \\ (\otimes \mathfrak{e}_6) \end{array} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} 2,2 \\ \otimes \mathfrak{e}_6 \end{array}, & \begin{array}{l} 31 \\ \mathfrak{e}_6 \end{array} \begin{array}{l} (2,2) \\ (\otimes \mathfrak{e}_6) \end{array} \begin{array}{l} 2,2 \quad 1 \\ \otimes \mathfrak{e}_6 \end{array} \begin{array}{l} 2,2 \\ \otimes \mathfrak{e}_6 \end{array} 1 \end{array}$$

$$\begin{array}{lll}
1 \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7 \overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7 \overset{I^{\oplus 2}}{\otimes} 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,2}{\otimes}, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \\
I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 3}}{\mathbf{e}}_7 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 2}}{\mathbf{e}}_7 I^{\oplus 2} &
\end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1,3,2}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1,3,3}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1^{\oplus 3}}{1} \\ 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1^{\oplus 2}}{I^{\oplus 2}} & & \end{array}$$

$$1\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8^{3,5} I^{\oplus 3}, \quad 1\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8^{3,5} I^{\oplus 2}, \quad 31\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8^{3,5} I^{\oplus 2}$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 2} \quad 1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 1}$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{3,3}{\mathbf{e}}_7 \otimes \overset{3,3}{\mathbf{e}}_7' \otimes \overset{3,3}{\mathbf{e}}_7 \otimes \overset{1}{\mathbf{e}}_7 1, \quad I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \otimes \overset{3,3}{\mathbf{e}}_7 I^{\oplus 2}$$

$$1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 3}, \quad 1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 2}$$

$$1 \overset{1}{e}_6 \overset{3,5}{\otimes} \overset{5,5}{e}_8 \overset{5,4}{\otimes} I^{\oplus 2} \overset{5,5}{e}_8 \overset{5,5}{\otimes} \overset{1}{e}_7 1,$$

$$1\mathfrak{so}(\overset{1}{\otimes}\overset{1,1}{\mathfrak{so}})^2\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_61$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}}_7)^2\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}}_7)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_61$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}}_6)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_71$$

2.5.14 SSgLgLIg(21)

$$1513221\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 2}3221\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\otimes}\mathfrak{e}_7,$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\mathfrak{e}_7$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\otimes}\mathfrak{e}'_7,$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_7,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\otimes}\mathfrak{e}'_7$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,3}{\otimes}\mathfrak{e}_6,$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_6,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,3}{\otimes}\mathfrak{e}_6$$

$$I^{\oplus 2}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$321\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\mathfrak{e}'_7$$

$$31\overset{1}{\mathfrak{e}}_6\overset{2,2}{\otimes}\overset{2,2}{\mathfrak{e}}_6\overset{2,2}{\otimes}\overset{1}{\mathfrak{e}}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$2321 \overset{1}{e_7} \otimes \overset{3,3}{e_7} \otimes \overset{3,3}{e_7} \otimes \overset{1}{e_7} \otimes \overset{3,3}{e_7}$$

$$2231513221 \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8}$$

$$1 \overset{1}{\mathbf{e}}_7 \otimes \overset{4,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 \otimes \overset{1}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_7$$

$$1\mathbf{e}_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} \overset{5,5}{\otimes} 1\mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7'$$

$$1\mathbf{e}_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} \overset{5,5}{\otimes} 1\mathbf{e}_8 \overset{5,3}{\otimes} \mathbf{e}_6$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{4,5}{\otimes} 1e_8 \overset{4,2}{\otimes} 50$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} e_6$$

2.5.15 SSgLgLIgLgS(43)

$$1 \overset{1}{\mathbf{e}}_8 \overset{5,5}{\otimes} \overset{5,5}{\mathbf{e}}_8 \overset{1}{\otimes} \overset{5,5}{\mathbf{e}}_8 122315132,$$

$$I^{\oplus 4} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,5,1} \otimes \mathbf{e}_8^{5,5,4,2},$$

$$513221 \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8 1223,$$

$$I^{\oplus 3} \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{5,5\ 1\ 5,5} \otimes \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{1223151},$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{e}_8,$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 4}$$

$$231513221 \overset{1}{\mathbf{e}}_8 \overset{5,5}{\otimes} \overset{5,5}{\mathbf{e}}_8 \overset{1}{\otimes} \overset{5,5}{\mathbf{e}}_8 \overset{5,5}{\otimes} \mathbf{e}_8 1,$$

$$\begin{matrix} 2,4 & 1 & 5,5 & & 5,5 & 1 & 5,5 \\ \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 \end{matrix} I^{\oplus 4},$$

$$I^{\oplus 2} \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{5,5} 12231513,$$

$$1513221 \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8 I^{\oplus 3},$$

$$I^{\oplus 2} \overset{3221}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3},$$

$$\begin{array}{cccccc} 2,4 & 1 & 5,5 & & 5,5 & 1 & 5,5 & & 4,2 \\ \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes \end{array}$$

$$3221 \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} 122315$$

$$31513221 \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} I^{\oplus 2}$$

$$I^{\oplus 4} \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8 I^{\oplus 4}$$

$$I^{\oplus 2} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \epsilon_8 1223151$$

$$\begin{matrix} 3,5 & 1 & 5,5 & & 5,5 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so}1, \end{matrix}$$

$$313221 \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1 \ 4,2} \oplus \mathfrak{so}1,$$

$$1513221 \overset{1}{e}_8 \overset{5,5}{\otimes} \overset{5,5}{e}_8 \overset{1}{\otimes} \overset{4,2}{e}_8 \text{ sol},$$

$$I^{\oplus 3} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \oplus \mathfrak{e}_8^{1,4,2} \oplus \mathfrak{so}1$$

$$\begin{matrix} 4,5 & 1 & 5,5 & & 5,5 & 1 & 4,2 \\ \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{so}1, \end{matrix}$$

$$I^{\oplus 5} \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5,1} \otimes \mathfrak{e}_8^{4,2},$$

$$I^{\oplus 2} \otimes^{2,4} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 \otimes^1 \mathfrak{e}_8 \otimes^{4,2} \mathfrak{so}1,$$

$$\begin{array}{cccc} 5,5 & 1 & 5,5 & 5,5 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so}1 \end{array}$$

$$21513221 \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{4,2}{\mathfrak{e}_8} \otimes \mathfrak{so}1$$

$$I^{\oplus 4} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \oplus \mathfrak{e}_8^{1,4,2} \oplus \mathfrak{so}1$$

$$1e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5} e_8 \otimes^{4,2},$$

$$\begin{matrix} 2,4 & 1 & 5,5 & & 5,5 & 1 & 5,4 \\ \otimes & \mathbf{e}_8 & \otimes & \mathbf{e}_8 & \otimes & \mathbf{e}_8 & \otimes & \mathbf{e}_7 1, \end{matrix}$$

$$1e_7^{1,4,5} \otimes e_8^{5,5} \otimes e_8^{1,5,5} \otimes e_8 I^{\oplus 4}$$

$$I^{\oplus 3} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 1$$

$$321e_7^1 \otimes^{3,3} e_7^{3,3} \otimes^{1,3} e_7^{3,3} \otimes e_7^1,$$

$$1e_7 \overset{1}{\otimes} \overset{5,5}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{4,2}{\otimes} so1$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{5,4}{\otimes} e_7 1$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes \text{sol}$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} e_7' 1$$

$$1e_6 \otimes^{1,3,5} e_8 \otimes^{5,5} e_8 \otimes^{1,4,2} \text{sol}$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}}_6 \overset{1}{\otimes} \overset{2,2}{\mathbf{e}}_6 \overset{1}{\otimes} \overset{2,2}{\mathbf{e}}_6 1$$

2.5.16 SSgLgLIgLIgS(15)

[illegible]

$$1\mathbf{e}_7 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 1 & 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} I^{\oplus 2} \\ I^{\oplus 2} \end{smallmatrix}$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^1$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} 1$$

$$I^{\oplus 2} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{13}$$

$$1 \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} \mathbf{e}_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_7 \overset{1}{\otimes} \overset{4,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{4,2}{\otimes} \mathfrak{so}1$$

$$1\mathbf{e}_7^{1\ 3,3} \otimes \mathbf{e}_7^{1\ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7' I^{\oplus 2}$$

$$1 \mathbf{e}_7 \overset{1 \ 3,3}{\otimes} \mathbf{e}_7 \overset{1 \ 3,3}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

2.5.19 SSgLIgLgLIgS(15)

$$I^{\oplus 2} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} I^{\oplus 3} I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_8 \otimes^{5,5} \mathbf{e}_8 \otimes^{5,5} \mathbf{e}_8 \otimes^{5,4} \mathbf{e}_7 1$$

$$1 \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} I_7^{\oplus 2} 1$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7'} 1$$

2.5.20 SSgLIgLIgLg(2)

$$I^{\oplus 2} \oplus I^{\oplus 2} \oplus \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,4,2} \oplus \mathfrak{so}_{50}$$

2.6 5 node(1332)

2.6.1 gLgLGgLG(17)

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7')^2$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7^{4,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,4}$$

$$\mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_6^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_6$$

2.6.2 gLgLIgLg(4)

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_7$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

2.6.3 gLIgLgLg(10)

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} (\overset{1,1}{\otimes} \mathfrak{so})^2$$

$$\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

$$\mathfrak{e}_7' \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

$$\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,5}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,4}{\otimes} \mathfrak{e}_6$$

$$51231513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6, \quad 31513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6, \quad 15131513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & I^{+6}\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 321513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \\ 51231513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & {}^131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 15131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_6 \end{array}, & \begin{array}{l} 3,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_6 \end{array}, & I^{\oplus 3} \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_6 \end{array} \\ I^{\oplus 2} \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_7 \end{array} (\otimes \mathfrak{e}_8)^3 \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_6 \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & \begin{array}{l} 3,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & I^{\oplus 3} \mathfrak{e}_7 \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} \\ I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^3 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2, \end{array} & \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2, \end{array} & 321 \mathbf{e}_7'(\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7' \otimes \mathbf{e}_7 \\ I^{\oplus 3} \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2 & & \end{array}$$

$$3221\mathfrak{e}_7(\otimes\mathfrak{e}_7)^2\otimes\mathfrak{e}_7'\otimes\mathfrak{e}_7, \quad 512321\mathfrak{e}_7(\otimes\mathfrak{e}_7)^2\otimes\mathfrak{e}_7'\otimes\mathfrak{e}_7, \quad 151321\mathfrak{e}_7(\otimes\mathfrak{e}_7)^2\otimes\mathfrak{e}_7'\otimes\mathfrak{e}_7$$

$$I^{\oplus 2}\mathfrak{e}_7'(\otimes\mathfrak{e}_7)^3\otimes\mathfrak{e}_7$$

$$321\mathfrak{e}_6^{(2,2)}(\otimes\mathfrak{e}_6)^3\otimes\mathfrak{e}_7, \quad 15131\mathfrak{e}_6^{(2,2)}(\otimes\mathfrak{e}_6)^3\otimes\mathfrak{e}_7, \quad \otimes\mathfrak{e}_6^{(2,2)}(\otimes\mathfrak{e}_6)^3\otimes\mathfrak{e}_7^{(3,4)}$$

$$\begin{array}{lll}
31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}, & \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}, & I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} \\
1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} & &
\end{array}$$

$$\begin{array}{lll}
1\epsilon_7^{3,3} \otimes \epsilon_7'({}^{3,3}(\otimes \epsilon_7)^2 \otimes \mathfrak{so}), & {}^{2,3} \otimes \epsilon_7'({}^{3,3}(\otimes \epsilon_7)^3 \otimes \mathfrak{so}), & {}^{3,3} \otimes \epsilon_7'({}^{3,3}(\otimes \epsilon_7)^3 \otimes \mathfrak{so}) \\
I^{\oplus 3} \epsilon_7'({}^{3,3}(\otimes \epsilon_7)^3 \otimes \mathfrak{so}) & &
\end{array}$$

$$\begin{array}{ccc} \begin{array}{c} 2,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array}, & \begin{array}{c} 3,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array}, & I^{\oplus 3} \mathbf{e}_7 \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array} \end{array}$$

$$\begin{array}{ccc} \begin{array}{c} 2,3 \\ \otimes \end{array} \epsilon_7 \begin{array}{c} 3,3 \\ (\otimes) \end{array} \epsilon_7^2 \begin{array}{c} 3,3 \\ \otimes \end{array} \epsilon_6 \begin{array}{c} 2,2 \\ \otimes \end{array} \epsilon_6, & \begin{array}{c} 3,3 \\ \otimes \end{array} \epsilon_7 \begin{array}{c} 3,3 \\ (\otimes) \end{array} \epsilon_7^2 \begin{array}{c} 3,3 \\ \otimes \end{array} \epsilon_6 \begin{array}{c} 2,2 \\ \otimes \end{array} \epsilon_6, & I^{\oplus 3} \epsilon_7 \begin{array}{c} 3,3 \\ (\otimes) \end{array} \epsilon_7^2 \begin{array}{c} 3,3 \\ \otimes \end{array} \epsilon_6 \begin{array}{c} 2,2 \\ \otimes \end{array} \epsilon_6 \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7'$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2$$

$$31 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 (\otimes \mathfrak{e}_7')^2$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$5131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$3221\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$321\mathfrak{e}_7\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_7'\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$23151321\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{4,5}{\otimes}\mathfrak{e}_8$$

$$2315131\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^3\overset{3,5}{\otimes}\mathfrak{e}_8$$

$$1\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}\overset{2,3}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2$$

$$1\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1\epsilon'_7 (\otimes \epsilon_7)^2 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$1\epsilon_7 \otimes \epsilon_6 (\otimes \epsilon_6)^2 \otimes \mathfrak{so}$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon'_7$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6 \otimes \mathfrak{so}$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2$$

$$1\epsilon_6 (\otimes \epsilon_7)^3 \otimes \epsilon_7$$

$$1\epsilon_6 (\otimes \epsilon_7)^2 (\otimes \epsilon'_7)^2$$

$$1\epsilon_6 (\otimes \epsilon_7)^2 \otimes \epsilon_6 \otimes \mathfrak{so}$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1513221\epsilon_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^3$$

2.6.6 SgLgLgLgLgS(407)

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 \otimes,$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 \otimes,$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 1231512,$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 \otimes,$$

$$I^{\oplus 4} \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 \otimes,$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 1232,$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 \otimes$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 12313$$

$$\otimes \epsilon_7 (\otimes \epsilon_7)^3 \otimes \epsilon_7 1232151$$

$$31513221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^3\overset{5,5}{\otimes}\mathbf{e}_812231513151, \quad 15131513221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^3\overset{5,5}{\otimes}\mathbf{e}_812231513151$$

$$\begin{array}{l}
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & 4,4 \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & \otimes
\end{array} \\
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & 12313
\end{array} \\
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & 1232151
\end{array} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \begin{array}{c} 4,3 \\ \otimes \end{array} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 I^{\oplus 4} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \begin{array}{c} 1 \\ 1232 \end{array}
\end{array}$$

$$\begin{array}{l}
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \\
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{12231512} \\
2,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
I^{\oplus 3} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
3,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,5} I^{\oplus 2}
\end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,3 \\ (\otimes \mathfrak{so})^3 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,4 \\ (\otimes \mathfrak{so})^3 & \otimes \mathfrak{so} & \otimes \end{array} \end{array}$$

$$\begin{array}{l} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ 1 \mathbf{e}_6 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \otimes \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ 1 \mathbf{e}_6 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7'{}^{4,5} \otimes \epsilon_8^{(5,5)}{}^2 \otimes \epsilon_8 122315123 \\ & I^{\oplus 2} \epsilon_7'{}^{4,5} \otimes \epsilon_8^{(5,5)}{}^2 \otimes \epsilon_8 12231513151 \\ & 1\epsilon_7'{}^{(5,5)} \otimes \epsilon_8^{(5,5)}{}^3 \otimes \epsilon_8 122315123 \\ & 1\epsilon_7'{}^{(5,5)} \otimes \epsilon_8^{(5,5)}{}^3 \otimes \epsilon_8 12231513151 \end{aligned}$$

$$\begin{array}{l} 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{4,4} \\ 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{12313} \\ 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1232151} \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,4}{\otimes} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 I^{\oplus 3} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7{}^{4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$1 \mathbf{e}'_7({}^{5,5} \otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$\begin{array}{c} 3,3 \\ \otimes \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 I^{\oplus 3} \\ 512321 \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 I^{\oplus 2} \end{array}$$

$$\begin{array}{c} 2,3 \\ \otimes e'_7 \end{array} \left(\begin{array}{c} 3,3 \\ \otimes e_7 \end{array} \right)^3 \begin{array}{c} 3,3 \\ \otimes e'_7 \end{array} \begin{array}{c} 3,3 \\ \otimes \end{array}$$

$$I^{\oplus 3} \mathbf{e}'_7 (\overset{(3,3)}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 122315123$$

$$31_{\mathfrak{so}} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13$$

$$2231so(\overset{1,1}{\otimes}so)^3\overset{2,4}{\otimes}e_81$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}$$

$$315131\mathbf{e}_6\left(\overset{2,2}{\otimes}\mathbf{e}_6\right)^3\overset{3,5}{\otimes}\mathbf{e}_81$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{2,2}{\otimes} \mathbf{e}_6 \right)^3 \overset{3,5}{\otimes} \mathbf{e}_8 I^{\oplus 3}$$

$$\begin{matrix} 2,2 & 3,5 & (5,5) & 5,5 \\ \otimes & \otimes & \otimes & \otimes \\ \mathbf{e}_6 & \mathbf{e}_8 & \mathbf{e}_8 & \mathbf{e}'_7 1 \end{matrix}$$

$$1e_6 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_7' 1$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,3}{\otimes} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$3150 \otimes^{2,4} e_8(\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 122315123$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 12231513151$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3} 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 1$$

2.6.7 SgLgLgLIg(27)

$$\otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$313221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$I^{\oplus 5} \mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$21513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$1513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$\otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$\otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$321\mathfrak{e}_7' (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 3} \mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$3221\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$512321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$151321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$1e_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,4}{\otimes} e_7$$

$$31e_6(\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{\otimes} e_7$$

$$1\mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\otimes} \mathbf{e}_7'$$

$$I^{\oplus 4} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \oplus \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \oplus \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}$$

2.6.9 SgLgLIgLgS(1)

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 501$$

2.6.10 SgLIgLgLgLg(12)

$$\begin{array}{lll} \begin{array}{cccccc} 2,4 & 5,5 & 1 & 5,5 & 4,2 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{array} & \begin{array}{cccccc} I^{\oplus 4} & 5,5 & 1 & 5,5 & 4,2 & 1,1 \\ \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{array} & \begin{array}{cccc} 2,4 & 1 & 5,5 & 5,3 \\ 1\mathfrak{so} & \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^2 & \otimes \mathfrak{so} \end{array} \end{array}$$

$$3221\mathbf{e}_8 \otimes \mathbf{e}_8^{5,5} (\otimes \mathbf{e}_8^{5,5})^2 \otimes \mathbf{e}_7^{5,4}, \quad I^{\oplus 3} \mathbf{e}_8 \otimes \mathbf{e}_8^{5,5} (\otimes \mathbf{e}_8^{5,5})^2 \otimes \mathbf{e}_7^{5,5}$$

$$3221\epsilon_8^{5,5} \otimes \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_6^{5,3}, \quad I^{\oplus 3} \epsilon_8^{5,5} \otimes \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_6^{5,4}$$

$$513221e_8 \otimes^{5,5} e_8 (\otimes^{5,5} e_8)^2 \otimes^{4,2} 50$$

$$3221\mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,5}{\otimes} \mathbf{e}_7$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \otimes^{1(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes^{5,5} \mathfrak{e}_6$$

$$1e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} 50 \overset{1,1}{\otimes} 50$$

$$1e_7 \otimes^{3,3} e_7 (\otimes^{3,3} e_7)^2 \otimes^{3,2} so$$

2.6.11 SgLIgLgLgLgS(43)

$$\begin{array}{lll}
3221\epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{4,2}}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,3}}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,4}} \\
I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,5}}, & 3221\epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes I^{\oplus 4}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 122313 \\
I^{\oplus 2} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 1223151315, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes I^{\oplus 5}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 12231512 \\
I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 12231511 & &
\end{array}$$

$$\begin{array}{lll} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{4,2} \mathfrak{so}1, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} 1223132, & I^{\oplus 4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{4,2} \mathfrak{so}1 \\ 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 I^{\oplus 6}, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 122315123, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 12231513215 \\ 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8^{12231513}, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8^{12231513151} \end{array}$$

$$\begin{aligned}
& 3221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1, & 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1232, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 123 \\
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}, & I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 1, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \\
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 13
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1
\end{aligned}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 1$$

2.6.12 SgLIgLgLIg(1)

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

2.6.13 SgLIgLgLIgS(1)

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1$$

$$31\mathfrak{e}_6(\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}_6})^3\overset{3,3}{\otimes}\mathfrak{e}_7,$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}_6})^3\overset{3,4}{\otimes}\mathfrak{e}_7$$

$$1513221\mathfrak{e}_8(\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}_8})^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 2}{}^{3221}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 2}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}_6}(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{2,2}{\otimes}\mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{4,4}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 2}\mathfrak{e}_7'(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{3,3}{\otimes}\mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2(\overset{3,3}{\otimes}\mathfrak{e}_7')^2$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$3221\mathfrak{e}_8\overset{1}{\otimes}\overset{4,2}{\mathfrak{so}}(\overset{1,1}{\otimes}\mathfrak{so})^3$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{4,2}{\otimes}\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}\overset{4,5}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}$$

$$\begin{aligned} I^{\oplus 4} \mathbf{e}_8^{\otimes 2} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{122315123} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513151} \end{aligned}$$

$$\begin{aligned}
& \otimes_{\mathfrak{e}_8}^{5,5} {}^1_1 (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2} \\
& 21513221 \otimes_{\mathfrak{e}_8}^{1,5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2} \\
& \otimes_{\mathfrak{e}_8}^{4,5} {}^1_1 (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\
& I^{\oplus 5} \otimes_{\mathfrak{e}_8}^{1,5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\
& \otimes_{\mathfrak{e}_8}^{3,5} {}^1_1 (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3} \\
& I^{\oplus 2} \otimes_{\mathfrak{e}_7}^{1,4,5} \otimes_{\mathfrak{e}_8}^{(5,5)} (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 1223132 \\
& I^{\oplus 2} \otimes_{\mathfrak{e}_7}^{1,4,5} \otimes_{\mathfrak{e}_8}^{(5,5)} (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 122315123 \\
& I^{\oplus 2} \otimes_{\mathfrak{e}_7}^{1,4,5} \otimes_{\mathfrak{e}_8}^{(5,5)} (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 12231513151 \\
& {}^1_{1513221} \otimes_{\mathfrak{e}_8}^{1,5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3} \\
& I^{\oplus 3} \otimes_{\mathfrak{e}_8}^{I^{\oplus 3}, 5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2} \\
& I^{\oplus 4} \otimes_{\mathfrak{e}_8}^{I^{\oplus 2}, 5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\
& I^{\oplus 3} \otimes_{\mathfrak{e}_8}^{I^{\oplus 3}, 5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3} \\
& {}^1_{1\mathfrak{e}_7} \otimes_{(\mathfrak{e}_8)}^{5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 122315123 \\
& {}^1_{1\mathfrak{e}_7} \otimes_{(\mathfrak{e}_8)}^{5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 12231513151 \\
& \otimes_{\mathfrak{e}_8}^{5,5} {}^1_1 (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,5} I^{\oplus 2} \\
& 21513221 \otimes_{\mathfrak{e}_8}^{1,5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,5} I^{\oplus 2} \\
& I^{\oplus 4} \otimes_{\mathfrak{e}_8}^{I^{\oplus 2}, 5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^3 \otimes_{\mathfrak{e}_7}^{5,5} I^{\oplus 2}
\end{aligned}$$

$$\begin{array}{l}
2,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,4 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 12313 \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1232151 \\
I^{\oplus 3} \quad 1 \quad 3,3 \quad 3,3 \quad 4,2 \\
\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,5 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1231512
\end{array}$$

$$\begin{array}{c} 31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 3} \\ 31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_6^{2,2} 13151^1 \end{array}$$

$$\begin{aligned}
& 1 \epsilon_7' \otimes \epsilon_8 (\otimes \epsilon_8)^2 \otimes \epsilon_8 122315123 \\
& 1 \epsilon_7' \otimes \epsilon_8 (\otimes \epsilon_8)^2 \otimes \epsilon_8 12231513151 \\
& \otimes \epsilon_8 (\otimes \epsilon_8)^3 \epsilon_7' I^{\oplus 2} \\
& 21513221 \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' I^{\oplus 2} \\
& I^{\oplus 4} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' I^{\oplus 2} \\
& \otimes \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1 \\
& I^{\oplus 5} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1 \\
& I^{\oplus 2} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1
\end{aligned}$$

$$\begin{aligned}
& \otimes_{\mathbf{e}_7}^{2,3} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} \\
& \otimes_{\mathbf{e}_7}^{2,3} \otimes_{\mathbf{e}_7}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} I^{\oplus 3} \\
& I^{\oplus 3} \otimes_{\mathbf{e}_7}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{4,4} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7} 12313 \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7} 1232151 \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^1 (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7'}^{3,2}
\end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \\ & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 12313 \\ & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 1232151 \\ & \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 \otimes^{2,2} \\ & \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2} \\ & I^{\oplus 2} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2} (\otimes \mathbf{e}_8)^{2 \oplus 5, 5} \otimes \mathbf{e}_7^{5, 4} \otimes \mathbf{e}_7^{3, 3} 1 \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4, 5} (\otimes \mathbf{e}_8)^{5, 5} \otimes \mathbf{e}_7^{5, 4} I^{\oplus 3} \\ & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5, 5} \otimes \mathbf{e}_7^{5, 4} I^{\oplus 3} \end{aligned}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ 21513221 \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array} \\ 21513221 \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \\ I^{\oplus 4} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array}$$

$$\begin{aligned} & 2321^1_{\mathbf{e}_7} \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_8 1 \\ & I^{\oplus 2} \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,4} \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,2} \\ & I^{\oplus 2} \mathbf{e}_7^1 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_8^{4,5} I^{\oplus 3} \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3} \\ & 1_{\mathbf{e}_7}^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \\ & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,4} \end{aligned}$$

$$1_{\mathbf{e}'_7} \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^2 \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_7 \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^2 \otimes \mathbf{e}'_7 1$$

$$\begin{aligned} & 1\epsilon_8^{1,5,3} \otimes \epsilon_6^{(2,2)} (\otimes \epsilon_6)^{2,2} \otimes \epsilon_6^{2,2} 13151 \\ & 31\epsilon_6^{1, (2,2)} (\otimes \epsilon_6)^3 \otimes \epsilon_8^{3,5} I^{\oplus 2} \end{aligned}$$

$$150 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ 50 \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix}$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,3} e_6 I^{\oplus 2}$$

$$1e_6^{(3,3)} \otimes e_7^{(3,3)} \otimes e_7' I^{\oplus 3}$$

$$I^{\oplus 3} \mathbf{e}_7^{1(3,3)} (\otimes \mathbf{e}_7)^3 \otimes \mathbf{so}^{3,2} \mathbf{l}$$

$$I^{\oplus 2} \mathfrak{e}_8^{1,2} \otimes \mathfrak{so}(\mathfrak{so}^{1,1})^2 \otimes \mathfrak{so}^{1,1}$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^3 \bigotimes \mathbf{e}'_7 \bigotimes$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^3\overset{2,2}{\otimes}\mathfrak{so}1$$

$$I^{\oplus 2} \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7'^{3,3} I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1e_8^{1\ 5,4} \otimes e_7^{(3,3)} (\otimes e_7)^2 \otimes e_6^{3,3} 13$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1\mathfrak{e}_7^{1,5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13,$$

$$1\mathfrak{e}_7^{1,5,5}(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,5}1$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_71$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_6^{1,2,2}(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}1$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 3} \mathbf{1} \\ I^{\oplus 3} \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{array}{l} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \epsilon_8^{5,5} 1\epsilon_8^{5,5} \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \epsilon_8^{5,5} 12231512 \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8^{\oplus 4} \end{array}$$

$$\begin{array}{l} 1_{\mathfrak{e}_6}^{1, 3, 5} \otimes \mathfrak{e}_8^{(5, 5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5, 5} 1_{\mathfrak{e}_8}^{5, 5} \\ 1_{\mathfrak{e}_6}^{1, 3, 5} \otimes \mathfrak{e}_8^{(5, 5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5, 5} 1_{\mathfrak{e}_8} 12231512 \\ 1_{\mathfrak{e}_6}^{1, 3, 5} \otimes \mathfrak{e}_8^{(5, 5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5, 5} I^{\oplus 2} \otimes \mathfrak{e}_8 I^{\oplus 4} \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_6 \binom{2,2}{\otimes \mathfrak{e}_6}^3 \binom{2,2}{\otimes \mathfrak{e}_6} I^{\oplus 2} \mathfrak{e}_6 1$$

$$I^{\oplus 2} \mathbf{e}_7' \begin{pmatrix} 3,3 \\ \otimes \mathbf{e}_7 \end{pmatrix}^3 \begin{pmatrix} 3,3 \\ \otimes \mathbf{e}_7 \end{pmatrix} I^{\oplus 3} \mathbf{1}$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_7^{1(5,5)} (\otimes \mathbf{e}_8)^3 \otimes^{5,5} \mathbf{e}_7 1$$

$$1 \overset{1}{e}'_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} \overset{1}{e}_7 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,5} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 2},$$

$$31\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_8^{3,5} \mathfrak{e}_8^1$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^3 \otimes \mathfrak{so}^{1,1} \mathfrak{so}^1$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7'^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 2}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

2.6.17 SSgLgLgLIgLg(9)

$$\otimes \mathfrak{e}_8^{2,4} \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4},$$

$$I^{\oplus 4} \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{3,3}$$

$$321\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3}$$

$$231513221\mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{5,5}{e_8} \overset{1}{\otimes} \overset{5,4}{e_7}$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{2}{\otimes} \overset{3,3}{e_7} \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3}{\otimes} e_7'$$

$$1\mathbf{e}_6^{1(2,2)}(\otimes \mathbf{e}_6)^{2(2,2)} \otimes \mathbf{e}_6^{1(2,2)} \otimes \mathbf{e}_6^{2,2}$$

2.6.18 SSgLgLgLIgLgS(25)

[illegible]

$$I^{\oplus 2} \mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus 1 \mathbf{e}_7^{1\ 5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus$$

$$I^{\oplus 2} \mathbf{e}_7^1 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^2 \overset{3,3}{\otimes} \mathbf{e}_7^1 \overset{3,3}{\otimes} \mathbf{e}_7^1$$

$$1_{\mathfrak{e}_7}^{1, 4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes 1_{\mathfrak{e}_8}^{1, 5,5} \otimes \mathfrak{e}_8^{5,5} I^{\oplus 3}$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_8 \otimes so1$$

$$1e_7' \begin{smallmatrix} 1 \\ (3,3) \end{smallmatrix} \otimes e_7 \begin{smallmatrix} 2 \\ (3,3) \end{smallmatrix} \otimes e_7 \begin{smallmatrix} 1 \\ (3,3) \end{smallmatrix} \otimes e_7 1$$

$$1e_6^{1,3,5} \otimes e_8^{5,5} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes so1$$

2.6.23 SSgLIgLgLIgS(6)

$$\begin{aligned} 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 1223151, & \quad I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8, & \quad 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 I^{\oplus 2} 1223 \\ I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 I^{\oplus 4}, & \quad I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 I^{\oplus 3} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1$$

2.6.24 SSgLIgLgLIg(1)

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8$$

2.7 6 node(1127)

2.7.1 g6(12)

$$\mathfrak{so} \left(\otimes \mathfrak{so} \right)^3 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8, \quad \mathfrak{so} \left(\otimes \mathfrak{so} \right)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}$$

$$\mathfrak{so} \left(\otimes \mathfrak{e}_7 \right)^4 \otimes \mathfrak{e}_7$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_7$$

$$\mathfrak{so} \left(\otimes \mathfrak{e}_6 \right)^4 \otimes \mathfrak{e}_7$$

$$\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_7' \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

$$\mathfrak{e}_6 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

2.7.2 Sg6(81)

$$\begin{aligned} & \otimes \epsilon_7^{2,4} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & \otimes \epsilon_7^{5,4} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & 2151321 \epsilon_7^{(3,3)} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & 151321 \epsilon_7^{(3,3)} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3} \end{aligned}$$

$$\begin{array}{l} \begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7^4 \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \\ I^{\oplus 4} \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7^4 \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \\ \begin{smallmatrix} 1 \\ 2321 \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7^4 \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \end{array}$$

$$\begin{array}{l} 4,4 \quad 3,3 \quad 3,3 \\ \otimes e_7 (\otimes e_7)^4 \otimes 50 \\ 31321 e_7 (\otimes e_7)^4 \otimes 50 \\ 1512321 e_7 (\otimes e_7)^4 \otimes 50 \end{array}$$

$$\begin{array}{l} 3,5 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, \\ 313221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, \\ 1513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \end{array}$$

$$\begin{array}{c} 4,5 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7, \\ I^{\oplus 5} \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7, \end{array}$$

$$\begin{array}{c} \begin{array}{ccccccc} 5,5 & & 5,5 & & 5,4 & & 3,3 \\ \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & & & \end{array} \\ 21513221 \mathbf{e}_8 \begin{array}{ccccccc} 5,5 & & 5,5 & & 5,4 & & 3,3 \\ \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 & & & \end{array} \end{array}$$

$$2313221\mathbf{e}_8(\otimes\mathbf{e}_8)^4\otimes\mathbf{e}_6,$$

$$51231513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4\otimes\mathbf{e}_6,$$

$$I^{\oplus 6} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$31513221 \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$\begin{array}{l} 321513221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,5}{\otimes} \mathfrak{e}_6 \\ 15131513221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,5}{\otimes} \mathfrak{e}_6 \end{array}$$

$$\begin{aligned} &2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}^{5,3}, \\ &51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}^{5,3}, \end{aligned}$$

$$I^{\oplus 6} \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^4 \otimes \mathfrak{so}^{5,3},$$

$$31513221 \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^4 \otimes \mathfrak{so}^{5,3},$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^4 \otimes \mathfrak{so}^{5,5} \otimes \mathfrak{so}^{5,3} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^4 \otimes \mathfrak{so}^{5,5} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{ccccccc} 2,3 & 4,5 & 5,5 & & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes & \mathbf{e}_6, & & \\ I^{\oplus 2} \mathbf{e}_7 & (\otimes \mathbf{e}_8)^4 & \otimes & \mathbf{e}_6 & & & \end{array}$$

$$\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 \begin{matrix} 4,5 \\ \otimes \end{matrix} \mathbf{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathbf{e}_8 \right)^3 \begin{matrix} 5,5 \\ \otimes \end{matrix} \mathbf{e}_6,$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\begin{array}{c} 2,3 \quad 4,5 \quad (5,5) \quad 5,3 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}, \\ I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so} \end{array}$$

$${}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \otimes \mathfrak{e}_8 \left({}^{5,5} \otimes \mathfrak{e}_8 \right)^3 \otimes {}^{5,3} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,3} \mathfrak{so}$$

$$\begin{array}{l} 3150 \otimes e_8(2,4) \otimes e_8(5,5)^3 \otimes e_6(5,5) \\ 1e_6(4,5) \otimes e_8(5,5) \otimes e_8(5,5)^3 \otimes so(5,3) \end{array}$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{matrix} 3,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \right)^3 \begin{matrix} 5,3 \\ \otimes \end{matrix} \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^3 \overset{5,3}{\otimes} \mathfrak{so}$$

$$\begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} so,$$

$$I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{3,3} \mathfrak{so}$$

$$\begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7,$$

$$\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7,$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^3 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}_7$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7'$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_6 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3}$$

2.7.3 Sg6S(376)

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^{1231512},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^{123151},$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^{1232},$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^{12313}$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^{1232151}$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$31321\mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3}$$

$$\begin{aligned}
& \otimes \overset{2,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} \overset{4,4}{e_7} \\
& \otimes \overset{2,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 12313 \\
& \otimes \overset{2,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 1232151 \\
& I^{\oplus 3} \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} \overset{4,2}{e_7} \\
& \otimes \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} \overset{4,5}{e_7} \\
& \otimes \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 1231512 \\
& \otimes \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 123151^1 \\
& I^{\oplus 3} \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} \overset{4,5}{e_7} \\
& I^{\oplus 3} \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 1231512 \\
& I^{\oplus 3} \overset{3,3}{e'_7}(\otimes \overset{3,3}{e_7})^4 \otimes \overset{3,3}{e_7} 123151^1
\end{aligned}$$

$$\begin{array}{l}
{}^{2,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
{}^{2,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
{}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
{}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
I^{\oplus 3} \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
I^{\oplus 3} \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
I^{\oplus 2} \mathfrak{e}_7 ({}^{5,5} \otimes \mathfrak{e}_8)^4 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
I^{\oplus 2} \mathfrak{e}_7 ({}^{5,5} \otimes \mathfrak{e}_8)^4 \otimes {}^{5,5} \mathfrak{e}_8 12231513151
\end{array}$$

$$\begin{aligned}
& 2313221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 122315123 \\
& 2313221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151 \\
& I^{\oplus 6}\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513215 \\
& 321513221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 122315123 \\
& 321513221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151 \\
& 15131513221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513215 \\
& 15131513221\mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151
\end{aligned}$$

$$\begin{array}{ccccccc} 2,2 & & 3,3 & & 3,3 & & 4,4 \\ \otimes & \mathbf{e}_6 & (\otimes & \mathbf{e}_7)^4 & \otimes & \mathbf{e}_7 & \otimes \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^3 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^3 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^3 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^4\overset{2,2}{\otimes}\mathfrak{e}_6\overset{3,4}{\otimes} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^4\overset{2,2}{\otimes}\mathfrak{e}_6I^{\oplus 3} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^4\overset{2,2}{\otimes}\mathfrak{e}_6131\overset{1}{5}1 \end{array}$$

$$I^{\oplus 2} e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$1 e_7' (\otimes^{5,5} e_8)^4 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$\begin{array}{l} 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5,5} 122315123 \\ 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5,5} 12231513151 \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 \bigotimes \\ I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 I^{\oplus 3} \end{array}$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & (5,5) & & & \\ \otimes \mathbf{e}_6 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}'_7 & 1 & & \\ 1\mathbf{e}_6 & (4,5) & (5,5) & (5,5) & & & \\ & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}'_7 & 1 & & \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,3}{\otimes} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 3},$$

$$231\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 1,$$

$$31\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$321\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1,$$

$$I^{\oplus 2}\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2},$$

$$31\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6)^4 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

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$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 5}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$21513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$1513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 4}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$I^{\oplus 3}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$\begin{array}{c} 5,5 \quad 1 \\ \otimes \mathfrak{e}_8 \end{array} \left(\begin{array}{c} 5,5 \\ \otimes \mathfrak{e}_8 \end{array} \right)^4 \begin{array}{c} 5,3 \\ \otimes \mathfrak{so} \end{array}$$

$$21513221\mathbf{e}_8\overset{1}{\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)}^4\overset{5,3}{\otimes}\mathbf{so}$$

$$I^{\oplus 4} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^4 \otimes \mathfrak{so}_{50}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes_{\mathbf{e}_7}^{3,3} \right)^4 \bigotimes_{\mathbf{e}_7}^{3,3} \mathbf{so}$$

$$\begin{array}{l} I^{\oplus 2} \mathbf{e}_6^{(2,2)} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{13151} \\ 3I \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{(2,2)} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,4} \\ 3I \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{(2,2)} \otimes \mathbf{e}_6^{2,2} \otimes I^{\oplus 3} \\ 3I \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{(2,2)} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{13151} \end{array}$$

$$\begin{aligned}
& 1 \overset{1}{\epsilon}'_7 \otimes \overset{4,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,5}{\otimes} \epsilon_8 122315123 \\
& 1 \overset{1}{\epsilon}'_7 \otimes \overset{4,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,5}{\otimes} \epsilon_8 12231513151 \\
& \overset{5,5}{\otimes} \epsilon_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,4}{\epsilon}'_7 I^{\oplus 2} \\
& 21513221 \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,4}{\otimes} \epsilon'_7 I^{\oplus 2} \\
& I^{\oplus 4} \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,4}{\otimes} \epsilon'_7 I^{\oplus 2} \\
& \overset{4,5}{\otimes} \epsilon_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,5}{\otimes} \epsilon'_7 1 \\
& I^{\oplus 5} \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,5}{\otimes} \epsilon'_7 1 \\
& I^{\oplus 2} \overset{2,4}{\otimes} \overset{5,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^4 \overset{5,5}{\otimes} \epsilon'_7 1
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \\
& I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7'
\end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^4 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \\ & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^4 \otimes \epsilon_7^{3,3} \epsilon_7^{12313} \\ & 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^4 \otimes \epsilon_7^{3,3} \epsilon_7^{1232151} \\ & \epsilon_7^{3,3} \epsilon_7^{1,3,3}(\otimes \epsilon_7)^4 \otimes \epsilon_6^{3,3} \otimes \epsilon_6^{2,2} \\ & \otimes \epsilon_7^{3,3} \epsilon_7^{1,3,3}(\otimes \epsilon_7)^4 \otimes \epsilon_6^{3,3} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^{3,3} \bigotimes \mathbf{e}_8^{4,5} I^{\oplus 2}$$

$$\begin{array}{lll} 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,3} \otimes \mathbf{e}_6^{2,2}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 2}, & 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,3} I^{\oplus 2} \\ 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} 1, & 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\mathfrak{e}_6^{(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_7^{3,2} \otimes, & 1\mathfrak{e}_6^{(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^{(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_7' I^{\oplus 3} \\ I^{\oplus 2} \mathfrak{e}_7^{(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_7^{(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_6 I^{\oplus 2} & \end{array}$$

$$\begin{array}{lll} \begin{array}{c} 2,3 \\ \otimes \mathfrak{e}_7 \end{array} \begin{array}{c} 1 \\ (\otimes \mathfrak{e}_7)^4 \end{array} \begin{array}{c} 3,3 \\ \otimes \mathfrak{so}1, \end{array} & \begin{array}{c} 3,3 \\ \otimes \mathfrak{e}_7 \end{array} \begin{array}{c} 1 \\ (\otimes \mathfrak{e}_7)^4 \end{array} \begin{array}{c} 3,2 \\ \otimes \mathfrak{so}1, \end{array} & I^{\oplus 3} \begin{array}{c} 1 \\ \mathfrak{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathfrak{e}_7)^4 \end{array} \begin{array}{c} 3,2 \\ \otimes \mathfrak{so}1 \end{array} \\ I^{\oplus 2} \begin{array}{c} 1 \\ \mathfrak{e}_7 \end{array} \begin{array}{c} I^{\oplus 2} \\ (\otimes \mathfrak{e}_7)^4 \end{array} \begin{array}{c} 3,3 \\ \otimes \mathfrak{so}1 \end{array} & & \end{array}$$

$$1\epsilon_8^1 \otimes \epsilon_6^{5,3} (\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{2,2}, \quad 1\epsilon_6^{1,2} (\otimes \epsilon_6^{2,2})^4 \otimes \epsilon_8 I^{\oplus 2}, \quad 1\epsilon_8^1 \otimes \epsilon_6^{5,3} (\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} I^{\oplus 2}$$

$$31\epsilon_6^{1,2} (\otimes \epsilon_6^{2,2})^4 \otimes \epsilon_8 1$$

$$\begin{array}{lll} 2,2 \text{ } 1 & 2,2 & 2,2 \\ \otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_6)^4 & \otimes \mathfrak{so}1, \end{array} \quad \begin{array}{lll} I^{\oplus 2} \mathfrak{e}_6^1 & 2,2 & 2,2 \\ & (\otimes \mathfrak{e}_6)^4 & \otimes \mathfrak{so}1, \end{array} \quad \begin{array}{lll} 31 & 2,2 & 2,2 \\ \mathfrak{e}_6 & (\otimes \mathfrak{e}_6)^4 & \otimes \mathfrak{so}1 \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\bigotimes_{3,3} \mathbf{e}_7 \right)^4 \bigotimes_{3,3} \mathbf{e}'_7 \bigotimes_{3,2}, \quad I^{\oplus 2} \mathbf{e}'_7 \left(\bigotimes_{3,3} \mathbf{e}_7 \right)^4 \bigotimes_{3,3} \mathbf{e}'_7 \bigotimes_{3,3}, \quad I^{\oplus 2} \mathbf{e}'_7 \left(\bigotimes_{3,3} \mathbf{e}_7 \right)^4 \bigotimes_{3,3} \mathbf{e}'_7 I^{\oplus 3}$$

$$1\mathbf{e}_6^{\overset{1}{\times}\overset{3,5}{\otimes}\overset{5,5}{\otimes}\overset{5,3}{\otimes}\overset{2,2}{\otimes}}\mathbf{e}_8(\otimes\mathbf{e}_8)^3\otimes\mathbf{e}_6\otimes, \quad 1\mathbf{e}_6^{\overset{1}{\times}\overset{3,5}{\otimes}\overset{5,5}{\otimes}\overset{5,3}{\otimes}}\mathbf{e}_8(\otimes\mathbf{e}_8)^3\otimes\mathbf{e}_6I^{\oplus 2}, \quad 1\mathbf{e}_6^{\overset{1}{\times}\overset{3,5}{\otimes}\overset{5,5}{\otimes}\overset{5,4}{\otimes}}\mathbf{e}_8(\otimes\mathbf{e}_8)^3\otimes\mathbf{e}_61$$

$$I^{\oplus 2} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} 1, \quad 1 \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} 13$$

$$I^{\oplus 2} \mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^3 \otimes \mathbf{so}13, \quad 1\mathbf{e}_7^{(5,5)} \otimes \mathbf{e}_8^4 \otimes \mathbf{so}13$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8)^{5,5}{}^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}, \quad 1\mathfrak{e}_7^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1e_7' \otimes e_8 \otimes e_8^3 \otimes e_7' I^{\oplus 2}, \quad 1e_7' \otimes e_8 \otimes e_8^3 \otimes e_7' 1$$

$$1\epsilon_7'(\otimes^{3,3}\epsilon_7)^4\otimes^{4,5}\epsilon_8I^{\oplus 2}, \quad 1\epsilon_8\otimes^{5,4}\epsilon_7(\otimes^{3,3}\epsilon_7)^3\otimes^{3,3}\epsilon_7'I^{\oplus 2}$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$\begin{array}{l} 3,5 \atop 1 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 2} I^{\oplus 4} \\ 4,5 \atop 1 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 3} I^{\oplus 3} \\ 313221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 2} I^{\oplus 4} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 2} I^{\oplus 4} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 3} I^{\oplus 3} \\ I^{\oplus 4} I^{\oplus 2} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_8 I^{\oplus 3} \end{array}$$

$$\begin{aligned}
& I^{\oplus 1} \mathbf{c}_7^{\overline{1}, 4, 5} \otimes \mathbf{c}_8^{\overline{5, 5}} \otimes^3 \mathbf{c}_8^{\overline{5, 5}, 1, 5, 5} \\
& I^{\oplus 2} \mathbf{c}_7^{\overline{1}, 4, 5} \otimes \mathbf{c}_8^{\overline{5, 5}} \otimes^3 \mathbf{c}_8^{\overline{5, 5}, 1} 12231512 \\
& I^{\oplus 2} \mathbf{c}_7^{\overline{1}, 4, 5} \otimes \mathbf{c}_8^{\overline{5, 5}} \otimes^3 \mathbf{c}_8^{\overline{5, 5}, 1} I^{\oplus 2} I^{\oplus 4} \\
& 1 \mathbf{c}_7^{\overline{1}, 5, 5} \otimes^4 \mathbf{c}_8^{\overline{5, 5}, 1, 5, 4} \\
& 1 \mathbf{c}_7^{\overline{1}, 5, 5} \otimes^4 \mathbf{c}_8^{\overline{5, 5}, 1} \mathbf{c}_8 I^{\oplus 5} \\
& 1 \mathbf{c}_7^{\overline{1}, 5, 5} \otimes^4 \mathbf{c}_8^{\overline{5, 5}} I^{\oplus 24, 2}
\end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}_7^1 \bigotimes \mathbf{e}_7^1 \bigotimes \mathbf{e}_7^1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}_7^1 I^{\oplus 3} \mathbf{e}_7^1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}_7^1 I^{\oplus 2} \mathbf{e}_7^1 I^{\oplus 2} \end{aligned}$$

$$\begin{array}{l} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} 1^{5,5} \otimes \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} 1^{5,5} 12231512 \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} I^{\oplus 2} \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} I^{\oplus 4} \end{array}$$

$$1_{\mathfrak{e}_6}^{1,3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5}$$

$$1_{\mathfrak{e}_6}^{1,3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{12231512}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} 1223151, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} 4,2, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} \\
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 3} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_6^{2,2} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} \\
& 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1, & 1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} 1$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^4 \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} 1$$

$$1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,4}{\otimes} \mathbf{e}_7' 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^3 \otimes \mathbf{e}_6^{5,3} 1$$

$$1 \mathbf{e}_6^1 (\otimes \mathbf{e}_7^{3,3})^4 \otimes \mathbf{e}_7'^1 I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{1}{\overset{(3,3)}{\otimes}} \mathbf{e}_7)^4 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

$$1\mathbf{e}_6\left(\overset{2,2}{\otimes}\mathbf{e}_6\right)^4\overset{3,5}{\otimes}\mathbf{e}_81$$

2.7.7 gLIg5(3)

$$50 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^2 \otimes^{5,4} e_7 \otimes^{3,3} e_7$$

$$e_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8)^2 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} so \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} so$$

$$e_7^{3,3} \otimes e_7^1 (e_7^{3,3} \otimes e_7)^3 \otimes e_7^{3,2} \otimes 50$$

2.7.8 SgLIg5(5)

$$I^{\oplus 3} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}, \quad \mathfrak{iso}^{2,4} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5}$$

$$I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,4}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_6$$

2.7.9 SgLIg3LIgLg(1)

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

2.7.10 SgLIg3LIgLgS(1)

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

2.7.11 SSgLIg5(3)

$$I^{\oplus 2} \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,5}$$

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,4}$$

2.7.12 SSgLIg3LIgLg(1)

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

2.7.13 SSgLIg5S(12)

$$I^{\oplus 2} \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,5}$$

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,4}$$

2.7.14 SSgLIg5IS(4)

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2}, \quad 1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} I^{\oplus 4}, \quad 1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} I^{\oplus 3}$$

$$1\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7 1$$

2.7.15 SSg4LIgLg(3)

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$31513221 \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5}$$

$$1\mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

2.7.16 SSg4LIgLgS(20)

$$I^{\oplus 2} \mathfrak{e}_7^{1, 4, 5} \otimes \mathfrak{e}_8^{5, 5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5, 5} \otimes \mathfrak{so}1^{1, 4, 2}, \quad 1 \mathfrak{e}_7^{5, 5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5, 5} \otimes \mathfrak{so}1^{1, 4, 2}$$

$$1e_7^{1,4,5} \otimes e_8^{5,5} (\otimes e_8)^2 \otimes e_8^{5,5} \otimes e_8^{1,5,5} I^{\oplus 2}$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 \otimes^{1,4,2} so1$$

$$1\mathfrak{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_8} \overset{4,2}{\otimes} \mathfrak{so}1$$

2.7.17 SSg4LIgLIgS(4)

$$2,4 \atop 1 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \atop 5,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1, \quad I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \atop 5,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1, \quad I^{\oplus 3} I^{\oplus 2} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \atop 5,5 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,5}{e_8} 1$$

2.7.18 Sg4LIgIg(7)

$$\begin{array}{ccc} \begin{array}{c} 2,3 \\ \otimes e_7 \end{array} (\begin{array}{c} 3,3 \\ \otimes e_7 \end{array})^3 \begin{array}{c} 3,3 \\ \otimes e_7 \end{array} \begin{array}{c} 1 \\ \otimes e_7 \end{array} \begin{array}{c} 3,3 \\ \otimes e_7 \end{array}, & \begin{array}{c} 3,3 \\ \otimes e_7 \end{array} (\begin{array}{c} 3,3 \\ \otimes e_7 \end{array})^3 \begin{array}{c} 3,3 \\ \otimes e_7 \end{array} \begin{array}{c} 1 \\ \otimes e_7 \end{array} \begin{array}{c} 3,3 \\ \otimes e_7 \end{array}, & I^{\oplus 3} e_7 (\begin{array}{c} 3,3 \\ \otimes e_7 \end{array})^3 \begin{array}{c} 3,3 \\ \otimes e_7 \end{array} \begin{array}{c} 1 \\ \otimes e_7 \end{array} \begin{array}{c} 3,3 \\ \otimes e_7 \end{array} \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^3 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}_7} \overset{3,3}{\otimes} \mathbf{e}_7$$

$$2321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathbf{e}_8$$

$$150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 \otimes^{1,5,4} e_7$$

$$1e_6(\overset{3,3}{\otimes} e_7)^3 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{\otimes} e_7$$

2.7.19 SgLIg5S(32)

$$\begin{aligned}
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 1223151315, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3}, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \\
& I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5}, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 122313, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 5} \\
& I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 12231512, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 1223151^1 & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 1223132, & I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} 1, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 6} \\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 122315123, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 12231513215, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 12231513^1 \\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} 12231513151 & & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2}, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3} \\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,5} I^{\oplus 2} & & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2}, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,4} 1 \\
\\
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} 123, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,5} 1 \\
\\
& I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} 1, & 1\mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} 13 & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}, & 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,5} 1 & \\
\\
& I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_7^{5,4} 1 & & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} 13 & & \\
\\
& 1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so} 1 & &
\end{aligned}$$

2.8 $n \geq 7$ node(10,882)

2.8.1 $g(n)(47)$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^9 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^8 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^7 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^6 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^{15} \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{14} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{13} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{12} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{11} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{10} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}_7 (\overset{3,3}{\otimes} \mathfrak{e}_7)^5 \overset{4,5}{\otimes} \mathfrak{e}_8$$

$$\mathfrak{so}(\overset{2,2}{\otimes} \mathfrak{e}_6)^6 \overset{3,5}{\otimes} \mathfrak{e}_8$$

$$\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^8 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^6 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{15} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{14} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{13} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{12} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{11} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{10} \overset{5,5}{\otimes} \mathbf{e}_6$$

2.8.2 Sg(n)(570)

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^9 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^9 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$2151321e_7(\otimes^{3,3} e_7)^9 \otimes^{3,3} \mathfrak{so},$$

$$151321e_7({}^{3,3}\otimes e_7)^9 {}^{3,3}\otimes \mathfrak{so}$$

$$\begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^9 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^9 \bigotimes^{3,3} \mathfrak{so},$$

$${}^1_{2321}e_7\left(\bigotimes^{3,3}e_7\right)^9\bigotimes^{3,3}so,$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathbf{e}_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 \right)^9 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{so}$$

$$31321e_7(\bigotimes^{3,3} e_7)^9 \bigotimes^{3,3} \mathfrak{so}$$

$$1512321e_7(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} so$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^8 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^8 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$2151321e_7(\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so},$$

$$151321e_7(3,3 \otimes e_7)^8 3,3 \otimes 50$$

$$\otimes^{3,4} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^8 \otimes^{3,3} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_7 \left(\bigotimes \mathfrak{e}_7 \right)^8 \bigotimes \mathfrak{so},$$

$${}^1_{2321}\mathfrak{e}_7({}^{3,3}\otimes \mathfrak{e}_7)^8 {}^{3,3}\otimes \mathfrak{so},$$

$${}^{4,4}_{\otimes} \mathfrak{e}_7 ({}^{3,3}_{\otimes} \mathfrak{e}_7)^8 {}^{3,3}_{\otimes} \mathfrak{so}$$

$$31321e_7(\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so}$$

$$1512321e_7(\otimes^{3,3} e_7)^8 \otimes^{3,3} so$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\otimes^{5,4} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^7 \otimes^{3,3} \mathfrak{so},$$

$$2151321e_7(\otimes^{3,3} e_7)^7 \otimes^{3,3} \mathfrak{so},$$

$$151321e_7\binom{3,3}{\otimes e_7}^7\binom{3,3}{\otimes 50}$$

$$\begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^7 \bigotimes^{3,3} \mathfrak{so},$$

$${}^1_{2321}e_7({}^{3,3}\otimes e_7)^7 {}^{3,3}\otimes \mathfrak{so},$$

$${}^{4,4}_{\otimes} \mathfrak{e}_7 ({}^{3,3}_{\otimes} \mathfrak{e}_7)^7 {}^{3,3}_{\otimes} \mathfrak{so}$$

$$31321e_7(\bigotimes^{3,3} e_7)^7 \bigotimes^{3,3} so$$

$$1512321e_7(\bigotimes^{3,3} e_7)^7 \bigotimes^{3,3} so$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^6 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\otimes^{5,4} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^6 \otimes^{3,3} \mathfrak{so},$$

$$\otimes^{3,4} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^6 \otimes^{3,3} \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^6 \bigotimes^{3,3} \mathfrak{so},$$

$${}^{4,4}_{\otimes} \mathfrak{e}_7 ({}^{3,3}_{\otimes} \mathfrak{e}_7)^6 {}^{3,3}_{\otimes} \mathfrak{so}$$

$$31321e_7(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} so$$

$$1512321e_7(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} so$$

$${}^{4,4} \otimes \mathfrak{e}_7 ({}^{3,3} \otimes \mathfrak{e}_7)^5 \otimes {}^{3,3} \otimes \mathfrak{so}$$

$$31321e_7(\overset{3,3}{\otimes} e_7)^5 \overset{3,3}{\otimes} so$$

$$1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^5\overset{3,3}{\otimes}\mathbf{so}$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^{15} \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so}$$

$$31321e_7(\overset{3,3}{\otimes} e_7)^{15} \overset{3,3}{\otimes} so$$

$$1512321e_7(\otimes^{3,3} e_7)^{15} \otimes^{3,3} so$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{e}_7 \right)^{14} \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so}$$

$$31321e_7(\otimes_{e_7}^{3,3})^{14} \otimes_{so}^{3,3}$$

$$1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathfrak{so}$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^{13} \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so}$$

$$31321e_7(\otimes_{e_7}^{3,3})^{13} \otimes_{50}^{3,3}$$

$$1512321e_7(\overset{3,3}{\otimes} e_7)^{13} \overset{3,3}{\otimes} so$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^{12} \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so}$$

$$31321e_7(\overset{3,3}{\otimes} e_7)^{12} \overset{3,3}{\otimes} 50$$

$$1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{12}\overset{3,3}{\otimes}\mathbf{so}$$

$$\begin{matrix} 4,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7^{11} \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so}$$

$$\begin{array}{l} 31321\mathfrak{e}_7(\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{so} \\ 1512321\mathfrak{e}_7(\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{l} 4,4 \\ \otimes e_7(\otimes e_7)^{10} \otimes so \\ 31321e_7(\otimes e_7)^{10} \otimes so \\ 1512321e_7(\otimes e_7)^{10} \otimes so \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes^{5,5}\epsilon_8)^9\otimes^{5,5}\epsilon_6 \\ 15131513221\epsilon_8(\otimes^{5,5}\epsilon_8)^9\otimes^{5,5}\epsilon_6 \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^9 \otimes^{5,3} \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^9 \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{array}{l} 321513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 15131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^8 \otimes^{5,3} \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^8 \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{array}{l} 321513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 15131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,5}{\otimes}\mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^7 \otimes^{5,3} \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^7 \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^6\overset{5,5}{\otimes}\epsilon_6 \\ 15131513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^6\overset{5,5}{\otimes}\epsilon_6 \end{array}$$

$$15131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$321513221\mathbf{e}_8\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)^{12}\overset{5,3}{\otimes}\mathbf{50}$$

$$321513221\mathbf{e}_8\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)^{11}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$321513221\mathbf{e}_8\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)^{11}\overset{5,3}{\otimes}\mathbf{50}$$

$$321513221 \mathbf{e}_8 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^{10} \bigotimes^{5,5} \mathbf{e}_6$$

$$321513221\mathbf{e}_8\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)^{10}\overset{5,3}{\otimes}\mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 \left(\otimes \mathbf{e}_8 \right)^9 \otimes \mathbf{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^9 \otimes^{5,3} \mathfrak{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,3}{\otimes} 50$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^6 \otimes^{5,3} \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^5 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^5 \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,3}{\otimes} \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{14} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{so}^{5,3}$$

[illegible]

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^8 \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^7 \overset{5,3}{\otimes} \mathbf{50}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^6 \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^5 \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^4 \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^{14} \overset{5,3}{\otimes} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{12} \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{11} \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{10} \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{5,5} \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{5,5} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_6 \binom{2,2}{\otimes \mathfrak{e}_6}^5 \otimes \mathfrak{e}_8,$$

$$15131 \mathfrak{e}_6 \binom{2,2}{\otimes \mathfrak{e}_6}^5 \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{3,3}{\otimes \mathfrak{e}_7}^6 \otimes \mathfrak{e}_8$$

$$321 \mathfrak{e}'_7 \binom{3,3}{\otimes \mathfrak{e}_7}^5 \otimes \mathfrak{e}_8$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^9 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^8 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^7 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^6 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^5 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^4 \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \otimes \mathfrak{so}$$

$$31 \mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \otimes \mathfrak{so}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \binom{5,3}{\otimes \mathfrak{so}}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \binom{5,3}{\otimes \mathfrak{so}}$$

$$31\mathfrak{so} \binom{1,1}{\otimes \mathfrak{so}}^6 \binom{2,4}{\otimes \mathfrak{e}_8}$$

$$31\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^5 \binom{4,5}{\otimes \mathfrak{e}_8}$$

$$231\mathfrak{so} \binom{1,1}{\otimes \mathfrak{so}}^5 \binom{2,4}{\otimes \mathfrak{e}_8}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \binom{2,4}{\otimes \mathfrak{e}_8}^{5,5} \binom{5,4}{\otimes \mathfrak{e}_7} \binom{3,3}{\otimes \mathfrak{e}_7}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \binom{2,3}{\otimes \mathfrak{e}_7}^{3,3} \binom{3,3}{\otimes \mathfrak{so}}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{array}{lll}
I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} 12231513215, & I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \overset{1}{\otimes} \mathbf{e}_8 12231513, & I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8 12231513151 \\
1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} 1223132, & 1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8 I^{\oplus 6}, & 1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8 122315123 \\
1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} 12231513215, & 1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \overset{1}{\otimes} \mathbf{e}_8 12231513, & 1 \mathbf{e}_6^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8 12231513151
\end{array}$$

[illegible]

$\begin{array}{l} 2,2 \quad 3,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 1223132, \\ \\ 2,2 \quad 3,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513215, \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 1223132, \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513215, \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 1223132, \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513215, \end{array}$	$\begin{array}{l} 2,2 \quad 3,5 \quad 5,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 I^{\oplus 6}, \\ \\ 2,2 \quad 3,5 \quad 5,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513, \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 I^{\oplus 6}, \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513, \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 I^{\oplus 6}, \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513, \end{array}$	$\begin{array}{l} 2,2 \quad 3,5 \quad 5,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 122315123 \\ \\ 2,2 \quad 3,5 \quad 5,5 \quad 5,5 \\ \otimes \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513151 \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 122315123 \\ \\ I^{\oplus 2} \mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513151 \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 122315123 \\ \\ 1\mathbf{e}_6 \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_8 12231513151 \end{array}$
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[illegible]

[illegible]

$$1e_6 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{10} \overset{5,5}{\otimes} e_8 12231513151$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^9 & \otimes \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^9 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,3 \\ (\otimes \mathfrak{so})^9 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,4 \\ (\otimes \mathfrak{so})^9 & \otimes \mathfrak{so} & \otimes \end{array} \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^8 & \otimes & \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^8 & \otimes & \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,3 \\ (\otimes \mathfrak{so})^8 & \otimes & \mathfrak{so} & \otimes & \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,4 \\ (\otimes \mathfrak{so})^8 & \otimes & \mathfrak{so} & \otimes & \end{array} \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & & 1,1 & & \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^7 & \otimes & \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & & 1,1 & & \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^7 & \otimes & \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} (\otimes \mathfrak{so})^7 \otimes \mathfrak{so} \otimes \\ I^{\oplus 2} \mathfrak{so} (\otimes \mathfrak{so})^7 \otimes \mathfrak{so} \otimes \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^6 & \otimes \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^6 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,3 \\ (\otimes \mathfrak{so})^6 & \otimes \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & 1,1 & 2,4 \\ (\otimes \mathfrak{so})^6 & \otimes \mathfrak{so} & \otimes \end{array} \end{array}$$

$$\begin{array}{l} \begin{array}{cccc} 2,2 & 1,1 & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^5 & \otimes & \mathfrak{so} \otimes \end{array} \\ \begin{array}{cccc} 3,2 & 1,1 & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^5 & \otimes & \mathfrak{so} \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccc} (1,1) & 1,1 & 2,3 \\ \otimes \mathfrak{so} & \otimes & \mathfrak{so} \otimes \end{array} \end{array}$$

$$I^{\oplus 2} \mathfrak{so}(\overset{1,1}{\otimes} \mathfrak{so})^5 \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes}$$

$$\begin{array}{l} 2,2 \quad (1,1) \quad 1,1 \quad 2,4 \\ \otimes \mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} \otimes \\ 3,2 \quad (1,1) \quad 1,1 \quad 2,3 \\ \otimes \mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} \otimes \\ I^{\oplus 2} \mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} \otimes \\ I^{\oplus 2} \mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} \otimes \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & & 1,1 & & 1,1 & & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^{14} & \otimes \mathfrak{so} & \otimes & \end{array} \\ \begin{array}{ccccc} 3,2 & & 1,1 & & 1,1 & & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^{14} & \otimes \mathfrak{so} & \otimes & \end{array} \\ I^{\oplus 2} \mathfrak{so} & (\otimes \mathfrak{so})^{14} & \otimes \mathfrak{so} & \otimes & \\ I^{\oplus 2} \mathfrak{so} & (\otimes \mathfrak{so})^{14} & \otimes \mathfrak{so} & \otimes & \end{array}$$

$$\begin{array}{l} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \\ \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \\ I^{\oplus 2} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \\ I^{\oplus 2} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^{12} & \otimes & \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^{12} & \otimes & \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \left(\begin{array}{ccccc} 1,1 & & & 1,1 & 2,3 \\ \otimes \mathfrak{so} & & & \mathfrak{so} & \otimes \end{array} \right)^{12} \\ I^{\oplus 2} \mathfrak{so} \left(\begin{array}{ccccc} 1,1 & & & 1,1 & 2,4 \\ \otimes \mathfrak{so} & & & \mathfrak{so} & \otimes \end{array} \right)^{12} \end{array}$$

$$\begin{array}{cc} \begin{array}{c} 2,2 \\ \otimes \end{array} \mathfrak{so} \left(\begin{array}{c} 1,1 \\ \otimes \end{array} \mathfrak{so} \right)^{11} & \begin{array}{c} 1,1 \\ \otimes \end{array} \mathfrak{so} \begin{array}{c} 2,4 \\ \otimes \end{array} \\ \begin{array}{c} 3,2 \\ \otimes \end{array} \mathfrak{so} \left(\begin{array}{c} 1,1 \\ \otimes \end{array} \mathfrak{so} \right)^{11} & \begin{array}{c} 1,1 \\ \otimes \end{array} \mathfrak{so} \begin{array}{c} 2,3 \\ \otimes \end{array} \end{array}$$

[illegible]

[illegible]

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_8^{3,2} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes I^{\oplus 3} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} \otimes I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_8^{3,2} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes I^{\oplus 3} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} \otimes I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_8^{3,2} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes I^{\oplus 3} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} \otimes I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_7^{3,2} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes I^{\oplus 3} \\ & I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} \otimes I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_7^{3,2} \\ I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} \otimes I^{\oplus 3} \\ I^{\oplus 3} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,5} \otimes I^{\oplus 2} \end{aligned}$$

$$\begin{array}{l} 150 \otimes_{\mathbf{e}_7}^{2,3} (\otimes_{\mathbf{e}_7}^{3,3})^9 \otimes_{\mathbf{e}_7}^{3,3} \otimes^{4,4} \\ 150 \otimes_{\mathbf{e}_7}^{2,3} (\otimes_{\mathbf{e}_7}^{3,3})^9 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7} 12313 \\ 150 \otimes_{\mathbf{e}_7}^{2,3} (\otimes_{\mathbf{e}_7}^{3,3})^9 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7} 1232151 \end{array}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}'_7 (\otimes^{3,3} \mathbf{e}_7)^{12} \otimes^{3,3} \mathbf{e}'_7 \otimes^{3,2} \\ I^{\oplus 3} \mathbf{e}'_7 (\otimes^{3,3} \mathbf{e}_7)^{12} \otimes^{3,3} \mathbf{e}'_7 I^{\oplus 3} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^{11} \bigotimes^{3,3} \mathbf{e}'_7 \bigotimes^{3,2} \\ I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^{11} \bigotimes^{3,3} \mathbf{e}'_7 I^{\oplus 3} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{(3,3)} \mathbf{e}_7 \right)^{10} \bigotimes^{(3,3)} \mathbf{e}'_7 \bigotimes^{(3,2)} \\ I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{(3,3)} \mathbf{e}_7 \right)^{10} \bigotimes^{(3,3)} \mathbf{e}'_7 I^{\oplus 3} \end{aligned}$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & & 5,5 & & \\ \otimes e_6 & \otimes e_8 & (\otimes e_8)^9 & \otimes & e_7' 1 & & \\ & 4,5 & (5,5) & & 5,5 & & \\ 1e_6 & \otimes e_8 & (\otimes e_8)^9 & \otimes & e_7' 1 & & \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & 5,5 \\ \otimes e_6 & \otimes e_8 & (\otimes e_8)^8 & \otimes e_7' 1 \\ 1e_6 & \otimes e_8 & (\otimes e_8)^8 & \otimes e_7' 1 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & 7 & 5,5 & & \\ \otimes \mathbf{e}_6 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8) & & \otimes \mathbf{e}'_7 & 1 & \\ & 4,5 & (5,5) & 7 & 5,5 & & \\ 1\mathbf{e}_6 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8) & & \otimes \mathbf{e}'_7 & 1 & \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{matrix} 2,2 & 3,5 & (5,5 & 5,5 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^6 & \otimes \mathfrak{e}'_7 1 \end{matrix}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{10} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$1_{50} \overset{1,1}{\otimes} 5_0 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^9 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^8 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$1_{50} \otimes^{1,1} 50 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$150 \otimes^{1,1} 50 \otimes^{2,4} e_8 \left(\otimes^{5,5} e_8 \right)^6 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$150 \otimes^{1,1} 50 \otimes^{2,4} e_8 \left(\otimes^{5,5} e_8 \right)^5 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$1_{50} \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^3 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^{13} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_7 I^{\oplus 3}$$

$$1_{50} \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{12} \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

[illegible]

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,4} e_6 1$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^6 \otimes^{5,4} \mathfrak{e}_6 1$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,4} e_6 1$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,4} e_6 1$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{14} \otimes^{5,4} \mathfrak{e}_6 1$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,4} e_6 1$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,4} e_6 1$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^{11} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 1$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^{10} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 1$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^9 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^8 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^7 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^6 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^5 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^4 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$1s0 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{14} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 I^{\oplus 3}$$

$$1s_0 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^{13} \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$1s_0 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^{12} \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$1s_0 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^{11} \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$1s_0 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^{10} \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^9 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^8 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^7 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^6 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^5 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,4}{\otimes} e_6 1$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^7 \overset{5,5}{\otimes} e'_7 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^6 \overset{5,5}{\otimes} e'_7 1$$

$$1_{50} \overset{1,1}{\otimes} 5_0 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^5 \overset{5,5}{\otimes} e_7' 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,5}{\otimes} e'_7 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} e_7' 1$$

$$1_{50} \overset{1,1}{\otimes} 5_0 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_7' 1$$

$$1_{\mathfrak{so}} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{12} \overset{5,5}{\otimes} \mathfrak{e}'_7 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^{11} \overset{5,5}{\otimes} e'_7 1$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{10} \otimes^{5,5} \mathfrak{e}'_7 1$$

$$31\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^5\overset{2,4}{\otimes}\mathfrak{e}_81$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^7 \otimes^{3,2} 501$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^9 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^8 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^7 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^6 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{15} \otimes \mathfrak{so}^{2,2} 1$$

2.8.4 SSg(n)(379)

[illegible]

$$I^{\oplus 4} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^7 \otimes \mathfrak{so}_{50}$$

$$\begin{array}{c} 5,5 \quad 1 \quad (5,5) \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_6 \\ \\ 21513221 \mathfrak{e}_8^1 (5,5) \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_6 \\ \\ I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} (5,5) \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{c} 5,5 \quad 1 \quad 5,5 \quad 5,3 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so} \\ I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{c} 5,5 \quad 1 \quad (5,5) \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}_6 \\ \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}_6 \\ \\ I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \quad 5,3 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so} \end{array} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so} \\ I^{\oplus 4} I^{\oplus 2} \begin{array}{c} 5,5 \quad 5,3 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so} \end{array} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 15 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \\ 21513221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 15 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \\ I^{\oplus 4} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 15 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \begin{array}{c} 6 \\ \otimes \end{array} \end{array}$$

$${}^{5,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{15} {}^{5,3} \otimes \mathfrak{so}$$

$$\begin{array}{l} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \end{array} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so} \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so} \end{array}$$

$$\begin{aligned} & 5,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6 \\ & 21513221 \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6 \\ & I^{\oplus 4} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6 \end{aligned}$$

$$\begin{array}{l} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so} \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so} \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so} \end{array}$$

$$I^{\oplus 3} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^9 \bigotimes^{3,3} \mathfrak{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^8 \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^7 \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^6 \bigotimes \mathfrak{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^5 \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{15} \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{14} \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{13} \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathfrak{e}_7 \left(\bigotimes \mathfrak{e}_7 \right)^{12} \bigotimes \mathfrak{so}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{11} \bigotimes \mathbf{so}$$

$$I^{\oplus 3} \mathfrak{e}_7^{(3,3)} (\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{so}^{3,3}$$

$$I^{\oplus 2} \mathfrak{e}_7^{\oplus 2} \left(\bigotimes_{\mathfrak{e}_7}^{3,3} \right)^{10} \bigotimes_{\mathfrak{e}_7}^{3,3} \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6,$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^{10} \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}^{5,3},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes 50$$

$$I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_6,$$

$$1e_7^{(5,5)}(\otimes e_8)^9 \otimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes^7 \mathbf{e}_6^{5,5},$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^8 \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so}^{5,3},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6,$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^7 \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes^{4,5} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^6 \otimes^{5,3} \mathfrak{so},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_6^{5,5},$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^6 \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so}^{5,3},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes^{4,5} \mathbf{e}_8 \left(\otimes^{5,5} \mathbf{e}_8 \right)^4 \otimes^{5,5} \mathbf{e}_6,$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^5 \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes^{4,5} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^4 \otimes^{5,3} \mathfrak{so},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_8^{14} \otimes \mathbf{e}_6^{5, 5},$$

$$1e_7^{(5,5)} \left(\bigotimes e_8 \right)^{15} \bigotimes e_6^{(5,5)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{14,5,3} \otimes \mathfrak{so},$$

$$1e_7^{(5,5)} \otimes e_8^{15} \otimes e_9^{5,3}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{15} \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{14} \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{13} \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{12} \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{e}_8$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{e}_8$$

$$321 \mathfrak{e}_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{e}_8$$

$$31 \mathfrak{e}_6 \binom{1}{2,2} (\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_8$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{1}{4,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{so}^{3,3}$$

$$1e_6^{(3,3)}(\otimes e_7)^6 \otimes so$$

$$1e_6^{(3,3)}(\otimes e_7)^5 \otimes^{(3,3)} \mathfrak{so}$$

$$1e_6^{(3,3)}(\otimes e_7)^{15} \otimes e_8^{(3,3)}$$

$$1e_6^{(3,3)} \left(\bigotimes e_7 \right)^{14} \bigotimes so^{(3,3)}$$

$$1e_6^{(3,3)}(\otimes e_7)^{13} \otimes 50$$

$$1e_6^{(1)} \otimes e_7^{(3,3)} \otimes 12 \otimes 50^{(3,3)}$$

$$1e_6(\overset{3,3}{\otimes} e_7)^{11} \overset{3,3}{\otimes} 50$$

$$1e_6^{(3,3)} \left(\bigotimes e_7 \right)^{10} \bigotimes^{3,3} 50$$

$$1\mathbf{e}_6 \binom{2,2}{\otimes \mathbf{e}_6}^6 \binom{3,5}{\otimes \mathbf{e}_8}$$

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$$\begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8)^9 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 1223132,$$

$$\otimes^{3,5}_1 \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^9 \otimes^{5,5} \mathfrak{e}_8 12231513215,$$

$$\begin{smallmatrix} 4,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^9 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 1223132,$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^9 \otimes {}^{5,5} \mathfrak{e}_8 12231513215,$$

$$\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^9 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 1223132,$$

$$\overset{5,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513215,$$

$$313221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^9\bigotimes^{5,5}\mathbf{e}_81223132,$$

$$1^1 5^1 1^1 3^1 2^1 2^1 \mathbf{e}_8 ({}^{5,5} \otimes \mathbf{e}_8)^9 {}^{5,5} \otimes \mathbf{e}_8 1^1 2^1 2^1 3^1 1^1 2^1,$$

$$313221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^9\bigotimes^{5,5}\mathbf{e}_812231513215,$$

$$\bigotimes^{3,5} \mathfrak{e}_8 \left(\bigotimes^{5,5} \mathfrak{e}_8 \right)^9 \bigotimes^{5,5} \mathfrak{e}_8 I^{\oplus 6},$$

$$\overset{3,5}{\otimes} \mathfrak{e}_8 \overset{1}{\left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^9} \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{12231513},$$

$$\bigotimes_{4,5}^1 \mathfrak{e}_8 \left(\bigotimes_{5,5} \mathfrak{e}_8 \right)^9 \bigotimes_{5,5} \mathfrak{e}_8 I^{\oplus 6},$$

$$\overset{4,5}{\otimes} \mathfrak{e}_8 \overset{1}{\left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^9} \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{12231513},$$

$$\begin{matrix} 5,5 & 1 & (5,5) \\ \otimes & \mathfrak{e}_8 & \otimes \mathfrak{e}_8 \end{matrix} \bigg)^9 \begin{matrix} 5,5 \\ \otimes & \mathfrak{e}_8 \end{matrix} I^{\oplus 6},$$

$$\overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_8} 12231513,$$

$$I^{\oplus 5} \mathbf{e}_8^1 (\bigotimes^{5,5} \mathbf{e}_8)^9 \bigotimes^{5,5} \mathbf{e}_8 1223132,$$

$$313221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^9\bigotimes^{5,5}\mathbf{e}_8I^{\oplus 6},$$

$$313221\mathbf{e}_8\overset{1}{\left(\overset{5,5}{\otimes}\mathbf{e}_8\right)}^9\overset{5,5}{\otimes}\mathbf{e}_812231\overset{1}{5}13,$$

$$\otimes_{\mathfrak{e}_8}^{3,5} (\otimes_{\mathfrak{e}_8}^{5,5})^9 \otimes_{\mathfrak{e}_8}^{5,5} 122315123$$

$$\begin{matrix} 3,5 & 1 & (5,5) \\ \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^9 & \otimes \mathfrak{e}_8 \end{matrix} 12231513151$$

$$\otimes_{\mathfrak{e}_8}^{4,5} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,5} 122315123$$

$$\otimes_{\mathfrak{e}_8}^{4,5} 1 \otimes_{\mathfrak{e}_8}^{5,5} 9 \otimes_{\mathfrak{e}_8}^{5,5} 12231513151$$

$$\otimes_{\mathfrak{e}_8}^{5,5} (\otimes_{\mathfrak{e}_8}^{5,5})^9 \otimes_{\mathfrak{e}_8}^{5,5} 122315123$$

$$\overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151$$

$$21513221\mathbf{e}_8^{(5,5)}(\otimes \mathbf{e}_8)^9 \otimes \mathbf{e}_8^{5,5}1223132$$

$$313221\mathbf{e}_8^1(\otimes \mathbf{e}_8^{5,5})^9 \otimes \mathbf{e}_8^{5,5}122315123$$

$$313221^1 \mathbf{e}_8 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^9 \bigotimes^{5,5} \mathbf{e}_8 12231513151$$

[illegible]

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \left(\bigotimes^{5,5} \mathfrak{e}_8 \right)^{12} \bigotimes^{5,5} \mathfrak{e}_8} 122315123$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \left(\bigotimes^{5,5} \mathfrak{e}_8 \right)^{12} \bigotimes^{5,5} \mathfrak{e}_8} 12231513151$$

[illegible]

$$\begin{array}{l} \begin{smallmatrix} 3,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 122315123 \\ \begin{smallmatrix} 3,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 12231513151 \\ \begin{smallmatrix} 4,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 122315123 \\ \begin{smallmatrix} 4,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 12231513151 \\ \begin{smallmatrix} 5,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 122315123 \\ \begin{smallmatrix} 5,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes & \end{smallmatrix} \epsilon_8 12231513151 \end{array}$$

[illegible]

$$\begin{aligned}
& \otimes_{\mathbf{e}_7}^{3,3,1} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,2} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1,3,3} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{4,2} \\
& \otimes_{\mathbf{e}_7}^{3,3,1} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1,3,3} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{4,5} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1,3,3} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{1231512} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1,3,3} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{123151} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7}^{1,3,3} (\otimes_{\mathbf{e}_7}^{3,3})^{13} \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3}
\end{aligned}$$

$$\begin{array}{l}
{}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7' \otimes {}^{3,2} \\
I^{\oplus 2} {}^{1} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7 \otimes {}^{4,2} \\
{}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7' I^{\oplus 3} \\
I^{\oplus 2} {}^{1} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7 \otimes {}^{4,5} \\
I^{\oplus 2} {}^{1} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7 1231512 \\
I^{\oplus 2} {}^{1} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7 123151 \\
I^{\oplus 2} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} \otimes {}^{3,3} \otimes \mathbf{e}_7' \otimes {}^{3,3}
\end{array}$$

$$\begin{array}{l}
{}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7' \otimes \\
I^{\oplus 2} {}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7 \otimes \\
{}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7' I^{\oplus 3} \\
I^{\oplus 2} {}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 2} {}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7 1231512 \\
I^{\oplus 2} {}^{3,3}{}_1 \otimes \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7 123151 \\
I^{\oplus 2} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{11} \otimes {}^{3,3} \otimes \mathbf{e}_7' \otimes
\end{array}$$

$$\begin{array}{c} 3,3 \quad 1 \quad (3,3) \\ \otimes e_7 \quad (\otimes e_7)^{10} \quad \otimes e'_7 \quad \otimes \end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{13,5,4} \otimes \mathbf{e}_6^1$$

$$\begin{aligned} & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_7 I^{\oplus 3} \\ & 1\mathbf{e}_7^{1,5,5}(\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}_6^{5,3} \otimes \mathbf{e}_6^{2,2} \\ & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_6^{5,4} 1 \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_7 I^{\oplus 3} \\ & 1_{\mathbf{e}_7}^{1,5,5} (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_6^{5,3} \otimes \mathbf{e}_6^{2,2} \\ & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_6^{5,4} \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{5,5} \otimes \mathbf{e}_7 I^{\oplus 3}{}^{5,4} \\ & 1_{\mathbf{e}_7}^{1,5,5} (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_6^{5,3} \otimes \mathbf{e}_6^{2,2} \\ & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{5,5} \otimes \mathbf{e}_6^{5,4} \end{aligned}$$

$$I^{\oplus 2} e_7^{1,4,5} \otimes e_8^{(5,5)} \otimes e_7^{5,4} I^{\oplus 3}$$

$$1 e_7^{1,5,5} \otimes e_8^{(10,5,4)} \otimes e_7 I^{\oplus 3}$$

$$I^{\oplus 2} e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_7^{5,4} I^{\oplus 3}$$

$$1 e_7^1 (\otimes e_8^{5,5})^9 \otimes e_7^{5,4} I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,2}, \\ & 1 \epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,5} \otimes I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{6,5,4} \otimes \mathbf{e}_7^{3,3}, \\ & 1 \mathbf{e}_7^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_7^{3,3}, \\ & 1 \mathbf{e}_7^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_7^{5,5} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \mathfrak{e}_7^{1, 4, 5} \otimes \mathfrak{e}_8^{(5, 5)} \otimes \mathfrak{e}_7^{5, 4} \otimes \mathfrak{e}_7^{3, 2}, \\ & 1 \mathfrak{e}_7^{(5, 5)} \otimes \mathfrak{e}_8^{5, 4} \otimes \mathfrak{e}_7^{3, 2}, \\ & I^{\oplus 2} \mathfrak{e}_7^{1, 4, 5} \otimes \mathfrak{e}_8^{(5, 5)} \otimes \mathfrak{e}_7^{5, 5} \otimes I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1 \epsilon_7^{1,5,5} (\otimes \epsilon_8)^{6,5,4} \otimes \epsilon_7^{3,3}, \\ & 1 \epsilon_7^{1,5,5} (\otimes \epsilon_8)^{6,5,5} \otimes \epsilon_7 I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{(6,5,4)} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \mathfrak{e}_7^{1, 4, 5} \otimes \mathfrak{e}_8^{(5, 5)} \otimes \mathfrak{e}_7^{5, 4} \otimes \mathfrak{e}_8^{3, 2}, \\ & 1 \mathfrak{e}_7^{1, 5, 5} \otimes \mathfrak{e}_8^{5, 4} \otimes \mathfrak{e}_7^{3, 2}, \\ & I^{\oplus 2} \mathfrak{e}_7^{1, 4, 5} \otimes \mathfrak{e}_8^{(5, 5)} \otimes \mathfrak{e}_7^{5, 5} \otimes I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} (\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_7^{5,4,3,3}, \\ & 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}_7^{5,4,3,3}, \\ & 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}_7^{5,5} I^{\oplus 2} \end{aligned}$$

$$\begin{array}{c} I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^4 \otimes \mathbf{e}_7^{5,4} I^{\oplus 3} \\ 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_7^{5,4} I^{\oplus 3} \end{array}$$

$$\begin{aligned} & I^{\oplus 2} e_7^{1, 4, 5} \otimes e_8 (\otimes e_8)^{14} \otimes e_7^{5, 4} \otimes e_8^{3, 2}, \\ & 1 e_7^{1, 5, 5} (\otimes e_8)^{15} \otimes e_7^{5, 4} \otimes e_8^{3, 2}, \\ & I^{\oplus 2} e_7^{1, 4, 5} \otimes e_8 (\otimes e_8)^{14} \otimes e_7^{5, 5} I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{(5, 5)} {}^{14} \otimes \epsilon_7^{5, 4} \otimes \epsilon_7^{3, 3} \\ & 1 \epsilon_7^{(5, 5)} \otimes \epsilon_8^{15} \otimes \epsilon_7^{5, 4} \otimes \epsilon_7^{3, 3}, \\ & 1 \epsilon_7^{(5, 5)} \otimes \epsilon_8^{15} \otimes \epsilon_7^{5, 5} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} e_7^{1, 4, 5} \otimes e_8^{(5, 5)} \otimes e_7^{5, 4} \otimes e_8^{3, 2}, \\ & 1 e_7^{1, 5, 5} \otimes e_8^{5, 4} \otimes e_7^{3, 2}, \\ & I^{\oplus 2} e_7^{1, 4, 5} \otimes e_8^{(5, 5)} \otimes e_7^{5, 5} I^{\oplus 2} \end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{(5, 5)} \otimes^{13} \otimes^{5, 4} \epsilon_7 \otimes^{3, 3} \\
& 1 \epsilon_7^{(5, 5)} \otimes^{14} \otimes^{5, 4} \epsilon_7 \otimes^{3, 3}, \\
& 1 \epsilon_7^{(5, 5)} \otimes^{14} \otimes^{5, 5} \epsilon_7 I^{\oplus 2}
\end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{13} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8^{5,5})^{14} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_7^{5, 4} \otimes \mathbf{e}_8^{3, 2}, \\ & \mathbf{1e}_7^{1, 5, 5} \otimes \mathbf{e}_8^{(5, 4)} \otimes \mathbf{e}_7^{3, 2}, \\ & I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_7^{5, 5} I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{(5, 5)} \otimes^{12} \otimes^{5, 4} \epsilon_7 \otimes^{3, 3} \\ & 1 \epsilon_7^{(5, 5)} \otimes^{13} \otimes^{5, 4} \epsilon_7 \otimes^{3, 3}, \\ & 1 \epsilon_7^{(5, 5)} \otimes^{13} \otimes^{5, 5} \epsilon_7 I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{13,5,4} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{(5, 5)} \otimes \epsilon_7^{5, 4} \otimes \epsilon_8^{3, 2}, \\ & 1 \epsilon_7^{1, 5, 5} \otimes \epsilon_8^{5, 4} \otimes \epsilon_7^{3, 2}, \\ & I^{\oplus 2} \epsilon_7^{1, 4, 5} \otimes \epsilon_8^{(5, 5)} \otimes \epsilon_7^{5, 5} I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7^{1 \ 4,5} \otimes \epsilon_8^{(5,5)}{}^{11} \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\ & 1 \epsilon_7^{1 \ (5,5)}{}^{12} \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1 \epsilon_7^{1 \ (5,5)}{}^{12} \otimes \epsilon_7^{5,5} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_8^{11, 5, 4} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{e}_7^{5, 4} \otimes \mathbf{e}_7^{3, 2},$$

$$I^{\oplus 2} \begin{smallmatrix} 1 & 4,5 \\ \mathbf{e}_7 & \otimes \mathbf{e}_8 \end{smallmatrix} \left(\begin{smallmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{smallmatrix} \right)^{10} \begin{smallmatrix} 5,4 & 3,3 \\ \otimes \mathbf{e}_7 & \otimes \end{smallmatrix}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{10} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3}$$

$$1\mathbf{e}_7^{1\ 5,5}(\otimes \mathbf{e}_8)^{11\ 5,4\ 3,2} \otimes, \\ I^{\oplus 2}\mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{5,5}(\otimes \mathbf{e}_8)^{10\ 5,5} \otimes \mathbf{e}_7 I^{\oplus 2},$$

$$\begin{array}{l} 1e_7^{(5,5)}(\otimes e_8)^{11} \otimes e_7^{5,4} \otimes e_7^{3,3}, \\ 1e_7^{(5,5)}(\otimes e_8)^{11} \otimes e_7^{5,5} I^{\oplus 2} \end{array}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{11} \otimes e_7^{5,4} I^{\oplus 3}$$

$$\begin{aligned} & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_7^{5,4} \otimes, \\ & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_7^{5,4} I^{\oplus 2}, \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_7^{5,4} I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_7^{5,4} \otimes \epsilon_8^{3,3}, \\ & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_7^{5,5} I^{\oplus 2}, \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_7^{5,5} 1 \end{aligned}$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^9 \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_7 \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^9 \otimes \mathbf{e}'_7 1$$

$$\begin{aligned} & 1_{\mathbf{e}'_7}{}^{1,4,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{5,5} {}^8 \otimes \mathbf{e}'_7 \otimes, \\ & I^{\oplus 2} \mathbf{e}'_7 {}^{1,4,5} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^{5,5} {}^8 \otimes \mathbf{e}'_7 I^{\oplus 2}, \\ & 1_{\mathbf{e}'_7}{}^{1,5,5} \otimes \mathbf{e}_8 {}^{5,4} \otimes \mathbf{e}'_7 I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_7^{5,5} I^{\oplus 2}, \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_7^{5,5} 1 \end{aligned}$$

$$\begin{aligned} & 1e_7' \otimes e_8 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} e_8 \otimes e_7 I^{\oplus 3} \\ & I^{\oplus 2} e_7 \begin{pmatrix} 4,5 \\ \otimes \end{pmatrix} e_8 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} e_8 \otimes e_7' 1 \end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^{1,4,5} \otimes \epsilon_8 (\otimes \epsilon_8)^{5,5} 7^{5,4} \otimes \epsilon_7^{3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8 (\otimes \epsilon_8)^{5,5} 7^{5,4} \otimes \epsilon_7' I^{\oplus 2}, \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^8 \otimes \epsilon_7' I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7^{5,5} I^{\oplus 2}, \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_7^{5,5} 1 \end{aligned}$$

$$1e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$I^{\oplus 2} e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,5} e'_7 1$$

$$\begin{aligned} & 1\epsilon_7^{1,4,5} \otimes \epsilon_8 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_7^{5,4} \otimes \epsilon_8^{3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_7^{5,4} I^{\oplus 2}, \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{7,5,4} \otimes \epsilon_7' I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}'_7} \otimes_{\mathbf{e}_8}^{1,4,5} (\otimes_{\mathbf{e}_8}^{5,5})^6 \otimes_{\mathbf{e}_7}^{5,4} \otimes_{\mathbf{e}_7}^{3,3}, \\ & 1_{\mathbf{e}'_7} \otimes_{\mathbf{e}_8}^{1,4,5} (\otimes_{\mathbf{e}_8}^{5,5})^6 \otimes_{\mathbf{e}_7}^{5,5} \mathbf{e}_7 I^{\oplus 2}, \\ & 1_{\mathbf{e}'_7} \otimes_{\mathbf{e}_8}^{5,5} (\otimes_{\mathbf{e}_8}^{5,5})^7 \otimes_{\mathbf{e}'_7}^{5,5} 1 \end{aligned}$$

$$1e_7' \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$I^{\oplus 2} e_7 \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,5} e_7' 1$$

$$\begin{aligned} & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \otimes (\mathbf{e}_8)^5 \otimes \mathbf{e}'_7 \otimes \mathbf{e}_8, \\ & I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}_8 \otimes (\mathbf{e}_8)^5 \otimes \mathbf{e}'_7 I^{\oplus 2}, \\ & 1\mathbf{e}'_7 \otimes (\mathbf{e}_8)^6 \otimes \mathbf{e}'_7 I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_7^{5,5} I^{\oplus 2}, \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_7^{5,5} 1 \end{aligned}$$

$$1\epsilon_7' \otimes^{4,5} \epsilon_8(\otimes^{5,5} \epsilon_8)^5 \otimes^{5,4} \epsilon_7 I^{\oplus 3}$$

$$I^{\oplus 2} \epsilon_7 \otimes^{4,5} \epsilon_8(\otimes^{5,5} \epsilon_8)^5 \otimes^{5,5} \epsilon_7' 1$$

$$\begin{aligned} & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^4 \otimes \epsilon_7^{5,4} \otimes^{3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^4 \otimes \epsilon_7^{5,4} I^{\oplus 2}, \\ & 1\epsilon_7^{(1,5,5)} (\otimes \epsilon_8)^5 \otimes \epsilon_7^{5,4} I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 1 \epsilon_7^{1,4,5} \otimes \epsilon_8 ({}^{5,5} \epsilon_8)^4 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3}, \\ & 1 \epsilon_7^{1,4,5} \otimes \epsilon_8 ({}^{5,5} \epsilon_8)^4 \otimes \epsilon_7^{5,5} I^{\oplus 2}, \\ & 1 \epsilon_7^{1,5,5} ({}^{5,5} \epsilon_8)^5 \otimes \epsilon_7^{5,5} 1 \end{aligned}$$

$$1e_7' \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$I^{\oplus 2} e_7 \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,5} e_7' 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{14} \otimes^{5,4} e_7 \otimes^{3,2},$$

$$1e_7' \otimes e_8 \left(\otimes e_8 \right)^{14} \otimes e_7 \otimes e_8,$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{14} \otimes^{5,4} e_7 I^{\oplus 3}$$

$$I^{\oplus 3} \mathfrak{e}_7^{(3,3)} (\otimes \mathfrak{e}_7)^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$I^{\oplus 3} \mathbf{e}_7^{1(3,3)} (\otimes \mathbf{e}_7)^{11} \otimes \mathbf{so}^{3,2} 1$$

$$I^{\oplus 3} \mathbf{e}_7^{1(3,3)} (\otimes \mathbf{e}_7)^{10} \otimes \mathbf{so}1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^9\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^8 \otimes \mathfrak{so}1^{2,2}$$

$$31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^7 \otimes \mathfrak{so}1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^6\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes} \mathfrak{e}_6)^5 \overset{2,2}{\otimes} \mathfrak{so}1$$

$$31\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{15} \overset{2,2}{\otimes} \mathbf{so}1$$

$$31\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{14} \overset{2,2}{\otimes} \mathfrak{so}1$$

$$31\mathfrak{e}_6(2,2 \otimes \mathfrak{e}_6)^{13} 2,2 \otimes \mathfrak{so}1$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{12} \otimes \mathfrak{so}1$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{so}1$$

[illegible]

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1 \mathbf{e}_6^1 \otimes^{3,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^6 \otimes^{5,4} \mathbf{e}_6 1$$

$$1 \overset{1}{e}_6 \otimes \overset{3,5}{e}_8 (\otimes \overset{5,5}{e}_8)^5 \otimes \overset{5,4}{e}_6 1$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{e}_6^{5,4} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^{13} \otimes \mathbf{e}_6^{5,4} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^{12} \otimes \mathbf{e}_6^{5,4} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^{11} \otimes \mathbf{e}_6^{5,4} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1e_7 \binom{5,5}{\otimes} e_8 \binom{10}{\otimes} 4,2 \otimes 1,1 \otimes 1$$

$$1e_7^{1(5,5)}(\otimes e_8)^{10} \otimes so_{13}^{4,2}$$

$$1e_7(\overset{1}{\otimes} \overset{5,5}{e_8})^9 \overset{4,2}{\otimes} so \overset{1,1}{\otimes} so1$$

$$1e_7^{(5,5)}(\otimes e_8)^9 \otimes so_{13}$$

$$1e_7 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} so \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} so 1$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^9 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6^{(3,3)} (\otimes \mathbf{e}_7)^8 \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6^{(3,3)} (\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6^{(3,3)} (\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^{15}\overset{3,3}{\otimes}\mathbf{e}_6I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_6 I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_7' \left(\bigotimes \mathbf{e}_7 \right)^7 \bigotimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^6 \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^5 \binom{4,5}{\otimes} \mathfrak{e}_8 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^5 \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{15} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{14} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{13} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{12} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{11} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{10} \binom{3,2}{\otimes} \mathfrak{so}1$$

$$1 \mathfrak{e}_8 \binom{1}{4,2} \otimes \mathfrak{so} \binom{1,1}{\otimes \mathfrak{so}}^4 \binom{1,1}{\otimes} \mathfrak{so}1$$

$$1 \mathfrak{e}'_7 \binom{1}{4,5} \otimes \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^9 \binom{4,2}{\otimes} \mathfrak{so} \binom{1,1}{\otimes} \mathfrak{so}1$$

$$1 \mathfrak{e}'_7 \binom{1}{4,5} \otimes \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^9 \binom{4,2}{\otimes} \mathfrak{so}13$$

$$1 \mathfrak{e}'_7 \binom{1}{4,5} \otimes \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^8 \binom{4,2}{\otimes} \mathfrak{so} \binom{1,1}{\otimes} \mathfrak{so}1$$

$$1 \mathfrak{e}'_7 \binom{1}{4,5} \otimes \mathfrak{e}_8 \binom{5,5}{\otimes \mathfrak{e}_8}^8 \binom{4,2}{\otimes} \mathfrak{so}13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,2} 1$$

$$1e_6^{(3,3)}(\otimes e_7)^8 \otimes^{3,2} \mathfrak{so}1$$

$$1e_6^{(3,3)}(\otimes e_7)^7 \otimes^{3,2} \mathfrak{so}1$$

$$1e_6^{(3,3)} \otimes e_7^{(3,2)} \otimes \mathfrak{so}1$$

$$1e_6^{(3,3)} \otimes e_7^{(3,2)} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{(3,3)} \otimes \mathfrak{e}_7^{15} \otimes \mathfrak{so}^{3,2} 1$$

$$1e_6(\overset{3,3}{\otimes} e_7)^{14} \overset{3,2}{\otimes} so1$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^{13} \otimes^{3,2} \mathfrak{so}1$$

$$1\mathfrak{e}_6^{(3,3)} \otimes \mathfrak{e}_7^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{so}^{3,2}1$$

$$1e_6^{(3,3)} \otimes e_7^{10} \otimes so1^{3,2}$$

$$1\mathbf{e}_6^{(1)}(\otimes^{(2,2)}\mathbf{e}_6)^5\otimes^{(3,5)}\mathbf{e}_81$$

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$$\begin{aligned} & \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1,5,3}{\otimes}, \\ & 313221 \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1,5,3}{\otimes}, \\ & \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{1223151}, \\ & 313221 \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1,5,4}{\otimes}, \\ & \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{1223151}, \end{aligned}$$

$$\begin{aligned} & \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,5,4 \\ \otimes \end{smallmatrix}, \\ & I^{\oplus 5} \begin{smallmatrix} 1,5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,5,3 \\ \otimes \end{smallmatrix}, \\ & \begin{smallmatrix} 4,5,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,5,4 \\ \otimes \end{smallmatrix}, \\ & I^{\oplus 5} \begin{smallmatrix} 1,5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,5,4 \\ \otimes \end{smallmatrix}, \\ & \begin{smallmatrix} 5,5,1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1,5,5 \\ \otimes \end{smallmatrix}, \end{aligned}$$

$$\begin{array}{l} 3,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8 \otimes \\ 3,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8 12231512 \\ 4,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \\ 4,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8 12231512 \\ 313221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \end{array}$$

$$\begin{aligned}
& 3,5 \overset{1}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3}, \\
& \overset{5,5}{\otimes} \mathfrak{e}_8 (\overset{1}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 2}, \\
& 313221 \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3}, \\
& \overset{1}{1513221} \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 4}, \\
& \overset{1}{1513221} \overset{1}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3}, \\
& I^{\oplus 2} \overset{2,4}{\otimes} \overset{5,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3}, \\
& I^{\oplus 3} \overset{3,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& 4,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 2} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 4}, \\
& 5,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 3} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 3}, \\
& I^{\oplus 5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 4} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 4}, \\
& I^{\oplus 5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 3} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 3}, \\
& I^{\oplus 2} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 24,2} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \otimes, \\
& I^{\oplus 4} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 2} \oplus \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 4},
\end{aligned}$$

$$\begin{array}{l}
4,5 \quad 1 \quad 5,5 \\
\otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 3} \quad I^{\oplus 3} \\
313221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 2} \quad I^{\oplus 4} \\
21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 2} \quad I^{\oplus 4} \\
21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 3} \quad I^{\oplus 3} \\
I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 24,2} \quad \otimes \\
I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^5 \quad 5,5 \quad I^{\oplus 3} \quad I^{\oplus 3}
\end{array}$$

[illegible]

[illegible]

[illegible]

$$\begin{matrix} 3,5 & 1 & (5,5) \\ \otimes & \mathbf{e}_8 & (\otimes \mathbf{e}_8)^{14} \end{matrix} \begin{matrix} 5,5 & 1 & 5,3 \\ \otimes & \mathbf{e}_8 & \otimes, \end{matrix}$$

$$\begin{matrix} 3,5 & 1 & (5,5) \\ \otimes & \mathbf{e}_8 & (\otimes \mathbf{e}_8)^{14} \end{matrix} \begin{matrix} 5,5 & 1 & 5,4 \\ \otimes & \mathbf{e}_8 & \otimes, \end{matrix}$$

$$\begin{matrix} 3,5 & 1 & (5,5) \\ \otimes & \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^{14} \end{matrix} \begin{matrix} 5,5 & 1 & 5,5 \\ \otimes & \mathfrak{e}_8 & \otimes \end{matrix}$$

$$\begin{array}{c} 3,5 \quad 1 \quad (5,5) \\ \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \quad \otimes \mathfrak{e}_8 \otimes \end{array}$$

$${}^{3,5}{}^1_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{11} {}^{5,5}{}^1_1 \otimes \mathfrak{e}_8 12231512$$

$$\begin{array}{c} 4,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \quad \otimes \mathfrak{e}_8 \otimes \end{array}$$

$$\overset{4,5}{\otimes} \overset{1}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_8} 12231512$$

$$313221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^{11}\otimes^{5,5}\mathbf{e}_8\otimes^{1,5,5}$$

$$\begin{array}{ccccccc} 5,5 & 1 & & 5,5 & & 5,5 & 1 \\ \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^{11} & \otimes \mathfrak{e}_8 & 1223151 \end{array}$$

$$313221\mathbf{e}_8^1(\otimes \mathbf{e}_8)^{5,5} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 12231512$$

$$I^{\oplus 5} \mathfrak{e}_8^1 (\bigotimes \mathfrak{e}_8^{5,5})^{11} \bigotimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 12231512$$

$$1513221\epsilon_8^1(\otimes^{5,5}\epsilon_8)^{11}\otimes^{5,5}\epsilon_8^112231512$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{11} {}^{5,5} I^{\oplus 2} {}^{4,2}_2 \otimes$$

$$I^{\oplus 5} \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{11} \bigotimes \mathbf{e}_8 \bigotimes$$

$${}^{3,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{11} {}^{5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 I^{\oplus 4}$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{11} {}^{5,5} I^{\oplus 3} \otimes \mathfrak{e}_8 I^{\oplus 3}$$

$$313221 \mathbf{e}_8^1 (\otimes \mathbf{e}_8^{5,5})^{11} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} I^{\oplus 4}$$

$$21513221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{11} \bigotimes \mathbf{e}_8 I^{\oplus 2} I^{\oplus 4}$$

$$21513221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^{11} \otimes^{5,5} \mathbf{e}_8 I^{\oplus 3}$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{p}_8 \end{smallmatrix} \left(\begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{p}_8 \end{smallmatrix} \right)^{11} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{p}_8 \end{smallmatrix} I^{\oplus 2} \begin{smallmatrix} 4,2 \\ \otimes \\ \mathfrak{p}_8 \end{smallmatrix}$$

$$\oplus_4 I^{\oplus 2}_{(5,5)} \oplus_{11} I^{\oplus 3}_{(5,5)} \oplus_3$$

$$1 \rightarrow \mathfrak{e}_8 \rightarrow (\oplus \mathfrak{e}_8) \rightarrow \oplus \mathfrak{e}_8 \rightarrow 1$$

$$\begin{array}{c} 3,5 \quad 1 \quad (5,5) \\ \otimes \mathfrak{e}_8 \quad (\otimes \mathfrak{e}_8)^{10} \quad \otimes \mathfrak{e}_8 \quad \otimes \end{array}$$

$${}^{3,5}{}^1_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^{10} {}^{5,5}{}^1_1 \otimes \mathfrak{e}_8 12231512$$

$$\begin{array}{c} 4,5 \quad 1 \quad (5,5) \\ \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \quad \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 \otimes \end{array} \end{array}$$

$$\otimes_{\mathfrak{e}_8}^{4,5} 1 \left(\otimes_{\mathfrak{e}_8}^{5,5} \right)^{10} \otimes_{\mathfrak{e}_8}^{5,5} 1 12231512$$

$$313221\mathbf{e}_8(\overset{1}{\otimes}\overset{5,5}{\mathbf{e}_8})^{10}\overset{5,5}{\otimes}\overset{1}{\mathbf{e}_8}\overset{5,5}{\otimes}$$

$$\begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \mathbf{e}_8 \end{array} (\begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \mathbf{e}_8 \end{array})^{10} \begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \mathbf{e}_8 \end{array} 1223151$$

$$313221\epsilon_8^1(\otimes \epsilon_8)^{5,5} 10 \otimes \epsilon_8^{5,5} 12231512$$

$$I^{\oplus 5} \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^{10} \bigotimes \mathfrak{e}_8 12231512$$

$${}^1 1513221 {}^1 \mathbf{e}_8 ({}^{5,5} \otimes \mathbf{e}_8)^{10} {}^{5,5} \otimes {}^1 \mathbf{e}_8 12231512$$

[illegible]

$$1\mathbf{e}_7 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} 12 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} I^{\oplus 2} 4,2 \begin{pmatrix} \\ \otimes \end{pmatrix}$$

$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{10} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^{1,5} \otimes \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{10} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^{1,5} 12231512 \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{10} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} \\
& 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^{1,5,4} \otimes \\
& 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^{1,5} I^{\oplus 5} \\
& 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2} \otimes
\end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^9 & \quad \mathbf{e}_7^{3,3} \bigotimes \mathbf{e}_7^{1,3,2} \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^9 & \quad \mathbf{e}_7^{3,3} I^{\oplus 3} \mathbf{e}_7^1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^9 & \quad \mathbf{e}_7^{3,3} I^{\oplus 2} \mathbf{e}_7^1 I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^8 & \quad \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1,3,2} \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^8 & \quad \mathbf{e}_7^{3,3} I^{\oplus 3} \mathbf{1} \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^8 & \quad \mathbf{e}_7^{3,3} I^{\oplus 2} \mathbf{e}_7 I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^7 \mathbf{e}_7^{3,3} \mathbf{e}_7^{1,3,2} \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^7 \mathbf{e}_7^{3,3} I^{\oplus 3} 1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^7 \mathbf{e}_7^{3,3} I^{\oplus 2} \mathbf{e}_7 I^{\oplus 2} \end{aligned}$$

$$\begin{aligned} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^6 \bigotimes \mathbf{e}_7^1 \mathbf{e}_7^2 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^6 \bigotimes \mathbf{e}_7^1 I^{\oplus 3} 1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^6 \bigotimes \mathbf{e}_7^1 I^{\oplus 2} I^{\oplus 2} \end{aligned}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{15} \otimes e_7^{1(5,5)}1$$

$$1\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}\mathbf{e}_71$$

$$1\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{13}\overset{5,5}{\otimes}\mathbf{e}_71$$

$$1\mathbf{e}_7^{1(5,5)}(\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}_7^{1(5,5)}1$$

$$1\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{11}\overset{5,5}{\otimes}\mathbf{e}_71$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^9 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^8 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^7 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^6 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^5 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$1 \overset{1}{e}'_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^4 \overset{5,5}{\otimes} \overset{1}{e}_7 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{14} \otimes^{5,5} e_7 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_7 1$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^{12} \overset{5,5}{\otimes} e_7 1$$

$$1\overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \otimes \overset{5,5}{\mathbf{e}}_7 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,5} e_7 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \mathbf{e}_6^{1 \ 3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_7^{5,5} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^{14} \otimes \mathbf{e}_7^1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{12} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$I^{\oplus 2} \mathbf{e}'_7 ({}^{3,3} \bigotimes \mathbf{e}_7)^9 {}^{3,3} \bigotimes \mathbf{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^8 \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^7 \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^6 \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^5 \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{15} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{14} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{13} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{12} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{11} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \binom{1}{3,3} (\otimes \mathfrak{e}_7)^{10} \otimes \binom{3,3}{3,3} I^{\oplus 2} \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \binom{1,1}{1,1} (\otimes \mathfrak{so})^9 \otimes \binom{1,1}{1,1} 1\mathfrak{so} 1$$

$$1\mathfrak{so} \binom{1,1}{1,1} (\otimes \mathfrak{so})^8 \otimes \binom{1,1}{1,1} 1\mathfrak{so} 1$$

$$1\mathfrak{so} \binom{1,1}{1,1} (\otimes \mathfrak{so})^7 \otimes \binom{1,1}{1,1} 1\mathfrak{so} 1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^6\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^5\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{15}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{14}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{13}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{12}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{11}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^9\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^6\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^5\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^4\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_7^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^9 \otimes \epsilon_7^{3,3} I^{\oplus 2} 1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^9\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^8\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^8\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^7\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^6\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^6\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

2.8.7 $\mathfrak{gLig}(\mathfrak{n}-1)(3)$

$$\mathfrak{e}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^5\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{e}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^5\overset{5,4}{\otimes}\mathfrak{e}_6$$

$$\mathfrak{e}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^5\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}$$

2.8.8 $\mathfrak{gLigLg}(\mathfrak{n}-4)\mathfrak{LigLg}(1)$

$$\mathfrak{so}\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^4\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8$$

2.8.9 $\mathfrak{SgLig}(\mathfrak{n}-1)(26)$

$$I^{\oplus 2}\mathfrak{e}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^3\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so},$$

$$1\mathfrak{so}\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^4\overset{5,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{e}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^4\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so},$$

$$1\mathfrak{so}\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}_8})^5\overset{5,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^9 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^9 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^8 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^8 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^7 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^7 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^6 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_7^{5,5'}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,4}$$

2.8.10 SgLig(n-3)LigLg(1)

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{1,5,5})^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

2.8.11 SgLig(n-3)LigLgS(11)

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^9 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^8 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^6 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^5 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{4,2}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{8(5,5)} \otimes e_8^{1(4,2)} \otimes \mathfrak{so}1$$

$$1e_7^{(5,5)}(\otimes e_8)^{7(5,5)} \otimes e_8^{(1,4,2)} \otimes \mathfrak{so}1$$

$$1e_7^{1(5,5)}(\otimes e_8)^{6(5,5)} \otimes e_8^{1(4,2)} \otimes so1$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^5 \otimes e_8^{(1)} \otimes sol^{(4,2)}$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^4 \otimes e_8^{(5,5)} \otimes so1^{(4,2)}$$

$$1e_7(\overset{5,5}{\otimes} e_8)^{14} \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} \mathfrak{so}1$$

$$1\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{13}\overset{5,5}{\otimes}\mathbf{e}_8\overset{4,2}{\otimes}\mathbf{so}1$$

$$1e_7^{(1)}(\otimes^{5,5} e_8)^{12} \otimes^{5,5} e_8^{(1)} \otimes^{4,2} \mathfrak{so}1$$

$$1e_7^{(1)}(\otimes^{5,5} e_8)^{11} \otimes^{5,5} e_8^{(1)} \otimes^{4,2} \mathfrak{so}1$$

$$1e_7' \otimes e_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes & \otimes & \end{smallmatrix} \text{sol}$$

$$1e_7' \otimes e_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \end{smallmatrix} e_8 \otimes sol$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,5} e_8 \otimes^{4,2} sol$$

$$1e_7' \otimes^{4,5} e_8(\otimes^{5,5} e_8)^5 \otimes^{5,5} e_8 \otimes^{4,2} sol$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^4 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^3 \otimes \epsilon_8 \otimes \epsilon_8 1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^3 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{13} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{12} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{11} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{10} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^9 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^8 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^5 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^4 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^3 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6^{1\ 3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{13} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^{1\ 3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{12} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^{1\ 3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^{1\ 3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 4,2} \otimes \mathfrak{so}1$$

2.8.16 Sg(n-2)LlgLg(15)

$$\otimes \epsilon_8^{3,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5},$$

$$\otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5},$$

$$\otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$313221\epsilon_8^{5,5} (\otimes \epsilon_8)^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5},$$

$$I^{\oplus 5} \epsilon_8^{5,5} (\otimes \epsilon_8)^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5},$$

$$21513221\epsilon_8^{5,5} (\otimes \epsilon_8)^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$1513221\epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5},$$

$$1\epsilon_7^{5,5} (\otimes \epsilon_8)^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$5131513221\epsilon_8^{5,5} (\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$321\epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$31\epsilon_6^{3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$1\epsilon_7'^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$1\epsilon_6^{3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\ 5,5}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{1,5,5} \mathfrak{e}_8 \right)^{14} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^{13} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{12} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^{11} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^{10} \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^9 \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^8 \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^7 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_6 1$$

$$150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^6 \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^5 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 1$$

$$1_{\mathfrak{so}} \otimes^{2,4} e_8(\otimes^{5,5} e_8)^4 \otimes^{5,4} e_6 1$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8)^{14} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_6 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{1,1}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{1,1}$$